Integer multiplication in $O(n \log n)$ time on a single two-dimensional grid, with a machine-checked proof*

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Abstract

We give an algorithm that multiplies two n-bit integers in $O(n \log n)$ steps on a deterministic multitape Turing machine, together with a machine-checked (Lean 4) proof of an $O(n \log n)$ step bound for an explicit eight-stack machine. The algorithm of Harvey and van der Hoeven reaches the same bound by routing a one-dimensional Fourier transform through a multidimensional transform of pairwise coprime lengths, which requires Chinese remaindering and a resampling step between grids of different sizes. Our algorithm works on a single two-dimensional power-of-two grid of the same size as the transform it computes. Cooley–Tukey twiddle factors are read as fractional row shifts. Each row is spread by a data-independent sinc-Gaussian kernel, the two-dimensional transform is computed by a multiplication-free polynomial transform and recursive one-dimensional transforms, and the spreading is undone by a two-term recurrence per column whose conditioning is uniform. Because the kernel's passband stays flat up to the Nyquist frequency, the working precision grows only additively along the recursion. A single-threaded C implementation using GMP for inner products agrees bit for bit with GMP on all tested inputs (one random operand pair per size, up to a billion bits), while running over a hundred times slower.

*This work was carried out with substantial assistance from AI agents (Gemini 4 and other models), including proof development, auditing, and the Lean 4 formalization; see Section 1.4.

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1 Introduction

1.1 Background

Fourier-based integer multiplication splits two n-bit nonnegative integers A and B into $N = \lceil n/b \rceil$ radix-2^b digits of $b = \Theta(\log n)$ bits. It chooses a transform length $R_0 = \Theta(n/\log n)$ with $R_0 \geq 2N$ and evaluates a cyclic convolution of length R_0 at fixed-point precision $p_0 = \Theta(\log n)$ bits. A one-dimensional radix-2 discrete Fourier transform (DFT) of length R_0 performs $\frac{1}{2} R_0 \log_2 R_0$ complex butterfly multiplications on p_0-bit operands. If each p_0-bit product costs $\mathsf{M}(p_0)$ steps, these twiddle multiplications alone cost $\Theta(R_0 \log R_0 \cdot \mathsf{M}(p_0)) = \Theta(n \cdot \mathsf{M}(\log n))$ steps, an extra factor of $\Theta(\log \log n)$ over $n \log n$ when $\mathsf{M}(p) = \Theta(p \log p)$ [HvdH21]. Schönhage and Strassen [SS71] achieved $O(n \log n \log \log n)$ steps on a multitape Turing machine by working in Fermat quotient rings $\mathbb{Z}/(2^{2^k} + 1)\mathbb{Z}$. Fürer [Für07, Für09] reduced the $\log \log n$ factor to $2^{O(\log^* n)}$ using synthetic roots in cyclotomic rings, and Harvey, van der Hoeven, and Lecerf [HvdHL16] sharpened that bound to $O(n \log n \cdot 8^{\log^* n})$ unconditionally.

A classical algebraic tool for eliminating butterfly multiplications is the polynomial transform of Nussbaumer and Quandalle [NQ79, Nus80, Nus82]. In the negacyclic quotient ring $\mathcal{R}_r := \mathbb{C}[y]/(y^r + 1)$ where r is a power of two, the indeterminate y satisfies $y^r \equiv -1$ and has multiplicative order $2r$. Multiplying an element of $\mathcal{R}_r$ by any power of y is a signed cyclic shift of its r complex coefficients and requires zero scalar multiplications. Combined with negacyclic-to-cyclic weighting [SS71, CF94], a polynomial transform of length t_1 over $\mathcal{R}_r$ ($t_1 \mid r$) reduces a two-dimensional transform $F_{t_1,r}$ to t_1 one-dimensional transforms F_r and an odd-stride permutation [NQ79, Nus82]. Integer multiplication, however, requires a one-dimensional transform $F_{t_1 r}$, which differs from $F_{t_1,r}$ by the Cooley–Tukey twiddle factors $e^{2\pi i j_1 u_2/(t_1 r)}$.

Harvey and van der Hoeven [HvdH21] achieved $O(n \log n)$ steps by bridging one-dimensional and power-of-two multidimensional transforms through Chinese remaindering. Their algorithm routes the length-R transform through a d-dimensional Good–Thomas transform of pairwise coprime lengths and resamples each axis to a power-of-two length by Gaussian convolution, a step related to nonuniform Fourier transform (NUFFT) gridding [DR93, PST01, GL04, BMK19]. Like the present paper, it evaluates the resulting power-of-two multidimensional transforms by Nussbaumer's polynomial transforms [NQ79, Nus80, Nus82]. The proof of their main theorem takes $d = 1729$ [HvdH21]. Because the source length and target power-of-two length on each axis differ, the resampling offset varies across every coordinate index and expands the grid at each recursive step. A recent preprint [OAI26] gives an $O(n(\log n)^{1-\kappa})$ algorithm on a multitape Turing machine by combining a Harvey–van der Hoeven-style architecture with finite linear network primitives; the present paper does not use it.

1.2 Main results

We show that $O(n \log n)$ integer multiplication requires neither Chinese remaindering nor resampling between grids of different sizes. Instead, we relate $F_{t_1 r}$ and $F_{t_1,r}$ directly on a single two-dimensional power-of-two grid $R = t_1 r$. Throughout, an n-bit integer means a nonnegative integer in $[0, 2^n)$ (signed n-bit integers reduce immediately in $O(n)$ steps).

Theorem 1. *There is a deterministic multitape Turing machine and a constant $C > 0$ such that, for every $n \geq 2$ and any two n-bit integers A and B, given the binary representations of A and B on the input tape, the machine writes the $2n$-bit binary product on the output tape in at most $Cn \log_2 n$ steps.*

The machine of Theorem 1 has the following structural properties. Every Fourier transform it performs has power-of-two length, and each top-level transform of length $R_0 = \Theta(n/\log n)$ recurses on $t_{1,\ell} \times r_\ell$ sub-grids of the same total size ($R_\ell = t_{1,\ell} r_\ell$) with no grid expansion, no Chinese remainder map, no prime transform length, and no resampling between grids of different sizes. Inner products of $O((\log n)^2)$-bit integers are delegated to the same algorithm one level down, backed at the bottom stage by a one-dimensional radix-2 Fourier multiplier using schoolbook arithmetic.

Theorem 2 (machine-checked)**.** *A fixed stack machine* `tmIntMul` *(Mathlib's* `Turing.FinTM2`*: eight binary stacks, 32 register states, a fixed finite control) satisfies the following, verified in Lean 4 [dMU21, Mat20]. For every $n \geq 2$ and all $A, B < 2^n$, started with the $2n$ input bits it halts with the $2n$ bits of $A \cdot B$ on its output stack within $C \cdot n\lfloor \log_2 n \rfloor$ steps, for a constant C (the formal statement is existential; its proof exhibits the explicit witness* `C_tm_total` *$\approx 5 \cdot 10^{92}$, Section 8,* `IntMul.tmIntMul_evalsToInTime_n_log_n`*). Every step executes $O(1)$ elementary stack operations (*`IntMul.tmIntMul_elementary_stack_ops_n_log_n`*). The proof uses no axioms beyond* `propext`*,* `Classical.choice`*, and* `Quot.sound`*.*

Table 1: Comparison of transform components in [HvdH21] and in this paper.

Component	[HvdH21]	This paper
CRT map	Good–Thomas embedding onto coprime axes	None
Coprime lengths	Pairwise coprime axis lengths	None (powers of two t_1, r only)
Resampling	Gaussian resampling between grids of different sizes	Same-grid row spreading by windowed sinc-Gaussian kernel
Multidimensional transform	Nussbaumer polynomial transforms	Nussbaumer polynomial transform over $\mathbb{C}[y]/(y^r + 1)$ (one two-dimensional level)
Dimension	High-dimensional ($d = 1729$ in main proof)	Two-dimensional ($t_1 \times r$ grid, $t_1 r = R$)
Working precision	$O(\log n)$ bits, fixed per transform; resampling conditioned by oversampling the target grid	$O(\log n)$ bits at every level, additive schedule $p_{\ell+1} = p_\ell + 2K_\ell + 12$; two-term solvers conditioned by at most $2.36\, t_1$, no oversampling

Table 1 summarizes what the reduction removes relative to [HvdH21]. The tools are classical; the contribution is combining a same-grid fractional-shift Poisson bridge with a windowed sinc-Gaussian kernel, uniform two-term cyclic and Nyquist deconvolution, an additive precision schedule, and a machine-checked $O(n \log n)$ bound for an explicit eight-stack machine. Specifically, the polynomial transform over $\mathbb{C}[y]/(y^r + 1)$ and its reduction of $F_{t_1,r}$ to t_1 calls of F_r with the odd-stride permutation are due to Nussbaumer and Quandalle [NQ79, Nus80, Nus82], and the negacyclic twist is classical [SS71, CF94]. Kernel spreading analyzed by Poisson summation comes from NUFFT gridding [DR93, PST01, GL04, BMK19], while row spreading uses Kronecker substitution [Sch82, Har09] with overlap-add alongside the classical two-call real convolution identity. Our written proof imports no external algorithmic subroutine as a black box, and its analytic inputs are standard real-analysis facts: Poisson summation for Schwartz functions, the convolution theorem and Parseval's identity, and elementary series and Gaussian tail estimates.

1.3 Comparison with related work

Harvey and van der Hoeven [HvdH21] route a length-R transform through a d-dimensional transform of pairwise coprime lengths via Chinese remaindering and resample each axis onto a larger power-of-two length by Gaussian convolution. Their recursion uses a large fixed dimension. Our algorithm works on a single two-dimensional power-of-two grid $R_\ell = t_{1,\ell} r_\ell$ at every recursion depth ℓ. The R_0/R_ℓ active subproblems at depth ℓ partition the original R_0 complex entries with zero grid expansion, while the flat passband of the windowed sinc-Gaussian kernel keeps the working precision $O(\log n)$ across all $O(\log \log n)$ levels.

A companion manuscript [CLW26] generalizes the same-grid fractional-shift bridge to d dimensions with a complex half-bin phase that removes the Nyquist special case, and combines it with the finite linear network primitives of [OAI26]; it claims an $o(n \log n)$ bound on a deterministic multitape Turing machine. Nothing in the present paper depends on [CLW26]. Its main theorem treats the inner multiplier on $O((\log n)^2)$-bit operands as a black box; its fully CRT-free instantiation and its machine-checked proof use (a formalization of) the present two-dimensional multiplier as that inner multiplier. The present paper is the self-contained two-dimensional, real-kernel instance: the one that is executed up to 10^9 bits in C/GMP and whose complexity bound is machine-checked as a statement about one concrete eight-stack machine.

1.4 Verification and executability

We formalize Theorem 2 in Lean 4 [dMU21] on Mathlib [Mat20] in the library `IntMul` [LeanMS26], comprising 27 files and 82,377 lines (Section 8). The development defines the explicit eight-stack binary machine `tmIntMul` and proves its bit-exactness and $O(n \log n)$ step and elementary-operation bounds for all $n \geq 2$. The analytic and fixed-point bounds on which the formal proof rests (Poisson identity, passband, stopband and conditioning bounds, spreading, multi-level accuracy, root recovery) are proved inside the same library; Table 3 lists the correspondence we established and the places where the formal constants are looser than the printed ones. Compile-time axiom checks verify that the top-level theorems use no axioms beyond `propext`, `Classical.choice`, and `Quot.sound`.

We also evaluate the C/GMP implementation `intmul_fastdft` (`algo=1`) of the algorithm of this paper, with the windowed sinc-Gaussian kernel and the additive schedule and inner products by GMP [CGMP26] (Section 9). Single-threaded on one Zen 3 core, the C/GMP program multiplies two 10^8-bit integers in 92.5 s excluding 5.5 s of one-time table setup (98.0 s including it), about 153 times slower than GMP's `mpz_mul` (0.606 s) on the same operands (153 to 323 times across the four measured sizes, 162 to 326 times including table setup). It agrees bit for bit with GMP at every size tested (one random operand pair per size), up to 10^9 bits.

Use of AI tools. This work was developed with substantial assistance from AI agents (Gemini 4 and other AI models), working under the direction of the authors. That assistance spanned the algorithm, the proofs in this paper, the C/GMP implementation and its benchmark harness, and the Lean 4 formalization. The authors have checked the written proofs in this paper. No author has checked the formal Lean proof line by line; the correctness of the formal development rests on verification by Lean's kernel.

The authors have checked the formal statements of the machine, its uniformity, and the step bound of Theorem 2 in `IntMul/TMEndToEnd.lean` and the encoding conventions in `IntMul/Defs.lean`. The further definitions and the elementary-operation theorem that fix the meaning of Theorem 2 (the `IsUniformFinTM2` predicate and the `CoeDep` instance in `IntMul/TMPrimitives.lean`, and `stmtMaxElemOps`, `FinTM2BoundedElemOps`, and `tmIntMul_elementary_stack_ops_n_log_n`

in `IntMul.lean`) are quoted verbatim in Section 8 so that readers can inspect them. Agreement among AI systems was used only as a working heuristic during exploration, never as mathematical evidence. Every theorem and lemma stated in this paper is established by a complete written proof, an explicit citation to the literature, or a kernel-checked formal theorem. Statements about the companion manuscript report its claims, standard facts about models of computation (such as the constant-factor simulation between stack machines and multitape Turing machines) are stated as such, and the measurements of Section 9 are read from the logs of the runs described there.

1.5 Organization

Section 2 gives an informal technical overview of the algorithm, the kernel, and the complexity accounting. Section 3 fixes notation, the multitape Turing machine model, fixed-point grids, elementary-function tables, tape routing, and the polynomial transform. Section 4 presents the Nussbaumer–Quandalle polynomial-transform reduction and proves the fractional-shift Poisson identity. Section 5 analyzes the windowed sinc-Gaussian passband, the two-term cyclic deconvolution, the Nyquist alternating-sum identity, and the finite-precision column solvers. Section 6 proves the Kronecker overlap-add spreading bound and the inductive forward-error closure under the additive precision schedule, and Section 7 gives the complete algorithm specification and proves Theorem 1. Section 8 describes the Lean 4 formalization (Theorem 2), Section 9 reports the C/GMP measurements, Section 10 discusses the remaining $\Theta(n \log n)$ term, and Appendices A–C contain the deferred proofs for table evaluation, breadth-first tape scheduling, tape routing, solver rounding, and filter norms.

2 Technical overview

This section gives a self-contained overview of the algorithm and its analysis, with forward pointers to the formal statements in Sections 3–8.

2.1 The algorithm in one box

At the top level, two n-bit integers A and B are split into $N = \lceil n/b \rceil$ radix-2^b digits $a_j, b_j \in [0, 2^b)$ with $b = \max(4, \lfloor \log_2 n \rfloor)$, zero-padded to a power of two $R_0 = 2^{K_0} \geq 2N$, and packed into one normalized complex vector $\tilde{x}_j = 2^{-b}(a_j + i b_j)$ at root precision $p_0 = 2b + 2K_0 + 6$. Two calls to the recursive transform `FastDFT` evaluate the cyclic digit convolution via the classical two-call real convolution identity (Lemma 3.4), after which nearest-integer rounding and radix-2^b carry propagation recover $A \cdot B$. Algorithm 1 summarizes the pipeline.

Algorithm 1 (at a glance). *Input:* n-bit integers A, B (if $n \le 64$, return $A \cdot B$ by schoolbook multiplication). Set $b = \max(4, \lfloor \log_2 n \rfloor)$, $N = \lceil n/b \rceil$, least power of two $R_0 = 2^{K_0} \ge 2N$, and $p_0 = 2b+2K_0+6$. Pack $\tilde{x}_j = 2^{-b}(a_j + ib_j)$ $(0 \le j < N$, zero-padded to $R_0)$, compute $\hat{y} = \texttt{FastDFT}(\tilde{x}, R_0, p_0)$, set $\hat{w} = \overline{\mathrm{RM}_{p_0+2}(\hat{y}, \hat{y})}$, compute $\hat{z} = \texttt{FastDFT}(\hat{w}, R_0, p_0)$, round $-2^{2b-1}R_0^2 \operatorname{Im}(\hat{z}_j)$ to integers, and propagate radix-2^b carries.

Recursive $\texttt{FastDFT}(x, R_\ell, p_\ell)$ *at* $R_\ell = 2^{K_\ell}$*:* If $R_\ell \le 4p_\ell^2$, run a radix-2 FFT at precision $q_b = p_\ell + K_\ell + 4$. Otherwise factor $R_\ell = t_1 r$ with $t_1 = 2^{\lfloor K_\ell/2 \rfloor}$, $r = 2^{\lceil K_\ell/2 \rceil}$, and set $p_{\mathrm{work},\ell} = p_\ell + K_\ell + 6$, $p_{\ell+1} = p_\ell + 2K_\ell + 12$, $q_{\mathrm{child}} = p_{\ell+1} + 2$:

(1) *Spread rows:* Accumulate the Nyquist alternating sum A_{N} from row $t_1/2$ (25), de-interleave x into t_1 rows of length r, and spread row j_1 by the windowed sinc-Gaussian kernel at offset j_1/t_1 via overlap-add and signed Kronecker substitution (Theorem 6.3).

(2) *Polynomial-transform 2-D DFT:* (2a) Pre-twist column k_2 by ζ^{k_2} $(\zeta = e^{\pi i/r})$. (2b) Apply the multiplication-free polynomial transform G_{t_1} over $\mathbb{C}[y]/(y^r+1)$ (Lemma 3.5). (2c) Post-twist by ζ^{-k_2} (Proposition 4.1). (2d) Run t_1 recursive calls to `FastDFT` of length r at precision $p_{\ell+1}$ and route their outputs by the odd-stride permutation $u_1 = (2u_2 - 1)v_1 \bmod t_1$ into column-major order (Lemma 3.3).

(3) *Deconvolve columns:* Solve $c_0(u_2)Z_{u_1} + c_1(u_2)Z_{(u_1-1) \bmod t_1} = Y_{u_1}$ on each column $u_2 \ne r/2$, solve $u_2 = r/2$ using A_{N}, quantize to $\widetilde{\mathbb{C}}_{p_\ell+2}$, and transpose to $u = u_2 + ru_1$ (Theorem 5.1, Lemma 5.2).

Inner products: $O((\log n)^2)$-bit integer products, delegated to the same algorithm one level down, backed at the bottom stage by a one-dimensional radix-2 Fourier multiplier using schoolbook arithmetic (Section 7).

2.2 Why one grid suffices

Consider a power-of-two transform length $R = t_1 r$ with $t_1 \mid r$ and $t_1 \in \{r, r/2\}$. Write the input index in strided form $j = j_1 + t_1 j_2$ and the output frequency in mixed-radix form $u = u_2 + ru_1$, where $0 \le j_1, u_1 < t_1$ and $0 \le j_2, u_2 < r$. Expanding our normalized one-dimensional DFT $(F_R x)_u = \frac{1}{R}\sum_{j=0}^{R-1} x_j e^{2\pi i j u/R}$ gives the Cooley–Tukey decomposition

$$(F_R x)_{u_2 + ru_1} = \frac{1}{t_1} \sum_{j_1=0}^{t_1-1} e^{2\pi i j_1 u_1/t_1} \left(\frac{1}{r} \sum_{j_2=0}^{r-1} x(j_1 + t_1 j_2)\, e^{2\pi i (j_2 + j_1/t_1) u_2/r} \right). \tag{1}$$

Without the fractional offset j_1/t_1 in the inner exponential, the right-hand side of (1) would be the two-dimensional DFT $(F_{t_1,r} U)_{u_1,u_2}$ of the array $U_{j_1,j_2} = x(j_1 + t_1 j_2)$. With that offset present, the inner sum is still a one-dimensional Fourier sum of length r along row j_1, except that every sample in row j_1 is shifted in position by the same fractional amount $j_1/t_1 \in [0, 1)$.

To transfer those fractional shifts onto the regular $t_1 \times r$ grid without changing its size, we convolve each row j_1 cyclically modulo r with a smooth, rapidly decaying kernel g evaluated at offset j_1/t_1:

$$U^{\circ}_{j_1,k_2} := \sum_{j_2=0}^{r-1} x(j_1 + t_1 j_2) \sum_{a \in \mathbb{Z}} g\!\left(j_2 + \frac{j_1}{t_1} - k_2 - ar \right).$$

Applying the one-dimensional Poisson summation formula along the row coordinate k_2 turns the periodic kernel convolution into a sum over integer frequency aliases $m \in \mathbb{Z}$. Because $(j_1/t_1)(u_2/r - m)$ supplies both the Cooley–Tukey twiddle $e^{2\pi i j_1 u_2/R}$ and the row-frequency shift $-m$ modulo t_1, the two-dimensional DFT of U° satisfies the exact identity (Theorem 4.2)

$$\left(F_{t_1,r} U^{\circ}\right)_{u_1,u_2} = \sum_{m \in \mathbb{Z}} \hat{g}\!\left(\frac{u_2}{r} - m \right) (F_R x)_{u_2 + r((u_1 - m) \bmod t_1)}. \tag{2}$$

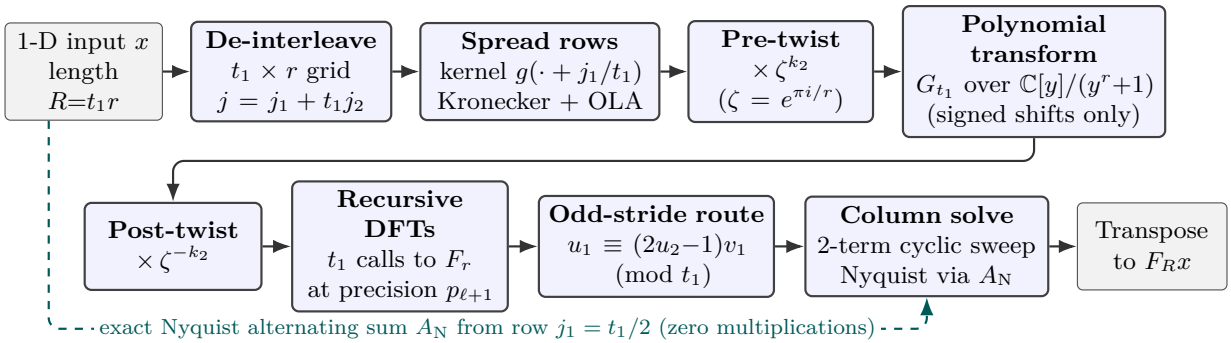


Figure 1: One recursive level of `FastDFT` on a single $t_1 \times r$ power-of-two grid ($R = t_1 r$). Row spreading turns Cooley–Tukey twiddles into a two-alias column mixture. The Nussbaumer–Quandalle polynomial transform over $\mathbb{C}[y]/(y^r + 1)$ and t_1 recursive length-r transforms evaluate $F_{t_1,r}$ without butterfly multiplications, and per-column two-term and Nyquist solvers recover $F_R x$.

For $y = u_2/r \in [0, 1)$, the arguments $y - m$ for $m \in \{0, 1\}$ lie in $[-1, 1)$, whereas all other aliases $m \notin \{0, 1\}$ satisfy $|y - m| \geq 1$. Whenever $\hat{g}$ is concentrated on $[-1, 1]$ up to exponentially small tails, each column u_2 of $F_{t_1,r}U^\circ$ reduces to a two-alias cyclic combination $c_0(u_2)Z_{u_1} + c_1(u_2)Z_{(u_1-1) \bmod t_1}$ of the target values $Z_{u_1} = (F_R x)_{u_2+ru_1}$ on the same $t_1 \times r$ grid.

Figure 1 shows the resulting single-level pipeline. Unlike [HvdH21], which embeds into pairwise coprime lengths by Chinese remaindering and resamples between grids of different sizes, the input, spread array, and output here occupy one $t_1 \times r$ power-of-two grid with no prime lengths and no grid expansion. Unlike general NUFFT gridding [DR93, PST01, GL04, BMK19], where arbitrary sample nodes require point-by-point interpolation onto an oversampled grid followed by diagonal division by $\hat{g}$, the shift j_1/t_1 here is constant along each row, so spreading is a shift-invariant cyclic convolution of length r and oversampling is replaced by a two-term cyclic solve per column.

2.3 Computing $F_{t_1,r}$ without multiplications

Evaluating $F_{t_1,r}U$ directly by row and column FFTs would still perform $\Theta(R \log t_1)$ complex butterfly multiplications along the first axis. Instead, we use the polynomial-transform reduction of Nussbaumer and Quandalle [NQ79, Nus80, Nus82] with negacyclic weighting [SS71, CF94] (Proposition 4.1). Because r is a power of two, the quotient ring $\mathcal{R}_r = \mathbb{C}[y]/(y^r + 1)$ contains the synthetic root y of order $2r$ ($y^r \equiv -1$). Since $t_1 \mid r$ with $\nu = r/t_1 \in \{1, 2\}$, the monomial $\omega_{t_1} := y^{2\nu}$ satisfies $\omega_{t_1}^{t_1/2} = y^r \equiv -1$ and has exact order t_1 in $\mathcal{R}_r$. The length-t_1 polynomial transform

$$(G_{t_1} P)_{v_1}(y) := \frac{1}{t_1} \sum_{j_1=0}^{t_1-1} P_{j_1}(y)\, y^{2\nu j_1 v_1} \pmod{y^r + 1}$$

can therefore be computed by $\log_2 t_1$ radix-2 butterfly stages over $\mathcal{R}_r$ using only additions, halvings, and signed cyclic shifts of r-coefficient polynomials, with zero scalar multiplications (Lemma 3.5).

To connect $\mathcal{R}_r$ to the column transforms of $F_{t_1,r}$, let $\zeta = e^{\pi i/r}$ ($\zeta^r = -1$). The r roots of $y^r + 1 = 0$ are the odd powers ζ^{2u_2-1} for $u_2 \in \{0, \ldots, r-1\}$, giving r ring homomorphisms $\phi_{u_2} \colon \mathcal{R}_r \to \mathbb{C}$, $y \mapsto \zeta^{2u_2-1}$. Pre-twisting column k_2 of row j_1 by ζ^{k_2} forms $P_{j_1}(y) = \sum_{k_2=0}^{r-1} \zeta^{k_2} U_{j_1,k_2} y^{k_2}$, whose image $\phi_{u_2}(P_{j_1}) = \sum_{k_2=0}^{r-1} U_{j_1,k_2} e^{2\pi i k_2 u_2/r}$ is the unnormalized length-r DFT along row j_1. Conversely, for each output polynomial $Q_{v_1} = (G_{t_1} P)_{v_1}$, post-twisting coefficient k_2 by ζ^{-k_2} and applying a recursive length-r transform F_r evaluates $\frac{1}{r}\phi_{u_2}(Q_{v_1})$ across all u_2 simultaneously. Finally, $\phi_{u_2}(y^{2\nu j_1 v_1}) =$

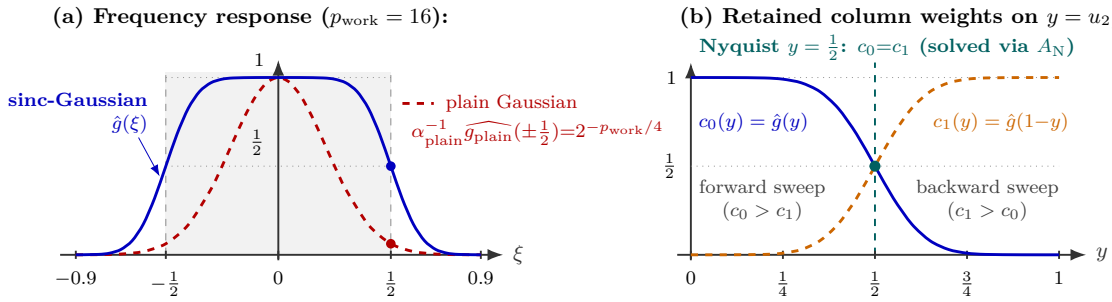


Figure 2: Passband behavior and two-term deconvolution weights. (a) At $\xi = \pm 1/2$, the normalized plain Gaussian drops to $2^{-p_{\text{work}}/4}$, whereas the windowed sinc-Gaussian kernel maintains $\hat{g}(\pm 1/2) \approx 1/2$. (b) Along each column $y = u_2/r \in [0, 1)$, the retained weights $c_0(y)$ and $c_1(y)$ cross only at the Nyquist column $y = 1/2$; non-Nyquist columns have $\|B^{-1}\|_{\infty\to\infty} = 1/|c_0 - c_1| < 2.36\, t_1$, and $y = 1/2$ is closed by the exact input alternating sum A_{N}.

$e^{2\pi i j_1(2u_2-1)v_1/t_1}$, and because $2u_2-1$ is odd while t_1 is a power of two, $v_1 \mapsto u_1 := (2u_2-1)v_1 \bmod t_1$ permutes $\{0, \ldots, t_1-1\}$ bijectively in every column u_2. This multiplication-free polynomial transform, together with the per-level tape transposes and odd-stride routing, is the only $\Theta(n \log n)$ work term in the entire algorithm.

2.4 The kernel and two-term deconvolution

The choice of spreading kernel g governs how precision grows with recursion depth. Suppose first that one spreads with a plain Gaussian $g_{\text{plain}}(t) = e^{-\pi t^2/\alpha_{\text{plain}}^2}$ scaled by $\pi\alpha_{\text{plain}}^2 = p_{\text{work},\ell} \ln 2$ so that excluded aliases decay to $O(2^{-p_{\text{work},\ell}})$. Its Fourier transform $\widehat{g_{\text{plain}}}(\xi) = \alpha_{\text{plain}} 2^{-p_{\text{work},\ell}\xi^2}$ peaks at $\xi = 0$ but falls at the Nyquist boundary $\xi = 1/2$ ($u_2 = r/2$) to $\alpha_{\text{plain}} 2^{-p_{\text{work},\ell}/4}$ (Figure 2(a)). Inverting the column system at $u_2 \approx r/2$ then divides by $O(2^{-p_{\text{work},\ell}/4})$; in the natural analysis (a heuristic estimate, not used elsewhere) this loses about $\frac{1}{4}p_{\text{work},\ell}$ bits of accuracy at every level and leads to a multiplicative precision schedule ($p_{\ell+1} \gtrsim \frac{5}{4}p_{\text{work},\ell}$) instead of an additive one.

We eliminate this Nyquist attenuation by multiplying the Gaussian by $\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$, obtaining the windowed sinc-Gaussian kernel (Theorem 4.2)

$$g(t) := \text{sinc}(t)\, e^{-\pi t^2/\alpha^2}, \qquad \alpha := 2\sqrt{\frac{p_{\text{work},\ell} \ln 2}{\pi}}. \tag{3}$$

In frequency space, multiplying by $\text{sinc}(t)$ convolves the Gaussian $\alpha e^{-\pi\alpha^2\xi^2}$ with the unit box $\mathbf{1}_{[-1/2,1/2]}$, yielding the erf-smoothed boxcar

$$\hat{g}(\xi) = \frac{1}{2}\left[\text{erf}\left(\sqrt{\pi}\alpha\left(\xi + \frac{1}{2}\right)\right) - \text{erf}\left(\sqrt{\pi}\alpha\left(\xi - \frac{1}{2}\right)\right)\right].$$

Because the shifted intervals $[y-m-1/2, y-m+1/2]$ tile $\mathbb{R}$, the weights satisfy the exact partition of unity $\sum_{m\in\mathbb{Z}} \hat{g}(y-m) = 1$. For every $y = u_2/r \in [0, 1)$, the two retained weights $c_0(u_2) = \hat{g}(y)$ and $c_1(u_2) = \hat{g}(y-1) = \hat{g}(1-y)$ satisfy $\max(c_0, c_1) \geq \hat{g}(1/2) = \frac{1}{2}\,\text{erf}(\sqrt{\pi}\alpha) > 0.49999999$ (for $p_{\text{work},\ell} \geq 8$), while all omitted aliases $m \notin \{0, 1\}$ telescope in closed form to $1 - c_0(u_2) - c_1(u_2) < 0.120 \cdot 2^{-p_{\text{work},\ell}}$ (Theorem 5.1(1)).

Figure 2(b) illustrates how each column $u_2 \in \{0, \ldots, r-1\}$ is deconvolved to recover $Z_{u_1} = (F_R x)_{u_2+ru_1}$ from $Y_{u_1} \approx c_0(u_2)Z_{u_1} + c_1(u_2)Z_{(u_1-1) \bmod t_1}$. Whenever $u_2 \neq r/2$, $|1/2 - y| \geq 1/r \geq$

$1/(2t_1)$ separates c_0 and c_1 enough that the damping power $a^{t_1} = (\min(c_0, c_1)/\max(c_0, c_1))^{t_1}$ is strictly below 0.14 (Theorem 5.1(2)). Sweeping forward when $u_2 < r/2$ ($c_0 > c_1$) or backward when $u_2 > r/2$ ($c_1 > c_0$) inverts the cyclic bidiagonal operator B in $O(t_1)$ scalar operations with ℓ_∞ inverse norm $1/|c_0(u_2) - c_1(u_2)| < 2.36\, t_1$.

At the single Nyquist column $u_2 = r/2$ ($y = 1/2$), symmetry gives $c_0(r/2) = c_1(r/2) = c \approx 1/2$. Because $t_1 \geq 2$ is even, the cyclic equations $c(Z_{u_1} + Z_{(u_1-1) \bmod t_1}) = Y_{u_1}$ sum with alternating signs $(-1)^{u_1}$ to zero and leave the alternating mode $\sum_{u_1=0}^{t_1-1} (-1)^{u_1} Z_{u_1}$ undetermined. By character orthogonality on $\mathbb{Z}_{t_1}$, however, that alternating sum collapses directly in the spatial domain to a single row of the input vector (Theorem 5.1(3), equation (25)):

$$A_{\mathrm{N}} := \sum_{u_1=0}^{t_1-1} (-1)^{u_1} Z_{u_1} = \frac{i}{r} \sum_{j_2=0}^{r-1} (-1)^{j_2} x\!\left(\frac{t_1}{2} + t_1 j_2\right).$$

Accumulating A_{N} from row $j_1 = t_1/2$ before spreading uses only additions, subtractions, and a power-of-two shift by r^{-1}, with zero scalar multiplications, and closes the open bidiagonal chain at $u_2 = r/2$ with inverse norm below $1.0000001\, t_1 < 2.36\, t_1$.

2.5 Additive precision schedule

Uniform $O(t_1)$ conditioning across all r columns yields a purely additive precision recurrence along the recursion tree (Lemma 6.5). At depth ℓ, a node of length $R_\ell = 2^{K_\ell} = t_{1,\ell} r_\ell$ receives an input $x \in \widetilde{\mathbb{C}}_{p_\ell+2}^{R_\ell}$ and must approximate $F_{R_\ell} x$ within ℓ_∞ error $2^{-p_\ell} \max(1, \|x\|_\infty)$. Two error groups enter before final quantization to $\widetilde{\mathbb{C}}_{p_\ell+2}$:

(1) *Excluded Poisson aliases:* Dropping modes $m \notin \{0, 1\}$ perturbs the ideal column right-hand side by at most $0.120 \cdot 2^{-p_{\mathrm{work},\ell}} \max(1, \|x\|_\infty)$, which multiplication by B^{-1} ($\|B^{-1}\|_{\infty\to\infty} < 2.36\, t_{1,\ell}$) amplifies to at most $\frac{1}{2} t_{1,\ell} 2^{-p_{\mathrm{work},\ell}} \max(1, \|x\|_\infty)$.

(2) *Spreading, polynomial transform, recursive calls, and column solvers.* Spatial window truncation and tap quantization in Step 1, rounding in the twists and $\log_2 t_{1,\ell}$ polynomial-transform stages of Steps 2a–2c, and the inductive error $2^{-p_{\ell+1}}$ of the $t_{1,\ell}$ child calls in Step 2d (on inputs of norm $O(r_\ell \max(1, \|x\|_\infty))$) perturb the column right-hand sides by $O(r_\ell 2^{-p_{\ell+1}} \max(1, \|x\|_\infty))$. Together with fixed-point rounding across the $O(t_{1,\ell})$-step column sweeps of Step 3, these terms sum after deconvolution to at most $200 R_\ell 2^{-p_{\ell+1}} \max(1, \|x\|_\infty)$.

Choosing the additive precision schedule (34),

$$p_{\mathrm{work},\ell} := p_\ell + K_\ell + 6, \qquad p_{\ell+1} := p_{\mathrm{work},\ell} + K_\ell + 6 = p_\ell + 2K_\ell + 12,$$

makes $\frac{1}{2} t_{1,\ell} 2^{-p_{\mathrm{work},\ell}} \leq 2^{-p_\ell - 7}$ and $200 R_\ell 2^{-p_{\ell+1}} \leq \frac{25}{2048} 2^{-p_\ell}$. Adding the final $\widetilde{\mathbb{C}}_{p_\ell+2}$ rounding error $\frac{\sqrt{2}}{8} 2^{-p_\ell}$ gives a total error below $0.197 \cdot 2^{-p_\ell} \max(1, \|x\|_\infty) < \frac{1}{4} 2^{-p_\ell} \max(1, \|x\|_\infty)$, closing the induction at every non-leaf node (and leaves $R_L \leq 4p_L^2$ close at precision $q_b = p_L + K_L + 4$).

Because $K_{\ell+1} = \lceil K_\ell/2 \rceil$, the log-length halves at each level and hits the leaf threshold $R_L \leq 4p_L^2$ in $L = O(\log K_0) = O(\log\log n)$ levels. Summing $2K_j + 12$ from 0 to $\ell - 1$ bounds the precision at every level by $p_\ell \leq p_0 + 4K_0 + 14\ell = \Theta(\log n)$, while the integer scalar height H_ℓ satisfies $H_\ell \leq 1 + K_0 + 2\ell = O(\log n)$ (Section 3). Every level therefore stores R_0 fixed-width complex records of $w_\ell = \Theta(\log n)$ bits, maintaining the invariant bit volume $R_0 p_\ell = \Theta(n)$ throughout the recursion.

2.6 Cost on a multitape Turing machine

With R_0 active entries of $\Theta(\log n)$ bits at every depth $\ell < L$, the multitape Turing machine cost breaks cleanly into one $\Theta(n \log n)$ movement-and-shift term and $o(n \log n)$ arithmetic and table work (Section 7):

- *Polynomial transform and tape routing:* At depth ℓ, each of the R_0/R_ℓ nodes performs $\log_2 t_{1,\ell} = \lfloor K_\ell/2 \rfloor$ radix-2 polynomial-transform stages over $\mathcal{R}_{r_\ell}$ (Lemma 3.5), two matrix transposes, and one odd-stride merge-sort column permutation $u_1 = (2u_2 - 1)v_1 \bmod t_{1,\ell}$ (Lemma 3.3), costing $O(R_\ell p_{\ell+1}(1 + \log_2 t_{1,\ell}))$ steps per node. Across all R_0/R_ℓ nodes, this is $O(n(1 + K_0/2^{\ell+1}))$ steps at depth ℓ, which sums geometrically over $\ell = 0, \ldots, L-1$ to $O(nK_0 + nL) = O(n \log n)$.
- *Kronecker spreading and window cancellation:* Truncating $g(d + j_1/t_{1,\ell})$ at radius $m_{1,\ell} = \Theta(p_{\ell+1})$ and folding modulo r_ℓ gives a filter of effective length $\tilde{W}_\ell = \min(2m_{1,\ell} + 1, r_\ell)$. Using overlap-add of block size $\tilde{W}_\ell$ and signed two's-complement Kronecker substitution [Sch82, Har09] at slot width $B_{\text{slot}} = \Theta(p_{\ell+1})$ reduces row spreading across a node to $O(R_\ell/\tilde{W}_\ell)$ integer products of bit-length $\Theta(\tilde{W}_\ell p_{\ell+1}) = O((\log n)^2)$ (Theorem 6.3). The window length $\tilde{W}_\ell$ cancels between the block count and the packed bit-length, so all multiplications in spreading, scalar twists, column sweeps, and leaf FFTs ($R_L \le 4p_L^2$) operate on $O((\log n)^2)$-bit integers, of total bit volume $O(n)$ per level for $\ell < L$ and $O(n \log \log n)$ at the leaf level (its $K_L = O(\log \log n)$ butterfly stages; Part 3(c) of the proof of Theorem 1).
- *Tables and non-circular bootstrap:* All shared per-level filter, twist, solver, and leaf-twiddle tables are evaluated from truncated Taylor and Machin series at guard precision in time $O(\sqrt{n}(\log n)^{7/2} \log \log n) = o(n)$ (equation (8)), and a breadth-first ping-pong schedule executes the recursion on $O(1)$ tapes (Appendix A). To avoid circularity in the asymptotic constant when multiplying $O((\log n)^2)$-bit integers on $O(1)$ tapes, the proof in Section 7 uses a fixed three-stage bootstrap. A base radix-2 FFT multiplier with schoolbook twiddles running in $O(N(\log 2N)^2)$ steps feeds a first recursive multiplier running in $O(N \log 2N)$ steps, which in turn multiplies all inner operands of the outer call in $O(n(\log \log n)^2)$ total steps.

2.7 What is verified and what is executed

Section 8 describes the Lean 4 formalization [LeanMS26] of Theorem 2, which constructs a single parameter-free eight-stack binary machine `tmIntMul` (which an eight-tape Turing machine simulates with constant-factor slowdown, one tape per stack; this standard simulation is not part of the formal development). From Lean's three standard logical axioms, it proves that `tmIntMul` halts with the exact $2n$-bit product within $C \cdot n\lfloor \log_2 n \rfloor$ steps and $O(n \log n)$ elementary stack operations for all $n \ge 2$. The analytic and fixed-point bounds on which the formal proof rests are proved inside the same library (Table 3). Section 9 reports the C/GMP measurements [CGMP26]: bit-exact agreement with GMP at every tested size up to 10^9 bits, at 153–323 times the single-threaded running time of `mpz_mul`.

3 Preliminaries

This section fixes global notation (Section 3.1), defines the multitape Turing machine model and dyadic fixed-point grids (Section 3.2), states the elementary-function table contract and tape schedule (Section 3.3), records the matrix transposition and odd-stride routing bounds (Section 3.4), and recalls our Discrete Fourier Transform and polynomial-transform conventions (Section 3.5).

3.1 Notation

Logarithms are natural (ln) unless a base is written ($\log_2$). For an integer $R \geq 1$, a vector $x = (x_0, \ldots, x_{R-1}) \in \mathbb{C}^R$ carries the Euclidean supremum norm $\|x\|_\infty := \max_{0 \leq j < R} |x_j|$, where $|a + ib| = \sqrt{a^2 + b^2}$ is the complex modulus, as well as the coordinatewise supremum norm $\|x\|_{\infty,\mathrm{coord}} := \max_{0 \leq j < R} \max(|\operatorname{Re}(x_j)|, |\operatorname{Im}(x_j)|)$. For a linear map $B \colon \mathbb{C}^m \to \mathbb{C}^m$, $\|B\|_{\infty \to \infty}$ denotes the induced ℓ_∞ operator norm (the maximum absolute row sum). Table 2 summarizes the notation used throughout the paper.

3.2 Multitape Turing machine and fixed-point grids

Following Schönhage and Strassen [SS71] and Harvey and van der Hoeven [HvdH21, Section 2.1], our model of computation is a deterministic multitape Turing machine with a fixed finite alphabet and a fixed finite number of tapes (all complexity bounds hold *a fortiori* on the standard $\Theta(\log n)$-bit Word RAM). Throughout the paper, a "bit operation" means a single Turing machine step (one state transition, tape-symbol read/write, and $\{-1, 0, +1\}$ head move on each of the $O(1)$ tapes), so all sequential tape scans, head movements, rewinds, and counter updates are charged explicitly. The input to a multiplication of two n-bit integers A, B is the concatenation of their fixed-width n-bit binary strings ($2n$ bits, so n is determined by the input) on a designated input tape, and the output is the fixed-width $2n$-bit binary string of $A \cdot B$ on a designated output tape; both are among the $O(1)$ tapes.

Definition 3.1 (Fixed-point grids and rounding)**.** For an integer $q \geq 0$, define the dyadic fixed-point real and complex grids

$$\widetilde{\mathbb{R}}_q := \{2^{-q} a : a \in \mathbb{Z}\}, \qquad \widetilde{\mathbb{C}}_q := \{2^{-q}(a + ib) : a, b \in \mathbb{Z}\}.$$

Here q counts fractional bits after the binary point, while the integer part is represented in signed two's-complement binary without modular wrap-around. A target accuracy parameter $p \in \mathbb{Z}_{\geq 1}$ uses the working grid $\widetilde{\mathbb{C}}_{p+2}$ ($q = p + 2$). For $x \in \mathbb{R}$, let $\mathcal{Q}_q(x) := 2^{-q} \operatorname{round}(2^q x) \in \widetilde{\mathbb{R}}_q$ denote nearest-grid rounding under any fixed deterministic tie-breaking rule on half-integers (such as ties to even, or $\lfloor 2^q x + 1/2 \rfloor$), and for $z = x + iy \in \mathbb{C}$, define coordinatewise rounding $\mathcal{Q}_q(z) := \mathcal{Q}_q(x) + i\mathcal{Q}_q(y) \in \widetilde{\mathbb{C}}_q$.

For every $x \in \mathbb{R}$ and $z \in \mathbb{C}$, regardless of which tie-breaking rule on half-integers is fixed in Definition 3.1, coordinatewise nearest rounding satisfies

$$|\mathcal{Q}_q(x) - x| \leq \frac{1}{2} 2^{-q}, \qquad |\mathcal{Q}_q(z) - z| \leq \frac{\sqrt{2}}{2} 2^{-q}. \tag{4}$$

On the grid $\widetilde{\mathbb{C}}_{p+2}$ ($q = p + 2$), real coordinate rounding error is at most $\frac{1}{8} 2^{-p} = 0.125 \cdot 2^{-p}$ and complex Euclidean rounding error is at most $\frac{\sqrt{2}}{8} 2^{-p} < 0.177 \cdot 2^{-p}$.

Scalar height and uniform fixed-width records. For $z \in \mathbb{C}$, $\|z\|_{\infty,\mathrm{coord}} \leq |z| \leq \sqrt{2} \|z\|_{\infty,\mathrm{coord}}$. If a scalar $z \in \widetilde{\mathbb{C}}_q$ satisfies $\|z\|_{\infty,\mathrm{coord}} \leq 2^H$ for an integer $H \geq 0$ (which holds whenever $|z| \leq 2^H$, or when z is produced by the polynomial transform G_{t_1} from inputs of coordinatewise magnitude at most 2^H by Lemma 3.5), then its scaled integer coordinates $2^q \operatorname{Re}(z), 2^q \operatorname{Im}(z) \in \mathbb{Z}$ have absolute value at most 2^{q+H}. As proved in Lemma 6.5, along any recursive call tree of `FastDFT` starting from a normalized root input with $\|x^{(0)}\|_\infty < \sqrt{2}$ ($H_0 := 1$), the computed child input vectors $\widehat{V}_{v_1,\cdot}$ at

Table 2: Summary of notation used throughout Sections 3–7 and Appendices A–C.

Symbol	Meaning
$n,\ b,\ N$	bit length of input integers $A, B \geq 0$, radix-2^b chunk width $b = \max(4, \lfloor \log_2 n \rfloor)$, and chunk count $N = \lceil n/b \rceil$
$R_0 = 2^{K_0},\ p_0$	root transform length ($2N \leq R_0 < 4N$) with $K_0 = \log_2 R_0$, and root target precision $p_0 = 2b + 2K_0 + 6$
$R_\ell = 2^{K_\ell} = t_{1,\ell} r_\ell$	transform length at recursion depth ℓ, with $K_\ell = \log_2 R_\ell$, $t_{1,\ell} = 2^{\lfloor K_\ell/2 \rfloor}$, $r_\ell = 2^{\lceil K_\ell/2 \rceil}$, and $\nu_\ell = r_\ell / t_{1,\ell} \in \{1, 2\}$
$k_{1,\ell},\ k_{2,\ell}$	$\log_2 t_{1,\ell}$ and $\log_2 r_\ell$ ($k_{1,\ell} + k_{2,\ell} = K_\ell$)
L	leaf recursion depth (first depth ℓ where $R_\ell \leq 4p_\ell^2$)
$p_\ell,\ p_{\mathrm{work},\ell},\ p_{\ell+1}$	target precision at depth ℓ, working cutoff $p_{\mathrm{work},\ell} = p_\ell + K_\ell + 6$, and child target precision $p_{\ell+1} = p_\ell + 2K_\ell + 12$
$\widetilde{\mathbb{R}}_q,\ \widetilde{\mathbb{C}}_q,\ \mathcal{Q}_q$	real and complex dyadic grids $2^{-q}\mathbb{Z}$ and $2^{-q}(\mathbb{Z} + i\mathbb{Z})$ ($q = p + 2$ for target precision p), and coordinatewise nearest-grid rounding
$q_{\mathrm{child}},\ u = 2^{-p_{\ell+1}}$	child fractional grid parameter $q_{\mathrm{child}} = p_{\ell+1} + 2$ and unit accuracy scale of child calls (the letter u also denotes the one-dimensional frequency index $u = u_2 + r u_1$ of (9); the meaning is fixed by context: u_2 indexes columns, $u = 2^{-p_{\ell+1}}$ is the accuracy scale)
$H_\ell,\ w_{\mathrm{coord},\ell},\ w_\ell$	integer coordinate height bound at depth ℓ, signed coordinate bit width, and complex record width $w_\ell = 2w_{\mathrm{coord},\ell}$
$\mathcal{R}_r = \mathbb{C}[y]/(y^r + 1),\ G_{t_1}$	negacyclic polynomial ring of rank r and normalized length-t_1 polynomial transform over $\mathcal{R}_r$
$P_{j_1}(y),\ Q_{v_1}(y)$	input and output polynomials in $\mathcal{R}_r$ of the polynomial transform G_{t_1}
$F_R,\ F_{t_1,r}$	normalized positive one-dimensional DFT of length R and two-dimensional DFT of shape $t_1 \times r$
$g(t),\ \hat{g}(\xi),\ \alpha_\ell$	windowed sinc-Gaussian kernel $\mathrm{sinc}(t)e^{-\pi t^2/\alpha_\ell^2}$, its continuous Fourier transform, and $\alpha_\ell = 2\sqrt{p_{\mathrm{work},\ell} \ln 2/\pi}$
$m_{1,\ell},\ W_\ell,\ \tilde{W}_\ell$	spatial window half-width, raw tap count $W_\ell = 2m_{1,\ell} + 1$, and modulo-r_ℓ folded filter length $\tilde{W}_\ell = \min(W_\ell, r_\ell)$
$c_0(u_2),\ c_1(u_2),\ M(u_2)$	retained Poisson alias weights $\hat{g}(u_2/r)$ and $\hat{g}(1 - u_2/r)$ in column u_2, and $M(u_2) = \max(c_0(u_2), c_1(u_2))$
$\rho(u_2),\ \theta(u_2),\ A_{\mathrm{N}}$	non-Nyquist recurrence ratio $-\min(c_0, c_1)/M(u_2)$, cyclic closure scalar $(1 - \rho(u_2)^{t_1})^{-1}$, and Nyquist spatial alternating sum
$\mathcal{M}_Y$	$\max(1, \|\widehat{Y}\|_\infty)$ for the routed child output $\widehat{Y}$ of a column (Lemma 5.2)

depth ℓ satisfy $\max(1, \|\widehat{V}_{v_1,\cdot}\|_\infty) \leq 2r_\ell \max(1, \|x^{(\ell)}\|_\infty)$, where $r_\ell = 2^{K_{\ell+1}}$. Consequently, the integer height bound $H_\ell := 1 + \sum_{j=0}^{\ell-1}(K_{j+1} + 1)$ satisfies $\log_2 \max(1, \|x^{(\ell)}\|_\infty) \leq H_\ell$ and

$$H_\ell = 1 + \sum_{j=0}^{\ell-1}(K_{j+1} + 1) \leq 1 + K_0 + 2\ell = O(\log n). \tag{5}$$

Because $K_{\ell+1} = \lceil K_\ell/2 \rceil$, the recursion reaches a leaf $R_L \leq 4p_L^2$ in $L \leq \lceil \log_2 K_0 \rceil$ levels, and $H_\ell < p_0 + \sum_{j=0}^{\ell-1}(2K_j + 12) = p_\ell$ for all $K_0 \geq 2$ and $\ell \geq 0$. Thus $p_\ell > H_\ell$ at all levels $\ell \geq 0$ under our schedule (34): every routed record at depth ℓ has coordinatewise magnitude at most $2r_\ell 2^{H_\ell} = 2^{H_{\ell+1}} \leq 2^{p_{\ell+1}}$ and Euclidean modulus at most $\sqrt{2} \cdot 2^{H_{\ell+1}} < 2^{H_{\ell+1}+1}$, while the intermediate prefix-sweep scalars $\tilde{P}_i, Z_i^{\mathrm{comp}}$ inside `CyclicSolve` have magnitude $< 2.37\, t_{1,\ell}\mathcal{M}_Y$ (where $\mathcal{M}_Y := \max(1, \|\widehat{Y}\|_\infty)$, including the rounding terms of Lemma 5.2); since $\mathcal{M}_Y \leq 2\max(1, \|x^{(\ell)}\|_\infty) \leq 2^{H_\ell+1}$ (shown in the proof of Lemma 6.5) and $t_{1,\ell} \leq r_\ell$, these scalars are bounded by $4.74\, r_\ell 2^{H_\ell} < 4 \cdot 2^{H_{\ell+1}} = 2^{H_{\ell+1}+2}$. Likewise, inside `NyquistSolve` the prefix sums satisfy $|\tilde{P}_i| \leq 2.0002\, t_{1,\ell}\mathcal{M}_Y$, the correction satisfies $|\tilde{\delta}| \leq 1.0001\, t_{1,\ell}\mathcal{M}_Y + |A_{\mathrm{N}}|/t_{1,\ell} + 2^{-p_{\ell+1}}$, and the outputs $Z_i^{\mathrm{comp}} = \tilde{P}_i + (-1)^i\tilde{\delta}$ satisfy $|Z_i^{\mathrm{comp}}| \leq 3.0003\, t_{1,\ell}\mathcal{M}_Y + |A_{\mathrm{N}}|/t_{1,\ell} + 2^{-p_{\ell+1}}$, where $|A_{\mathrm{N}}| \leq \|x^{(\ell)}\|_\infty \leq 2^{H_\ell}$ by (25); hence $|Z_i^{\mathrm{comp}}| < 6.1\, r_\ell 2^{H_\ell} < 2^{H_{\ell+1}+2}$ as well.

To ensure that all tape routing, matrix transposition, and merge-sort primitives operate on uniform fixed-width records, every complex scalar on $\widetilde{\mathbb{C}}_q$ ($q \le q_{\text{child}} = p_{\ell+1} + 2$) at recursion depth ℓ is padded by two's-complement sign extension and low-bit zero-filling to the uniform record width

$$w_\ell := 2w_{\text{coord},\ell}, \qquad w_{\text{coord},\ell} := q_{\text{child}} + H_{\ell+1} + 3 = \Theta(p_{\ell+1}) = \Theta(\log n) \tag{6}$$

bits (two $w_{\text{coord},\ell}$-bit signed two's-complement integers, where the $+3$ headroom bits accommodate the sign bit, the $\sqrt{2}$ Euclidean-to-coordinatewise margin, and the $2^{H_{\ell+1}+2}$ prefix-sweep bound). Re-padding the R_0 active records when moving between depth ℓ and depth $\ell+1$ is a single sequential scan costing $O(R_0 w_{\ell+1}) = O(n)$ Turing machine steps per level. Explicit bit-width bounds for all intermediate scratch and accumulator registers are listed in Appendix A.2.

Rounded arithmetic primitives. Addition and subtraction on $\widetilde{\mathbb{C}}_q$ are exact and take $O(q+H+1)$ bit operations. For $a, b \in \widetilde{\mathbb{C}}_q$ (or $a \in \widetilde{\mathbb{R}}_q, b \in \widetilde{\mathbb{C}}_q$), the rounded product $\mathrm{RM}_q(a,b) := \mathcal{Q}_q(ab)$ computes the exact integer product of the scaled numerators $N_a = 2^q a$ and $N_b = 2^q b$ (four real integer multiplications and two additions for complex-by-complex, or two real integer multiplications for real-by-complex) and rounds each coordinate once from fractional precision $2q$ to q, incurring error $|\mathrm{RM}_q(a,b) - ab| \le \frac{\sqrt{2}}{2} 2^{-q}$. For $z \in \widetilde{\mathbb{C}}_q$ and $d \in \widetilde{\mathbb{R}}_q \setminus \{0\}$, the rounded quotient $\mathrm{RD}_q(z,d) := \mathcal{Q}_q(z/d)$ shifts $N_z := 2^q z \in \mathbb{Z}[i]$ by 2^q to form $2^{2q} z$ and performs integer Euclidean division with remainder by $N_d := 2^q d \in \mathbb{Z} \setminus \{0\}$ to round $2^q(z/d)$ onto $\widetilde{\mathbb{C}}_q$, incurring $|\mathrm{RD}_q(z,d) - z/d| \le \frac{\sqrt{2}}{2} 2^{-q}$.

3.3 Elementary function evaluation and shared level tables

To avoid assuming unit-cost real-RAM oracles for transcendental constants and functions $(\pi, \ln 2, \sqrt{\cdot}, \exp, \sin, \cos, \mathrm{erf})$, Lemma 3.2 specifies how all level tables are deterministically computed on a multitape Turing machine (the table expressions are enumerated in Appendix A.1). Its proof appears in Appendix A.1.

Lemma 3.2 (Fixed-point evaluation of elementary functions and faithful tables)**.** *Let $s \ge 8$ be a target fractional precision and let $D := \max(s, K_0) = O(\log n)$. Work on the internal dyadic grid $\widetilde{\mathbb{R}}_S$ at precision $S := s + 15D + 64 = \Theta(D)$ with truncated series of $N_{\text{terms}} := 2S + 36D = \Theta(D)$ terms, where both the exact real argument x and the grid argument $\tilde{x} \in \widetilde{\mathbb{R}}_S$ lie in the stated domain (with $2^S \tilde{x}$ a signed integer of bit-length at most $S + 2D + 8 = O(D)$) and satisfy $|\tilde{x} - x| \le \delta_{\text{in}} \le 2^{-s-20}$ (with $\delta_{\text{in}} = 0$ for exact dyadic rational arguments $t = d + j_1/t_{1,\ell} \in \widetilde{\mathbb{R}}_{K_0} \subseteq \widetilde{\mathbb{R}}_S$). Each primitive (i)–(v) uses intermediate integers of bit-length at most $2S + 4D + 16 = O(D)$ and outputs $\tilde{y} \in \widetilde{\mathbb{R}}_S$ satisfying $|\tilde{y} - f(\tilde{x})| < 2^{-s-D-23} < 2^{-s-30}$ and $|\tilde{y} - f(x)| < 2^{-s-D-23} + L_f \delta_{\text{in}}$ (where $L_{\sqrt{\cdot}} \le 2^{D/2-1}$ on $[2^{-D}, 2^{2D}]$, $L_{\exp}, L_{\sin}, L_{\cos} \le 1$, $L_{\mathrm{erf}} = 2/\sqrt{\pi} < 1.13$, and $L_{\mathrm{sinc}} = 4/\pi < 1.28$ on $[-1/2, 1/2]$) in $O(D^3)$ schoolbook bit operations; every $\sqrt{\cdot}$ evaluation of the algorithm has its argument in $[1, 2^{2D}]$, where $L_{\sqrt{\cdot}} \le \frac{1}{2}$ (Appendix A.1). Evaluating any real or complex table expression of our algorithm at target precision $s := q + 10$ (where $q = q_{\text{child}} \ge p_{\text{work},\ell}$ for the α_ℓ-dependent tables of a non-leaf level, so $p_{\text{work},\ell} < s \le D$ there; the leaf twiddle tables use $q = q_b$ and involve no α_ℓ) produces $\tilde{v} \in \widetilde{\mathbb{R}}_S$ (resp. $\widetilde{\mathbb{C}}_S$) within $< 2^{-s-20} < 2^{-q-10}$ of the exact value v, so rounding once via $\bar{v} := \mathcal{Q}_q(\tilde{v})$ satisfies the 1-ulp faithful table rounding contract:*

$$|\bar{v} - v| \le \left(\frac{1}{2} + 2^{-10}\right) 2^{-q} < 2^{-q} \quad (\textit{real}), \qquad |\bar{v} - v| \le \sqrt{2}\left(\frac{1}{2} + 2^{-10}\right) 2^{-q} < \sqrt{2} \cdot 2^{-q} \quad (\textit{complex}). \tag{7}$$

On $\widetilde{\mathbb{C}}_{q_{\text{child}}}$ ($q_{\text{child}} = p_{\ell+1} + 2$, $u = 2^{-p_{\ell+1}}$), (7) gives $|\bar{v} - v| \le \frac{1}{4} u = 0.25u$ for real table entries and $|\bar{v} - v| \le \frac{\sqrt{2}}{4} u < 0.354u$ for complex table entries, where the procedures are:

(i) Evaluate π via Machin's identity $\frac{\pi}{4} = 4\arctan(1/5) - \arctan(1/239)$ with alternating remainder $\le 5(2N_{\text{terms}}+1)^{-1}5^{-(2N_{\text{terms}}+1)}$ for $\pi/4$ (both tails), hence $\le 20(2N_{\text{terms}}+1)^{-1}5^{-(2N_{\text{terms}}+1)} < 2^{-S-16}$ for π, and $\ln 2 = 2\sum_{k=0}^{\infty}\frac{1}{(2k+1)3^{2k+1}}$ with geometric remainder $\le \frac{3}{4(2N_{\text{terms}}+1)}3^{-2N_{\text{terms}}} < 2^{-S-16}$. Let $\tilde{\pi}, \tilde{\ell}_2 \in \widetilde{\mathbb{R}}_S$ denote these approximations and $\tilde{h} := \mathcal{Q}_S(\tilde{\pi}/2)$.

(ii) Evaluate square roots $\sqrt{x}$ for $x \in [2^{-D}, 2^{2D}]$ via exact integer square root $\lfloor\sqrt{2^{2S}\tilde{x}}\rfloor$.

(iii) Evaluate $\exp(x)$ for $x \in [-\pi(D+2)^2, 0]$, returning 0 when $\tilde{x} < -(S+16)\tilde{\ell}_2$ (since then $e^{\tilde{x}} < 2^{-S-15}$) and otherwise dividing $-\tilde{x} = m\tilde{\ell}_2 + \tilde{r}$ with $m \in \{0, \dots, S+16\}$, $\tilde{r} \in [0, \tilde{\ell}_2)$ and evaluating $2^{-m}\sum_{k=0}^{N_{\text{terms}}-1}(-\tilde{r})^k/k!$ (remainder $\le \tilde{r}^{N_{\text{terms}}}/N_{\text{terms}}! < 2^{-S-16}$).

(iv) Evaluate $\sin(x), \cos(x)$ for $|x| \le \pi(D+2)$ via range reduction $\tilde{x} = m\tilde{h} + \tilde{r}$ with integer $m := \lfloor \tilde{x}/\tilde{h} + 1/2 \rfloor$ ($|m| \le 2D+5$) and $|\tilde{r}| \le \tilde{h}/2 < 0.79 < 1$ and alternating Taylor series with remainder $\le |\tilde{r}|^{2N_{\text{terms}}}/(2N_{\text{terms}})! < 2^{-S-16}$ (and for $|t| \le 1/2$, evaluate $\operatorname{sinc}(t) = \sum_{k=0}^{\infty}(-1)^k\frac{(\pi t)^{2k}}{(2k+1)!}$ directly by its N_{terms}-term truncation with alternating remainder $\le \frac{(\pi/2)^{2N_{\text{terms}}}}{(2N_{\text{terms}}+1)!} < 2^{-S-16}$, avoiding division by small πt).

(v) Evaluate $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}}e^{-z^2}\sum_{n=0}^{\infty}\frac{2^n z^{2n+1}}{1\cdot 3\cdots(2n+1)}$ on $z \in [0, 3\sqrt{D}]$, returning 1 when $\mathrm{RM}_S(\tilde{z}, \tilde{z}) > (s+D+24)\tilde{\ell}_2$ (since then $\operatorname{erfc}(\tilde{z}) < 2^{-s-D-23}$) and otherwise summing the first N_{terms} series terms on $\widetilde{\mathbb{R}}_S$.

Shared level tables and breadth-first tape layout. For a top-level multiplication of n-bit integers, the dimensions $R_\ell = 2^{K_\ell}$ and precisions $(p_\ell, p_{\text{work},\ell}, p_{\ell+1})$ at each depth $\ell \in \{0, \dots, L\}$ depend only on (R_0, p_0) and ℓ. Before the recursive transforms begin, the algorithm writes once per level ℓ onto a table tape T_{tab} the folded filter taps and their Kronecker-packed integers, the twist roots, the column solver tuples $(\mu(u_2), \hat{\rho}(u_2), \hat{\theta}(u_2)) \in \widetilde{\mathbb{R}}^3_{q_{\text{child}}}$, and at $\ell = L$ the per-stage leaf twiddle lists. Since $t_{1,\ell}, r_\ell \le \sqrt{2R_0}$, $m_{1,\ell} \le p_L$ (Lemma 6.5), and $R_L \le 4p_L^2$, Lemma 3.2 with $D = O(p_L) = O(\log n)$ builds all shared level tables in

$$\mathcal{C}_{\text{tables}}(n) = O\big((L+1)(\sqrt{R_0}\,p_L^4 + p_L^5)\big) = O\big(\sqrt{n}(\log n)^{7/2}\log\log n\big) = o(n) \tag{8}$$

Turing machine steps. The recursion tree is then executed breadth-first on $O(1)$ tapes using, besides the input and output tapes, two alternating data tapes $T^{(0)}_{\text{data}}, T^{(1)}_{\text{data}}$, four scratch tapes $T_1, \dots, T_4$, the table tape T_{tab}, a level-segmented stack tape T_{stack} storing each node's Nyquist alternating sum, and dedicated work tapes for the nested inner multiplier. Total table replay and level-sweep coordination across all $L = O(\log\log n)$ levels take $O(n\log\log n) = o(n\log n)$ steps; the full tape schedule is given in Appendix A.2.

3.4 Matrix transposition and odd-stride column routing

The following lemma bounds the two data-routing primitives used at each recursion level; its proof (by multi-pass split-and-interleave scans and column-parallel bottom-up merge sort) is given in Appendix B.

Lemma 3.3 (Multitape transposition and odd-stride column routing)**.** *Let $t_1 \mid r$ be powers of two ($t_1 = 2^{k_1}$, $r = 2^{k_2}$), and let each matrix entry be a uniform fixed-width record of $w = \Theta(q)$ bits with $q \ge k_2 \ge k_1$. On a deterministic multitape Turing machine with a constant number of tapes:*

(i) Transposing a $t_1 \times r$ matrix between row-major order (j_1, j_2) and column-major order (j_2, j_1), including de-interleaving $x \in \widetilde{\mathbb{C}}_q^{t_1 r}$ indexed by $j = j_1 + t_1 j_2$ into t_1 contiguous rows of length r or transposing $(\widehat{Z}_{u_1,u_2})$ from column-major order (u_2, u_1) back to natural frequency order $u = u_2 + r u_1$, takes $O\big(t_1 r q (1 + \log_2 t_1)\big)$ Turing machine steps.

(ii) Given t_1 child DFT output vectors $\widehat{C}_{v_1} \in \widetilde{\mathbb{C}}_q^r$ in row order $v_1 \in \{0, \ldots, t_1 - 1\}$, routing all entries according to the odd-stride column permutation

$$(v_1, u_2) \longmapsto (u_1, u_2), \qquad u_1 := (2u_2 - 1)v_1 \bmod t_1,$$

into column-major order (u_2, u_1) takes $O\big(t_1 r q (1 + \log_2 t_1)\big)$ Turing machine steps.

3.5 Discrete Fourier transforms, two-call convolution, and polynomial transforms

For $R \in \mathbb{Z}_{\geq 1}$, the normalized one-dimensional Discrete Fourier Transform $F_R \colon \mathbb{C}^R \to \mathbb{C}^R$ is defined for $u \in \{0, \ldots, R-1\}$ by

$$(F_R x)_u := \frac{1}{R} \sum_{j=0}^{R-1} x_j \, e^{2\pi i j u / R}. \tag{9}$$

By the triangle inequality, $\|F_R x\|_\infty \leq \frac{1}{R} \sum_{j=0}^{R-1} |x_j| \leq \|x\|_\infty$. Under this normalization, $\big(F_R(\overline{F_R w})\big)_j = \frac{1}{R} \overline{w_j}$ for every $w \in \mathbb{C}^R$, and the cyclic convolution $(a \circledast b)_j := \sum_{k=0}^{R-1} a_k b_{(j-k) \bmod R}$ satisfies $F_R(a \circledast b) = R \cdot (F_R a) \odot (F_R b)$, where $\odot$ denotes pointwise multiplication. Packing two real sequences into one complex sequence is a classical textbook identity (see, e.g., [Nus82]); we record the exact two-call form in our normalization.

Lemma 3.4 (Two-call real cyclic convolution)**.** *For any two real vectors $a, b \in \mathbb{R}^R$, set $z := a + ib \in \mathbb{C}^R$. Then the real cyclic convolution $a \circledast b \in \mathbb{R}^R$ is given in terms of two evaluations of F_R by*

$$a \circledast b = -\frac{R^2}{2} \operatorname{Im}\Big(F_R\big(\overline{(F_R z)^2}\big) \Big), \tag{10}$$

where $(F_R z)^2 := (F_R z) \odot (F_R z)$ denotes the pointwise square.

Proof. By linearity of F_R and the convolution theorem,

$$(F_R z)^2 = (F_R a + i F_R b)^2 = (F_R a \odot F_R a - F_R b \odot F_R b) + 2i(F_R a \odot F_R b) = \frac{1}{R} F_R(d - e + 2ic),$$

where $d := a \circledast a \in \mathbb{R}^R$, $e := b \circledast b \in \mathbb{R}^R$, and $c := a \circledast b \in \mathbb{R}^R$. Applying $F_R(\overline{F_R w}) = \frac{1}{R} \overline{w}$ to $w := \frac{1}{R}(d - e + 2ic)$ yields

$$F_R\big(\overline{(F_R z)^2}\big) = \frac{1}{R^2} \overline{d - e + 2ic} = \frac{1}{R^2}(d - e - 2ic),$$

because $d, e, c \in \mathbb{R}^R$ are real. Taking the imaginary part and multiplying by $-R^2/2$ recovers $c = a \circledast b$. □

For dimensions (t_1, r), the normalized two-dimensional Discrete Fourier Transform $F_{t_1,r} \colon \mathbb{C}^{t_1 \times r} \to \mathbb{C}^{t_1 \times r}$ is defined for $(u_1, u_2) \in \{0, \ldots, t_1 - 1\} \times \{0, \ldots, r - 1\}$ by

$$(F_{t_1,r} U)_{u_1,u_2} := \frac{1}{t_1 r} \sum_{j_1=0}^{t_1-1} \sum_{j_2=0}^{r-1} U_{j_1,j_2} \, e^{2\pi i j_1 u_1 / t_1} \, e^{2\pi i j_2 u_2 / r}. \tag{11}$$

Let $r = 2^{k_2}$ and $t_1 = 2^{k_1}$ be powers of two with $t_1 \mid r$, and set $\nu := r/t_1 \in \{1, 2\}$. In the negacyclic quotient ring $\mathcal{R}_r := \mathbb{C}[y]/(y^r+1)$, the monomial $\omega_{t_1} := y^{2\nu}$ satisfies $\omega_{t_1}^{t_1/2} = y^{\nu t_1} = y^r \equiv -1 \pmod{y^r+1}$, so ω_{t_1} is a primitive t_1-th root of unity in $\mathcal{R}_r$. Following Nussbaumer [Nus80, Nus82], for $P = (P_0(y), \dots, P_{t_1-1}(y)) \in (\mathcal{R}_r)^{t_1}$ we define the normalized polynomial transform $G_{t_1} \colon (\mathcal{R}_r)^{t_1} \to (\mathcal{R}_r)^{t_1}$ for $v_1 \in \{0, \dots, t_1 - 1\}$ by

$$(G_{t_1} P)_{v_1}(y) := \frac{1}{t_1} \sum_{j_1=0}^{t_1-1} P_{j_1}(y)\, y^{2\nu j_1 v_1} \pmod{y^r + 1}. \tag{12}$$

Lemma 3.5 (Radix-2 polynomial transform G_{t_1})**.** *Let $t_1 = 2^{k_1} \geq 2$ ($k_1 \geq 1$) and $r = 2^{k_2}$ with $t_1 \mid r$. The radix-2 decimation-in-time Cooley–Tukey factorization of G_{t_1} over $\mathcal{R}_r$ bit-reverses the t_1 input polynomials and executes $k_1 = \log_2 t_1$ butterfly stages. In stage $s \in \{1, \dots, k_1\}$, each butterfly updates a pair of polynomials $A(y), B(y) \in \widetilde{\mathbb{C}}_q[y]/(y^r + 1)$ via*

$$A^{\mathrm{new}}(y) := \mathcal{Q}_q\!\left(\frac{1}{2}(A(y) + y^{2m} B(y))\right), \qquad B^{\mathrm{new}}(y) := \mathcal{Q}_q\!\left(\frac{1}{2}(A(y) - y^{2m} B(y))\right) \pmod{y^r + 1}.$$

On a deterministic multitape Turing machine, for inputs on $\widetilde{\mathbb{C}}_q$ ($q = p_{\ell+1} + 2 \geq k_2$) with coordinatewise magnitude $\|P\|_{\infty,\mathrm{coord}} \leq 2^H$ stored as uniform fixed-width records of width $w = \Theta(q + H + 1)$, computing G_{t_1} performs zero scalar multiplications, takes $O(t_1 r(q + H + 1)(1 + \log_2 t_1))$ Turing machine steps, preserves the coordinatewise magnitude bound $\|\cdot\|_{\infty,\mathrm{coord}} \leq 2^H$ at every stage (while for any Euclidean input bound $\|P\|_\infty \leq V$, stage s satisfies $\|P^{(s)}\|_\infty \leq V + s\frac{\sqrt{2}}{2} 2^{-q}$), and outputs $\widehat{Q} = (\widehat{Q}_0, \dots, \widehat{Q}_{t_1-1}) \in (\widetilde{\mathbb{C}}_q[y]/(y^r + 1))^{t_1}$ in natural index order $v_1 \in \{0, \dots, t_1 - 1\}$ satisfying

$$\|\widehat{Q} - G_{t_1} P\|_\infty \leq k_1 \frac{\sqrt{2}}{2} 2^{-q} = k_1 \frac{\sqrt{2}}{8} 2^{-p_{\ell+1}} < 0.177\, k_1\, 2^{-p_{\ell+1}}. \tag{13}$$

Proof. First, apply the bit-reversal permutation to the $t_1 = 2^{k_1}$ input polynomials $P_0, \dots, P_{t_1-1}$, each stored as a contiguous block of r records of width w, in $k_1 = \log_2 t_1$ sequential split-and-concatenate passes. In pass $s \in \{1, \dots, k_1\}$, within each of the 2^{s-1} segments of $t_1/2^{s-1}$ polynomials, copy even-indexed polynomials to auxiliary tape T_1 and odd-indexed polynomials to T_2 and concatenate T_1 and T_2, costing $O(t_1 r w)$ steps per pass and $O(t_1 r w \log_2 t_1)$ steps in total.

Next, in stage $s \in \{1, \dots, k_1\}$ of decimation-in-time Cooley–Tukey, the tape consists of $t_1/2^s$ contiguous blocks of 2^s polynomials. Within each block, the first 2^{s-1} polynomials $(A_j(y))_{j=0}^{2^{s-1}-1}$ are paired with the second 2^{s-1} polynomials $(B_j(y))_{j=0}^{2^{s-1}-1}$ with twiddle exponent

$$\delta_j := 2m_j = 2\nu j \cdot 2^{k_1 - s} \in \{0, 2, \dots, r-2\}, \qquad 0 \leq 2m_j \leq r - 2\nu 2^{k_1-s} < r,$$

since $j < 2^{s-1}$ and $\nu t_1 = r$. In one linear scan of the block, copy $(A_j)_{j=0}^{2^{s-1}-1}$ to T_1 and $(B_j)_{j=0}^{2^{s-1}-1}$ to T_2 ($O(2^s r w)$ steps), maintaining $\delta_j \in \{0, \dots, r-2\}$ by adding the stage stride $2\nu \cdot 2^{k_1-s}$ as j increments ($O(\log r) \leq O(w)$ steps per polynomial). For each pair j, multiplying $B_j(y) = \sum_{k_2=0}^{r-1} b_{k_2} y^{k_2}$ by $y^{\delta_j} \pmod{y^r + 1}$ yields $\sum_{k_2=0}^{r-1} \tilde{b}_{k_2} y^{k_2}$ with

$$\tilde{b}_{k_2} = \begin{cases} -b_{k_2+r-\delta_j} & \text{for } 0 \leq k_2 < \delta_j, \\ +b_{k_2-\delta_j} & \text{for } \delta_j \leq k_2 < r. \end{cases}$$

To stream $(\tilde{b}_{k_2})_{k_2=0}^{r-1}$ in natural order on a Turing machine, scan the first $r - \delta_j$ coefficient records $b_0, \dots, b_{r-\delta_j-1}$ of B_j on T_2 (counted by decrementing a k_2-bit counter initialized to $r - \delta_j$ in $O(r + \log r)$

total counter steps) and copy them to T_3. This leaves T_2 at $b_{r-\delta_j}$, after which T_3 is rewound to b_0. Then scan $a_0, \dots, a_{r-1}$ on T_1 in lockstep with the δ_j negated records $-b_{r-\delta_j}, \dots, -b_{r-1}$ on T_2 followed by the $r - \delta_j$ records on T_3, writing $A_j^{\mathrm{new}}(y)$ to the upper half of the block on the main tape and $B_j^{\mathrm{new}}(y)$ to T_4, and finally copying T_4 to the lower half of the block. Each polynomial pair in stage s takes $O(rw + \log r) = O(rw)$ steps, and all k_1 stages take $O(t_1 rw(1 + \log_2 t_1))$ steps.

For the norm and error bounds, the signed cyclic shift $y^{2m_j} B_j(y) \pmod{y^r + 1}$ is exact on $\widetilde{\mathbb{C}}_q$, and for each real or imaginary coordinate $x, y \in \widetilde{\mathbb{R}}_q \cap [-2^H, 2^H]$, the exact average $\frac{1}{2}(x \pm y)$ lies in $[-2^H, 2^H]$. Since $\pm 2^H \in \widetilde{\mathbb{R}}_q$ are exact grid points and $\mathcal{Q}_q$ is monotone on $\mathbb{R}$, $\mathcal{Q}_q(\frac{1}{2}(x \pm y)) \in [-2^H, 2^H]$, preserving $\|\cdot\|_{\infty,\mathrm{coord}} \le 2^H$. Simultaneously, for each complex coefficient k_2, rounding $\frac{1}{2}(a_{k_2} \pm \tilde{b}_{k_2})$ via $\mathcal{Q}_q$ incurs Euclidean error at most $\frac{\sqrt{2}}{2} 2^{-q}$ by (4). Because the exact butterfly $(A, B) \mapsto \frac{1}{2}(A + y^{2m} B, A - y^{2m} B)$ is a contraction in the complex Euclidean ℓ_∞ norm, the stage-s array $P^{(s)}$ and forward error e_s relative to the exact stage-s array satisfy

$$\|P^{(s)}\|_\infty \le \|P^{(s-1)}\|_\infty + \frac{\sqrt{2}}{2} 2^{-q}, \qquad e_s \le e_{s-1} + \frac{\sqrt{2}}{2} 2^{-q},$$

giving $e_{k_1} \le k_1 \frac{\sqrt{2}}{2} 2^{-q} < 0.177\, k_1\, 2^{-p_{\ell+1}}$. □

4 The two-dimensional bridge on a power-of-two grid

We next combine two ingredients on the same $t_1 \times r$ grid: the classical polynomial-transform reduction of a two-dimensional power-of-two DFT $F_{t_1,r}$ to one polynomial transform G_{t_1} and t_1 recursive one-dimensional DFTs F_r (Section 4.1), and the fractional-shift Poisson identity connecting $F_{t_1,r}$ of a row-spread array directly to the one-dimensional DFT $F_{t_1 r}$ (Section 4.2).

4.1 Polynomial-transform reduction ($F_{t_1,r} \to t_1 F_r$)

Reducing a two-dimensional power-of-two DFT to a polynomial transform over $\mathbb{C}[y]/(y^r + 1)$ and t_1 one-dimensional transforms via an odd-stride index permutation is due to Nussbaumer and Quandalle [NQ79, Nus82], using standard negacyclic-to-cyclic root weighting $y \mapsto \zeta^{2u_2 - 1}$ (cf. [SS71, CF94]).

Proposition 4.1 (Polynomial-transform reduction; Nussbaumer–Quandalle [NQ79, Nus82])**.** *Let $t_1 \mid r$ be powers of two, $\nu = r/t_1 \in \{1, 2\}$, $R = t_1 r$, and $\zeta := e^{\pi i / r}$ (so $\zeta^r = -1$). Given any $U \in \mathbb{C}^{t_1 \times r}$, perform the following five steps:*

1. ***Pre-twist:*** *For $(j_1, k_2) \in \{0, \dots, t_1 - 1\} \times \{0, \dots, r - 1\}$, set $\tilde{U}_{j_1,k_2} := \zeta^{k_2} U_{j_1,k_2}$ and form $P_{j_1}(y) := \sum_{k_2=0}^{r-1} \tilde{U}_{j_1,k_2} y^{k_2} \in \mathcal{R}_r$.*
2. ***Polynomial transform:*** *Compute $Q = G_{t_1} P \in (\mathcal{R}_r)^{t_1}$ via (12), and write $Q_{v_1}(y) = \sum_{k_2=0}^{r-1} q_{v_1,k_2} y^{k_2}$ for its unique representative of degree $< r$.*
3. ***Post-twist:*** *For $(v_1, k_2) \in \{0, \dots, t_1 - 1\} \times \{0, \dots, r - 1\}$, set $\tilde{q}_{v_1,k_2} := \zeta^{-k_2} q_{v_1,k_2}$.*
4. ***Recursive 1D DFTs:*** *For each $v_1 \in \{0, \dots, t_1 - 1\}$, compute the 1D DFT $\hat{q}_{v_1} := F_r(\tilde{q}_{v_1,\cdot}) \in \mathbb{C}^r$.*
5. ***Odd-stride permutation:*** *For each $v_1 \in \{0, \dots, t_1 - 1\}$ and $u_2 \in \{0, \dots, r - 1\}$, set*

$$u_1 := (2u_2 - 1)v_1 \bmod t_1, \qquad Y_{u_1,u_2} := (\hat{q}_{v_1})_{u_2}. \tag{14}$$

Then $Y_{u_1,u_2} = (F_{t_1,r}U)_{u_1,u_2}$ *for all* $(u_1,u_2) \in \{0,\dots,t_1-1\} \times \{0,\dots,r-1\}$.

Proof. We include the short proof in our normalization. Fix any $u_2 \in \{0,\dots,r-1\}$. Because $2u_2-1$ is odd and $\zeta^r = e^{\pi i} = -1$, the complex number $\xi_{u_2} := \zeta^{2u_2-1}$ satisfies

$$\xi_{u_2}^r = (\zeta^r)^{2u_2-1} = (-1)^{2u_2-1} = -1, \quad \text{so} \quad \xi_{u_2}^r + 1 = 0.$$

Therefore, the polynomial evaluation map

$$\phi_{u_2} \colon \mathbb{C}[y]/(y^r+1) \longrightarrow \mathbb{C}, \qquad H(y) \bmod (y^r+1) \longmapsto H(\xi_{u_2})$$

is a well-defined ring homomorphism.

Fix $v_1 \in \{0,\dots,t_1-1\}$ and $u_2 \in \{0,\dots,r-1\}$. Since $\zeta^{-k_2} e^{2\pi i k_2 u_2/r} = e^{-\pi i k_2/r} e^{2\pi i k_2 u_2/r} = \zeta^{(2u_2-1)k_2} = \xi_{u_2}^{k_2}$, Step 4 evaluates to

$$(\hat{q}_{v_1})_{u_2} = \frac{1}{r} \sum_{k_2=0}^{r-1} \tilde{q}_{v_1,k_2} e^{2\pi i k_2 u_2/r} = \frac{1}{r} \sum_{k_2=0}^{r-1} q_{v_1,k_2} \xi_{u_2}^{k_2} = \frac{1}{r} Q_{v_1}(\xi_{u_2}) = \frac{1}{r} \phi_{u_2}(Q_{v_1}(y)).$$

Because ϕ_{u_2} is a ring homomorphism on $\mathcal{R}_r$, applying ϕ_{u_2} directly to the sum defining $Q_{v_1}(y)$ in Step 2 gives

$$(\hat{q}_{v_1})_{u_2} = \frac{1}{t_1 r} \sum_{j_1=0}^{t_1-1} \phi_{u_2}\big(P_{j_1}(y)\big)\, \phi_{u_2}\big(y^{2\nu j_1 v_1}\big) = \frac{1}{t_1 r} \sum_{j_1=0}^{t_1-1} \left(\sum_{k_2=0}^{r-1} \zeta^{k_2} U_{j_1,k_2}\, \zeta^{(2u_2-1)k_2} \right) \zeta^{2\nu(2u_2-1)j_1 v_1}.$$

Noting that $\zeta^{k_2}\zeta^{(2u_2-1)k_2} = \zeta^{2u_2 k_2} = e^{2\pi i k_2 u_2/r}$ and $\zeta^{2\nu} = e^{2\pi i (r/t_1)/r} = e^{2\pi i/t_1}$, this becomes

$$(\hat{q}_{v_1})_{u_2} = \frac{1}{t_1 r} \sum_{j_1=0}^{t_1-1} \sum_{k_2=0}^{r-1} U_{j_1,k_2}\, e^{2\pi i k_2 u_2/r}\, e^{2\pi i j_1((2u_2-1)v_1)/t_1}.$$

Since $e^{2\pi i/t_1}$ is t_1-periodic in the exponent, setting $u_1 := (2u_2-1)v_1 \bmod t_1$ gives $(\hat{q}_{v_1})_{u_2} = (F_{t_1,r}U)_{u_1,u_2}$. Finally, since $2u_2-1$ is odd and t_1 is a power of two, $\gcd(2u_2-1, t_1) = 1$, so $v_1 \mapsto (2u_2-1)v_1 \bmod t_1$ is a bijection on $\{0,\dots,t_1-1\}$ for every u_2. □

4.2 Fractional-shift Poisson identity ($F_{t_1 r} \leftrightarrow F_{t_1,r}$)

Approximating fractional shifts by convolving with a localized smooth kernel and analyzing the resulting alias series via Poisson summation is the standard gridding mechanism of the nonuniform discrete Fourier transform (NUFFT) [DR93, PST01, GL04, BMK19], and is related to the Gaussian resampling step of [HvdH21]. Here, because $F_{t_1 r}$ and $F_{t_1,r}$ act on the same $t_1 \times r$ grid and the fractional shift j_1/t_1 is constant across row j_1, row-wise spreading requires no grid oversampling and produces a cyclic alias mixture along each column u_2 that is inverted by a two-term recurrence.

Theorem 4.2 (Fractional-shift Poisson identity)**.** *Let* $R = t_1 r$ *with* $t_1, r \in \mathbb{Z}_{\geq 1}$*, and for* $\alpha > 0$ *define the windowed sinc-Gaussian kernel*

$$g(t) := \operatorname{sinc}(t)\, e^{-\frac{\pi}{\alpha^2} t^2} = \frac{\sin(\pi t)}{\pi t}\, e^{-\frac{\pi}{\alpha^2} t^2} \quad (g(0) := 1), \tag{15}$$

whose continuous Fourier transform $\hat{g}(\xi) := \int_{-\infty}^{\infty} g(t) e^{-2\pi i \xi t}\, dt$ *is*

$$\hat{g}(\xi) = \int_{-1/2}^{1/2} \alpha\, e^{-\pi \alpha^2 (\xi - \xi')^2}\, d\xi' = \frac{1}{2} \left[\operatorname{erf}\left(\sqrt{\pi}\alpha \left(\xi + \frac{1}{2} \right) \right) - \operatorname{erf}\left(\sqrt{\pi}\alpha \left(\xi - \frac{1}{2} \right) \right) \right]. \tag{16}$$

For any input vector $x \in \mathbb{C}^R$, define the ideal spread array $U^\circ \in \mathbb{C}^{t_1 \times r}$ by convolving each row $j_1 \in \{0, \dots, t_1 - 1\}$ cyclically along $j_2 \in \{0, \dots, r-1\}$ with the real r-periodic filter $h^{(j_1),\mathrm{per}}$:

$$U^\circ_{j_1,k_2} := \sum_{j_2=0}^{r-1} x(j_1 + t_1 j_2)\, h^{(j_1),\mathrm{per}}_{(j_2-k_2) \bmod r}, \qquad h^{(j_1),\mathrm{per}}_d := \sum_{a \in \mathbb{Z}} g\Big(d + \frac{j_1}{t_1} - ar\Big). \tag{17}$$

Then for all $(u_1, u_2) \in \{0, \dots, t_1 - 1\} \times \{0, \dots, r-1\}$, the 2D DFT $F_{t_1,r} U^\circ$ and the 1D DFT $F_R x$ satisfy the exact identity

$$\big(F_{t_1,r} U^\circ\big)_{u_1,u_2} = \sum_{m \in \mathbb{Z}} \hat{g}\Big(\frac{u_2}{r} - m\Big)\, (F_R x)_{u_2 + r((u_1 - m) \bmod t_1)}. \tag{18}$$

Proof. Since $g \in \mathcal{S}(\mathbb{R})$ is a Schwartz function (sinc is smooth with $|\mathrm{sinc}^{(k)}| \le \pi^k$ on $\mathbb{R}$ for every k, being the inverse Fourier transform of the bounded, compactly supported $\mathbf{1}_{[-1/2,1/2]}$, and the product of such a function with the Gaussian $e^{-\pi t^2/\alpha^2}$ is a Schwartz function), the periodized sum defining $h^{(j_1),\mathrm{per}}_d$ converges absolutely. Expanding $F_{t_1,r} U^\circ$ gives

$$\big(F_{t_1,r} U^\circ\big)_{u_1,u_2} = \frac{1}{t_1 r} \sum_{j_1=0}^{t_1-1} e^{2\pi i j_1 u_1 / t_1} \sum_{j_2=0}^{r-1} x(j_1 + t_1 j_2) \left(\sum_{k_2=0}^{r-1} \sum_{a \in \mathbb{Z}} g\Big(j_2 + \frac{j_1}{t_1} - (k_2 + ar)\Big)\, e^{2\pi i k_2 u_2 / r} \right).$$

Because $e^{2\pi i (k_2 + ar) u_2 / r} = e^{2\pi i k_2 u_2 / r}$ for all $a \in \mathbb{Z}$, as k_2 ranges over $\{0, \dots, r-1\}$ and a ranges over $\mathbb{Z}$, the integer $k := k_2 + ar$ ranges over $\mathbb{Z}$ bijectively. Hence the inner parenthesized sum equals

$$S(j_1, j_2, u_2) := \sum_{k \in \mathbb{Z}} g(z - k) e^{2\pi i k u_2 / r}, \qquad \text{where } z := j_2 + \frac{j_1}{t_1}.$$

Because $\mathcal{F}\{\mathrm{sinc}\}(\xi) = \mathbf{1}_{[-1/2,1/2]}(\xi)$ in $L^2(\mathbb{R})$ and $\mathcal{F}\{e^{-\pi t^2/\alpha^2}\}(\xi) = \alpha e^{-\pi \alpha^2 \xi^2}$, the convolution theorem gives (16) (sinc is in L^2 but not L^1; the identity $\mathcal{F}(\mathrm{sinc} \cdot \phi) = \mathcal{F}(\mathrm{sinc}) * \mathcal{F}(\phi)$ holds for Schwartz ϕ, and both sides of (16) are continuous, so it holds pointwise). Fix $z \in \mathbb{R}$ and $\xi := u_2/r$, and define $f(t) := g(z - t) e^{2\pi i \xi t} \in \mathcal{S}(\mathbb{R})$. Substituting $s = z - t$ gives the continuous Fourier transform of f at frequency $m \in \mathbb{Z}$:

$$\hat{f}(m) = \int_{-\infty}^{\infty} g(z - t) e^{2\pi i (\xi - m) t}\, dt = e^{2\pi i (\xi - m) z} \int_{-\infty}^{\infty} g(s) e^{-2\pi i (\xi - m) s}\, ds = \hat{g}(\xi - m) e^{2\pi i (\xi - m) z}.$$

By the one-dimensional Poisson summation formula $\sum_{k \in \mathbb{Z}} f(k) = \sum_{m \in \mathbb{Z}} \hat{f}(m)$, we obtain

$$S(j_1, j_2, u_2) = \sum_{m \in \mathbb{Z}} \hat{g}\Big(\frac{u_2}{r} - m\Big)\, e^{2\pi i \left(j_2 + \frac{j_1}{t_1}\right)\left(\frac{u_2}{r} - m\right)}.$$

Since $j_2, m \in \mathbb{Z}$, $e^{-2\pi i j_2 m} = 1$. For each $m \in \mathbb{Z}$, let $\tilde{u}_1 := (u_1 - m) \bmod t_1 \in \{0, \dots, t_1 - 1\}$ so that $u_1 - m \equiv \tilde{u}_1 \pmod{t_1}$ and $e^{2\pi i j_1 (u_1 - m)/t_1} = e^{2\pi i j_1 \tilde{u}_1 / t_1}$. Then

$$\frac{j_1 \tilde{u}_1}{t_1} + \frac{\left(j_2 + \frac{j_1}{t_1}\right) u_2}{r} = \frac{j_1 r \tilde{u}_1 + (j_1 + t_1 j_2) u_2}{t_1 r} \equiv \frac{(j_1 + t_1 j_2)(u_2 + r\tilde{u}_1)}{R} \pmod 1,$$

because $t_1 j_2 r \tilde{u}_1 / (t_1 r) = j_2 \tilde{u}_1 \in \mathbb{Z}$. As (j_1, j_2) ranges over $\{0, \dots, t_1 - 1\} \times \{0, \dots, r-1\}$, $j := j_1 + t_1 j_2$ ranges over $\{0, \dots, R-1\}$, so the sum over (j_1, j_2) equals $(F_R x)_{u_2 + r\tilde{u}_1}$, proving (18). □

Theorem 4.2 holds for arbitrary positive integers $t_1, r \ge 1$; for the remainder of the paper, $t_1 \mid r$ are powers of two with $t_1 \ge 2$.

5 Sinc-Gaussian deconvolution and column solvers

5.1 Passband, stopband, conditioning, and the Nyquist identity

Theorem 5.1 (Two-term deconvolution and the Nyquist identity)**.** *Let $t_1 \mid r$ be powers of two with $t_1 \geq 2$ (so t_1 is even) and $t_1 \in \{r, r/2\}$, and let $R = t_1 r$. For a working cutoff parameter $p_{\mathrm{work}} \in \mathbb{Z}_{\geq 8}$, set $\alpha := 2\sqrt{\frac{p_{\mathrm{work}} \ln 2}{\pi}}$ (so $\pi\alpha^2 = 4p_{\mathrm{work}} \ln 2$). For each fixed column $u_2 \in \{0, \dots, r-1\}$, define $y := u_2/r \in [0,1)$, $Y^{\circ}_{u_1} := (F_{t_1,r} U^{\circ})_{u_1,u_2}$, $Z_{u_1} := (F_R x)_{u_2 + r u_1}$, and*

$$c_0(u_2) := \hat{g}(y), \qquad c_1(u_2) := \hat{g}(1-y) = \hat{g}(y-1), \qquad M(u_2) := \max(c_0(u_2), c_1(u_2)). \tag{19}$$

Then:

1. ***Partition of unity, passband, and telescoping stopband alias:*** *For every $u_2 \in \{0, \dots, r-1\}$, $\sum_{m \in \mathbb{Z}} \hat{g}(y-m) = 1$, the dominant coefficient $M(u_2)$ satisfies*

$$\frac{1}{2} - 0.08 \cdot 2^{-4p_{\mathrm{work}}} \leq \frac{1}{2}\operatorname{erf}(\sqrt{\pi}\alpha) = \hat{g}\left(\frac{1}{2}\right) \leq M(u_2) < 1 \qquad (M(u_2) > 0.49999999 \text{ for } p_{\mathrm{work}} \geq 8), \tag{20}$$

and the excluded Poisson stopband aliases ($m \notin \{0,1\}$) telescope in closed form to

$$\begin{aligned} \Psi(y) := \sum_{m \notin \{0,1\}} \hat{g}(y-m) &= \frac{1}{2}\operatorname{erfc}\left(\sqrt{\pi}\alpha\left(y + \frac{1}{2}\right)\right) + \frac{1}{2}\operatorname{erfc}\left(\sqrt{\pi}\alpha\left(\frac{3}{2} - y\right)\right) \\ &\leq \Psi(0) < 0.120 \cdot 2^{-p_{\mathrm{work}}} < 0.15 \cdot 2^{-p_{\mathrm{work}}}. \end{aligned} \tag{21}$$

2. ***Cyclic bidiagonal system for $u_2 \neq r/2$:*** *For all $u_2 \neq r/2$ and $t_1 \in \{r, r/2\}$, let $a := \min(c_0, c_1)/M(u_2) \in (0,1)$ and $\rho := -a \in (-1, 0)$. Because $t_1 \geq 2$ is even, $\rho^{t_1} = a^{t_1}$, and*

$$a^{t_1} \leq e^{-1.9999 \alpha t_1 |1/2 - y|} \leq e^{-0.99995\alpha} < 0.1400, \qquad \Gamma(u_2) := M(u_2)(1 - a^{t_1}) > 0.424. \tag{22}$$

The two-term cyclic bidiagonal operator $(BZ)_{u_1} := c_0(u_2) Z_{u_1} + c_1(u_2) Z_{(u_1 - 1) \bmod t_1} = Y_{u_1}$ ($u_1 \in \mathbb{Z}_{t_1}$) is invertible with exact closed form

$$Z_{u_1} = \begin{cases} \dfrac{1}{c_0(1-\rho^{t_1})} \displaystyle\sum_{s=0}^{t_1-1} \rho^s Y_{(u_1 - s) \bmod t_1}, & \text{if } c_0 > c_1 \ (u_2 < r/2), \\ \dfrac{1}{c_1(1-\rho^{t_1})} \displaystyle\sum_{s=0}^{t_1-1} \rho^s Y_{(u_1 + 1 + s) \bmod t_1}, & \text{if } c_1 > c_0 \ (u_2 > r/2), \end{cases} \tag{23}$$

and its $\ell_\infty \to \ell_\infty$ inverse operator norm satisfies

$$\|B^{-1}\|_{\infty \to \infty} = \frac{1}{|c_0(u_2) - c_1(u_2)|} = \frac{1 + a + \cdots + a^{t_1 - 1}}{M(u_2)(1 - a^{t_1})} \leq \frac{t_1}{\Gamma(u_2)} < 2.36\, t_1. \tag{24}$$

3. ***Nyquist alternating-sum identity at $u_2 = r/2$:*** *At the Nyquist column $u_2 = r/2$ ($y = 1/2$), where $c := c_0(r/2) = c_1(r/2) = \frac{1}{2}\operatorname{erf}(\sqrt{\pi}\alpha) \in (0.49999999, 1/2)$, the alternating sum of $Z_{u_1} = (F_R x)_{r/2 + r u_1}$ is given in terms of the input vector x with zero scalar multiplications by*

$$A_{\mathrm{N}} := \sum_{u_1=0}^{t_1-1} (-1)^{u_1} Z_{u_1} = \frac{i}{r} \sum_{j_2=0}^{r-1} (-1)^{j_2} x\left(\frac{t_1}{2} + t_1 j_2\right). \tag{25}$$

Replacing the redundant equation at $u_1 = 0$ by (25) yields a uniquely solvable linear system whose inverse has ℓ_∞ operator norm $\frac{t_1 - 1}{2c} < 1.0000001\, t_1 < 2.36\, t_1$ on $(Y_1, \dots, Y_{t_1 - 1})$ and $\frac{1}{t_1}$ on A_{N}.

Proof. **Part 1.** Because the unit intervals $[y-m-1/2, y-m+1/2]$ ($m \in \mathbb{Z}$) partition $\mathbb{R}$ up to endpoints of measure zero,

$$\sum_{m\in\mathbb{Z}} \hat{g}(y-m) = \sum_{m\in\mathbb{Z}} \int_{y-m-1/2}^{y-m+1/2} \alpha e^{-\pi\alpha^2\xi'^2}\, d\xi' = \int_{-\infty}^{\infty} \alpha e^{-\pi\alpha^2\xi'^2}\, d\xi' = 1.$$

For $y \in [0,1)$, let $\eta := \min(y, 1-y) \in [0, 1/2]$, so $M(u_2) = \hat{g}(\eta)$ since $\hat{g}$ is even. For $\eta > 0$,

$$\hat{g}'(\eta) = \alpha\Big(e^{-\pi\alpha^2(\eta+1/2)^2} - e^{-\pi\alpha^2(\eta-1/2)^2}\Big) < 0,$$

so $\hat{g}$ is strictly decreasing on $[0,\infty)$ and $M(u_2) \geq \hat{g}(1/2) = \frac{1}{2}\operatorname{erf}(\sqrt{\pi}\alpha) = \frac{1}{2}(1 - \operatorname{erfc}(\sqrt{\pi}\alpha))$. Since $\pi\alpha^2 = 4p_{\text{work}}\ln 2$ and $\operatorname{erfc}(z) \leq \frac{e^{-z^2}}{\sqrt{\pi}z}$ for $z > 0$, at $p_{\text{work}} \geq 8$ ($\sqrt{\pi}\alpha = \sqrt{4p_{\text{work}}\ln 2} \geq \sqrt{32\ln 2} > 4.7096$) we have $\frac{1}{2}\operatorname{erfc}(\sqrt{\pi}\alpha) \leq \frac{2^{-4p_{\text{work}}}}{2\sqrt{4\pi p_{\text{work}}\ln 2}} < 0.060 \cdot 2^{-4p_{\text{work}}} < 0.08 \cdot 2^{-4p_{\text{work}}}$ and $\hat{g}(1/2) > 0.499999999 > 0.49999999$.

Removing $m \in \{0,1\}$ from $\mathbb{R}$ leaves $(-\infty, y-3/2] \cup [y+1/2, \infty)$, which directly yields the formula for $\Psi(y)$ in (21). Since $\frac{d^2}{dz^2}\operatorname{erfc}(z) = \frac{4z}{\sqrt{\pi}}e^{-z^2} > 0$ for all $z > 0$ and both $y + 1/2 \geq 1/2 > 0$ and $3/2 - y \geq 1/2 > 0$ on $y \in [0,1]$, $\Psi(y)$ is strictly convex on $[0,1]$ and symmetric under $y \mapsto 1-y$. Hence $\max_{y\in[0,1]}\Psi(y) = \Psi(0) = \Psi(1)$. At $y = 0$, $z_1 := \sqrt{\pi}\alpha(1/2) = \sqrt{p_{\text{work}}\ln 2}$ and $z_2 := \sqrt{\pi}\alpha(3/2) = 3\sqrt{p_{\text{work}}\ln 2}$. Using $\operatorname{erfc}(z) \leq \frac{e^{-z^2}}{\sqrt{\pi}z}$ (or the three-term integration-by-parts bound $\operatorname{erfc}(z) \leq \frac{e^{-z^2}}{\sqrt{\pi}z}(1 - \frac{1}{2z^2} + \frac{3}{4z^4})$, which gives $\Psi(0) < 0.112 \cdot 2^{-p_{\text{work}}}$ for $p_{\text{work}} \geq 8$, matching $\Psi(0) \approx 0.1111 \cdot 2^{-p_{\text{work}}}$ at $p_{\text{work}} = 8$), we have for all $p_{\text{work}} \geq 8$:

$$\begin{aligned}\Psi(0) &= \frac{1}{2}\operatorname{erfc}\big(\sqrt{p_{\text{work}}\ln 2}\big) + \frac{1}{2}\operatorname{erfc}\big(3\sqrt{p_{\text{work}}\ln 2}\big) \\ &\leq \frac{2^{-p_{\text{work}}}}{2\sqrt{\pi p_{\text{work}}\ln 2}} + \frac{2^{-9p_{\text{work}}}}{6\sqrt{\pi p_{\text{work}}\ln 2}} < 0.1199 \cdot 2^{-p_{\text{work}}} < 0.120 \cdot 2^{-p_{\text{work}}}.\end{aligned}$$

Part 2. Fix $u_2 \neq r/2$ and let $\varepsilon := |1/2 - y| \geq 1/r > 0$. Then $M(u_2) = \hat{g}(1/2-\varepsilon)$ and $\min(c_0, c_1) = \hat{g}(1/2+\varepsilon)$. For $s \in [0, \varepsilon]$, let $\phi(s) := \ln\hat{g}(1/2-s) - \ln\hat{g}(1/2+s)$. Since $\hat{g}$ is decreasing on $[0,\infty)$, we have $-\hat{g}'(1/2-s) \geq 0$ for all $s \in [0,\varepsilon] \subseteq [0, 1/2]$, while $-\hat{g}'(1/2+s) = \alpha e^{-\pi\alpha^2 s^2}(1 - e^{-\pi\alpha^2(1+2s)}) \geq \alpha e^{-\pi\alpha^2 s^2}(1 - 2^{-4p_{\text{work}}})$, and for $s \geq 0$,

$$\hat{g}(1/2+s) = \int_0^1 \alpha e^{-\pi\alpha^2(\xi'+s)^2}\, d\xi' \leq e^{-\pi\alpha^2 s^2}\int_0^1 \alpha e^{-\pi\alpha^2\xi'^2}\, d\xi' = \hat{g}(1/2)e^{-\pi\alpha^2 s^2}.$$

Hence $\phi'(s) \geq -\hat{g}'(1/2+s)/\hat{g}(1/2+s) \geq \frac{\alpha(1-2^{-4p_{\text{work}}})}{\hat{g}(1/2)} = \frac{2\alpha(1-2^{-4p_{\text{work}}})}{\operatorname{erf}(\sqrt{\pi}\alpha)} > 1.9999\alpha$ for $p_{\text{work}} \geq 8$. Integrating $\phi'(s) > 1.9999\alpha$ over $s \in [0,\varepsilon]$ yields $a = e^{-\phi(\varepsilon)} \leq e^{-1.9999\alpha\varepsilon}$. Since $t_1 \in \{r, r/2\}$ and $\varepsilon \geq 1/r$, $t_1\varepsilon \geq \frac{r}{2}\cdot\frac{1}{r} = \frac{1}{2}$, so $a^{t_1} \leq e^{-1.9999\alpha t_1\varepsilon} \leq e^{-0.99995\alpha}$. For $p_{\text{work}} \geq 8$, $\alpha = 2\sqrt{p_{\text{work}}\ln 2/\pi} \geq 2\sqrt{8\ln 2/\pi} > 2.6571$, so $e^{-0.99995\alpha} < 0.0702 < 0.1400$ and $\Gamma(u_2) = M(u_2)(1-a^{t_1}) > 0.49999999 \times (1 - 0.1400) > 0.424$.

When $c_0 > c_1$, $Z_{u_1} - \rho Z_{u_1-1} = Y_{u_1}/c_0$ ($\rho = -c_1/c_0$), and unrolling t_1 steps around $\mathbb{Z}_{t_1}$ gives $(1-\rho^{t_1})Z_{u_1} = c_0^{-1}\sum_{s=0}^{t_1-1}\rho^s Y_{(u_1-s) \bmod t_1}$. When $c_1 > c_0$, dividing $c_1 Z_{u_1} + c_0 Z_{u_1+1} = Y_{u_1+1}$ by c_1 gives $Z_{u_1} - \rho Z_{u_1+1} = Y_{u_1+1}/c_1$ ($\rho = -c_0/c_1$), and unrolling t_1 steps backward gives $(1-\rho^{t_1})Z_{u_1} = c_1^{-1}\sum_{s=0}^{t_1-1}\rho^s Y_{(u_1+1+s) \bmod t_1}$ (with the $+1$ index shift). Because $t_1 \geq 2$ is even, $1-\rho^{t_1} = 1-a^{t_1} > 0.86$. In both cases, the absolute row sum of B^{-1} is $\frac{1+a+\cdots+a^{t_1-1}}{M(u_2)(1-a^{t_1})} = \frac{1}{|c_0-c_1|} \leq \frac{t_1}{\Gamma(u_2)} < 2.36\, t_1$.

Part 3. Since $t_1 \geq 2$ is a power of two, t_1 is even and $t_1/2 \in \{1, \ldots, t_1-1\}$ is an integer row index. At $u_2 = r/2$, writing $j = j_1 + t_1 j_2$ and $u = r/2 + r u_1$ gives $\frac{ju}{t_1 r} = \frac{j_1}{2t_1} + \frac{j_2}{2} + \frac{j_1 u_1}{t_1} + j_2 u_1 \equiv \frac{j_1}{2t_1} + \frac{j_2}{2} + \frac{j_1 u_1}{t_1} \pmod 1$. Summing $(-1)^{u_1} = e^{\pi i u_1}$ over $u_1 \in \{0, \ldots, t_1 - 1\}$ applies character orthogonality on $\mathbb{Z}_{t_1}$:

$$\sum_{u_1=0}^{t_1-1} (-1)^{u_1} e^{2\pi i j j_1 u_1/t_1} = \sum_{u_1=0}^{t_1-1} e^{2\pi i (j_1 - t_1/2) u_1/t_1} = t_1\, \delta_{j_1,\, t_1/2}.$$

Substituting $j_1 = t_1/2$ gives $e^{2\pi i (t_1/2)/(2t_1)} e^{\pi i j_2} = e^{\pi i/2}(-1)^{j_2} = i(-1)^{j_2}$ and $\frac{t_1}{t_1 r} = \frac{1}{r}$, proving (25). Setting $z'_{u_1} := (-1)^{u_1} Z_{u_1}$, the equations $z'_{u_1} - z'_{u_1 - 1} = (-1)^{u_1} Y_{u_1}/c$ $(1 \leq u_1 < t_1)$ and $\sum_{u_1=0}^{t_1-1} z'_{u_1} = A_{\mathrm{N}}$ have unique solution $z'_{u_1} = p_{u_1} + \frac{1}{t_1}(A_{\mathrm{N}} - \sum_{v=0}^{t_1-1} p_v)$ where $p_{u_1} = \sum_{k=1}^{u_1} (-1)^k Y_k/c$. The explicit inverse expresses z'_{u_1} as $\frac{1}{t_1} A_{\mathrm{N}} + \frac{1}{c} \sum_{k=1}^{t_1-1} w_{u_1,k} (-1)^k Y_k$ with $w_{u_1,k} = \frac{k}{t_1}$ for $k \leq u_1$ and $w_{u_1,k} = \frac{k}{t_1} - 1$ for $k > u_1$, whose absolute row sum over Y is $\frac{1}{c} \sum_{k=1}^{t_1-1} |w_{u_1,k}| \leq \frac{t_1 - 1}{2c} < 1.0000001\, t_1$. □

5.2 Finite-precision column solvers

We now specify the finite-precision procedures `CyclicSolve` ($u_2 \neq r/2$) and `NyquistSolve` ($u_2 = r/2$) on the grid $\widetilde{\mathbb{C}}_{q_{\mathrm{child}}}$ ($q_{\mathrm{child}} = p_{\ell+1} + 2$, $u := 2^{-p_{\ell+1}}$) and state their forward rounding bound relative to the exact solution $Z^\dagger$ of the retained two-term system with right-hand side $\widehat{Y}_{\cdot,u_2}$. Here $\theta(u_2) := (1 - \rho(u_2)^{t_1})^{-1}$ denotes the exact cyclic closure scalar, and $\hat{\theta}(u_2) \in \widetilde{\mathbb{R}}_{q_{\mathrm{child}}}$ its 1-ulp faithful precomputed table entry.

Procedure `CyclicSolve` ($u_2 \neq r/2$). Given $\widehat{Y} = (\widehat{Y}_0, \ldots, \widehat{Y}_{t_1-1}) \in \widetilde{\mathbb{C}}^{t_1}_{q_{\mathrm{child}}}$ and the 1-ulp faithful precomputed table scalars $\mu = \mu(u_2), \hat{\rho} = \hat{\rho}(u_2), \hat{\theta} = \hat{\theta}(u_2) \in \widetilde{\mathbb{R}}_{q_{\mathrm{child}}}$ from Section 3.3 (satisfying $|\mu - M(u_2)^{-1}|, |\hat{\rho} - \rho|, |\hat{\theta} - \theta(u_2)| \leq \frac{1}{4}u = 0.25u$, and branching directly on $u_2 < r/2$ versus $u_2 > r/2$):

- *Case 1: $u_2 < r/2$ ($c_0(u_2) > c_1(u_2)$, forward sweeps):*
  1. *Prescale (t_1 real-by-complex products):* Set $\tilde{B}_i := \mathrm{RM}_{q_{\mathrm{child}}}(\mu, \widehat{Y}_i)$ for $i = 0, \ldots, t_1 - 1$.
  2. *Forward prefix sweep ($t_1 - 1$ real-by-complex products):* Set $\tilde{P}_0 := \tilde{B}_0$, and for $i = 1, \ldots, t_1 - 1$, set $\tilde{P}_i := \tilde{B}_i + \mathrm{RM}_{q_{\mathrm{child}}}(\hat{\rho}, \tilde{P}_{i-1})$ (so in exact arithmetic $P_{t_1-1} = \sum_{s=0}^{t_1-1} \rho^s B_{t_1-1-s} = (1 - \rho^{t_1}) Z^\dagger_{t_1-1}$).
  3. *Wrap-around closure and forward recurrence pass (t_1 real-by-complex products):* Set

     $$Z^{\mathrm{comp}}_{t_1-1} := \mathrm{RM}_{q_{\mathrm{child}}}(\hat{\theta}, \tilde{P}_{t_1-1}), \qquad Z^{\mathrm{comp}}_0 := \tilde{B}_0 + \mathrm{RM}_{q_{\mathrm{child}}}(\hat{\rho}, Z^{\mathrm{comp}}_{t_1-1}),$$

     and for $i = 1, \ldots, t_1 - 2$, set $Z^{\mathrm{comp}}_i := \tilde{B}_i + \mathrm{RM}_{q_{\mathrm{child}}}(\hat{\rho}, Z^{\mathrm{comp}}_{i-1})$.
- *Case 2: $u_2 > r/2$ ($c_1(u_2) > c_0(u_2)$, backward sweeps):*
  1. *Prescale (t_1 real-by-complex products):* Set $\tilde{B}_i := \mathrm{RM}_{q_{\mathrm{child}}}(\mu, \widehat{Y}_{(i+1) \bmod t_1})$ for $i = 0, \ldots, t_1 - 1$.
  2. *Backward prefix sweep ($t_1 - 1$ real-by-complex products):* Set $\tilde{P}_{t_1-1} := \tilde{B}_{t_1-1}$, and for $i = t_1 - 2, \ldots, 0$, set $\tilde{P}_i := \tilde{B}_i + \mathrm{RM}_{q_{\mathrm{child}}}(\hat{\rho}, \tilde{P}_{i+1})$ (so in exact arithmetic $P_0 = \sum_{s=0}^{t_1-1} \rho^s B_s = (1 - \rho^{t_1}) Z^\dagger_0$).
  3. *Wrap-around closure and backward recurrence pass (t_1 real-by-complex products):* Set

     $$Z^{\mathrm{comp}}_0 := \mathrm{RM}_{q_{\mathrm{child}}}(\hat{\theta}, \tilde{P}_0), \qquad Z^{\mathrm{comp}}_{t_1-1} := \tilde{B}_{t_1-1} + \mathrm{RM}_{q_{\mathrm{child}}}(\hat{\rho}, Z^{\mathrm{comp}}_0),$$

     and for $i = t_1 - 2, \ldots, 1$, set $Z^{\mathrm{comp}}_i := \tilde{B}_i + \mathrm{RM}_{q_{\mathrm{child}}}(\hat{\rho}, Z^{\mathrm{comp}}_{i+1})$.

In both cases, `CyclicSolve` performs $t_1 + (t_1 - 1) + t_1 = 3t_1 - 1$ real-by-complex rounded multiplications and $2t_1 - 2$ complex additions.

Procedure `NyquistSolve` ($u_2 = r/2$) and final output transpose. Given $\widehat{Y} \in \widetilde{\mathbb{C}}^{t_1}_{q_{\mathrm{child}}}$, the exact input alternating sum $A_{\mathrm{N}} \in \widetilde{\mathbb{C}}_{q_{\mathrm{child}}}$ from (25) (which is exact on $\widetilde{\mathbb{C}}_{q_{\mathrm{child}}}$ because $x \in \widetilde{\mathbb{C}}^{R_\ell}_{p_\ell+2}$ and $q_{\mathrm{child}} \geq p_\ell + 2 + \log_2 r$), and the 1-ulp faithful Nyquist reciprocal $\mu_{\mathrm{N}} = \mu(r/2) \in \widetilde{\mathbb{R}}_{q_{\mathrm{child}}}$ ($|\mu_{\mathrm{N}} - c^{-1}| \leq \frac{1}{4}u$):

1. Set $\tilde{P}_0 := 0$, and for $i = 1, \ldots, t_1 - 1$, compute $\tilde{D}_i := \mathrm{RM}_{q_{\mathrm{child}}}(\mu_{\mathrm{N}}, \widehat{Y}_i)$ and $\tilde{P}_i := \tilde{D}_i - \tilde{P}_{i-1}$ ($t_1 - 1$ real-by-complex products).

2. Compute the alternating sum $\tilde{S} := \sum_{i=0}^{t_1-1} (-1)^i \tilde{P}_i$ in a $(w_{\mathrm{coord},\ell} + k_{1,\ell} + 1)$-bit scratch accumulator, form $\tilde{\delta} := \mathcal{Q}_{q_{\mathrm{child}}}((A_{\mathrm{N}} - \tilde{S})/t_1)$, and set $Z^{\mathrm{comp}}_i := \tilde{P}_i + (-1)^i \tilde{\delta}$ for $i = 0, \ldots, t_1 - 1$.

Once all r columns $u_2 \in \{0, \ldots, r-1\}$ have been solved in column-major order (u_2, u_1) and quantized to $\widehat{Z}_{u_1,u_2} = \mathcal{Q}_{p_\ell+2}(Z^{\mathrm{comp}}_{u_1,u_2})$, a single transpose from column-major order (u_2, u_1) to row-major order (u_1, u_2) (Lemma 3.3(i), costing $O(R_\ell w_\ell (1 + \log_2 t_{1,\ell})) = O(R_\ell p_{\ell+1}(1 + \log_2 t_{1,\ell}))$ bit operations on the uniform records of (6)) places the output in natural one-dimensional frequency order $u_2 + r u_1$.

Lemma 5.2 (Forward error of the column solvers)**.** *Let $\mathcal{M}_Y := \max(1, \|\widehat{Y}\|_\infty)$ and $u = 2^{-p_{\ell+1}}$ ($q_{\mathrm{child}} = p_{\ell+1} + 2$). For every column $u_2 \in \{0, \ldots, r-1\}$ with $t_1 \geq 2$ and $t_1 u \leq 2^{-20}$, under the hypotheses of Theorem 5.1 ($p_{\mathrm{work}} \geq 8$), the computed vector Z^{comp} and the exact solution $Z^\dagger$ of the retained two-term system with right-hand side $\widehat{Y}$ satisfy*

$$\|Z^{\mathrm{comp}} - Z^\dagger\|_\infty \leq 80\, t_1^2\, u\, \mathcal{M}_Y. \tag{26}$$

The proof of Lemma 5.2 (given in Appendix C.1) establishes the tighter bound $\|Z^{\mathrm{comp}} - Z^\dagger\|_\infty < 1.85\, t_1^2\, u\, \mathcal{M}_Y$ for $u_2 \neq r/2$ and $\leq 5\, t_1\, u\, \mathcal{M}_Y$ for $u_2 = r/2$. Since $5t_1 \leq 2.5\, t_1^2$ for $t_1 \geq 2$, both cases are below $2.5\, t_1^2\, u\, \mathcal{M}_Y$; we state the generous constant 80, which is the value used in Lemma 6.5.

6 Spreading, forward error, and the additive schedule

6.1 Filter estimates and modulo-r folding

At recursion depth ℓ, let $R_\ell = 2^{K_\ell} = t_{1,\ell} r_\ell$, with working cutoff $p_{\mathrm{work},\ell}$ and child precision $p_{\ell+1}$. We distinguish four filter objects for each row $j_1 \in \{0, \ldots, t_{1,\ell} - 1\}$.

(a) The *unfolded sample sequence* is $\kappa^{(j_1)}_d := g(d + j_1/t_{1,\ell})$ for $d \in \mathbb{Z}$.

(b) The *exact r_ℓ-periodic filter* is $h^{(j_1),\mathrm{per}}_d := \sum_{a \in \mathbb{Z}} \kappa^{(j_1)}_{d + a r_\ell}$ for $d \in \{0, \ldots, r_\ell - 1\}$.

(c) The *truncated modulo-r_ℓ folded filter* is $h^{(j_1),\mathrm{tr}}_d := \sum_{|e| \leq m_{1,\ell},\, e \equiv d \pmod{r_\ell}} \kappa^{(j_1)}_e$, supported on at most $\tilde{W}_\ell := \min(W_\ell, r_\ell)$ indices where $W_\ell := 2m_{1,\ell} + 1$.

(d) The *guarded q_{child}-grid tap vector* $\bar{H}^{(j_1)} \in \widetilde{\mathbb{R}}^{\tilde{W}_\ell}_{q_{\mathrm{child}}}$ and its *re-indexed r_ℓ-periodic filter* $\bar{h}^{(j_1)} \in \widetilde{\mathbb{R}}^{r_\ell}_{q_{\mathrm{child}}}$ are formed by evaluating $\kappa^{(j_1)}_e$ ($|e| \leq m_{1,\ell}$) via Lemma 3.2 at guard precision $v_{\mathrm{tap}} := q_{\mathrm{child}} + \lceil \log_2(W_\ell + 1) \rceil + 10$, summing residue classes $a = (-e) \bmod r_\ell$ when $W_\ell > r_\ell$ (giving cyclic offset $o := 0$, or reversing the W_ℓ contiguous taps $\kappa^{(j_1)}_{m_{1,\ell}-a}$ with cyclic offset $o := m_{1,\ell}$ when $W_\ell \leq r_\ell$), rounding each tap once via $\mathcal{Q}_{q_{\mathrm{child}}}$ to obtain $\bar{H}^{(j_1)}_a$, and setting $\bar{h}^{(j_1)}_d := \bar{H}^{(j_1)}_a$ if $d \equiv o - a \pmod{r_\ell}$ for $a \in \{0, \ldots, \tilde{W}_\ell - 1\}$ and $\bar{h}^{(j_1)}_d := 0$ otherwise.

The proof of Lemma 6.1 is given in Appendix C.2.

Lemma 6.1 (Filter estimates)**.** *Let $p_{\mathrm{work},\ell} \geq 8$, $\alpha_\ell = 2\sqrt{p_{\mathrm{work},\ell} \ln 2/\pi}$, $p_{\ell+1} \geq \max(p_{\mathrm{work},\ell}, 20)$, and $m_{1,\ell} \geq \left\lceil \alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi} \right\rceil$. For every row $j_1 \in \{0, \ldots, t_{1,\ell} - 1\}$:*

1. Unfolded ℓ_1 bound:

$$\begin{aligned} \|\kappa^{(j_1)}\|_1 = \sum_{d\in\mathbb{Z}} \left| g\left(d + \frac{j_1}{t_{1,\ell}}\right)\right| &\leq \frac{2}{\pi}\ln\alpha_\ell + \frac{5.25 - \ln\pi + E_1(1)}{\pi} \\ &\leq \frac{2}{\pi}\ln\alpha_\ell + \frac{5.25 - \ln\pi + \frac{7}{6e}}{\pi} < \frac{2}{\pi}\ln\alpha_\ell + 1.444 < \frac{2}{\pi}\ln\alpha_\ell + 1.62, \end{aligned} \tag{27}$$

where $E_1(1) = \int_1^\infty e^{-t}t^{-1}\,dt < 0.2194$.

2. Spatial tail bound:

$$\tau := \sum_{|d|>m_{1,\ell}} \left| g\left(d + \frac{j_1}{t_{1,\ell}}\right)\right| \leq \left(\frac{1}{\pi m_{1,\ell}} + \frac{1}{\pi p_{\ell+1}\ln 2}\right) 2^{-p_{\ell+1}} < 0.077 \cdot 2^{-p_{\ell+1}} < 0.10 \cdot 2^{-p_{\ell+1}}. \tag{28}$$

3. Periodic and folded filter bounds:

$$\begin{aligned} \|h^{(j_1),\mathrm{per}}\|_2 &\leq 1, & \|h^{(j_1),\mathrm{per}}\|_1 &\leq \sqrt{r_\ell}, \\ \|h^{(j_1),\mathrm{tr}} - h^{(j_1),\mathrm{per}}\|_1 &\leq \tau, & \|\bar{h}^{(j_1)} - h^{(j_1),\mathrm{tr}}\|_1 &\leq \tilde{W}_\ell 2^{-p_{\ell+1}-1}. \end{aligned} \tag{29}$$

6.2 Spreading by overlap-add and Kronecker substitution

To evaluate each row's length-r_ℓ cyclic convolution with its short filter $\bar{H}^{(j_1)}$ of length $\tilde{W}_\ell \leq r_\ell$, we combine classical overlap-add block convolution [Nus82] with Kronecker substitution for polynomial multiplication [Sch82, Har09] in a signed two's-complement radix representation.

Lemma 6.2 (Signed radix Kronecker packing and extraction)**.** *Let $s \geq 1$ and $B_{\mathrm{slot}} \geq 4$ be integers, and let $a_0, \ldots, a_{s-1} \in \mathbb{Z}$ satisfy $|a_j| < 2^{B_{\mathrm{slot}}-1}$. Define the signed radix-$2^{B_{\mathrm{slot}}}$ evaluation*

$$\mathrm{Pack}_{B_{\mathrm{slot}}}(a_0, \ldots, a_{s-1}) := \sum_{j=0}^{s-1} a_j\, 2^{jB_{\mathrm{slot}}} \in \mathbb{Z}. \tag{30}$$

(i) Given the signed binary representations of $a_0, \ldots, a_{s-1}$, the integer $\mathrm{Pack}_{B_{\mathrm{slot}}}(a)$ has bit-length at most $sB_{\mathrm{slot}} + 1$ and can be assembled in $O(sB_{\mathrm{slot}})$ bit operations on a multitape Turing machine via a single low-to-high streaming carry pass.

(ii) Conversely, if $c_0, \ldots, c_{m-1} \in \mathbb{Z}$ satisfy $|c_k| < 2^{B_{\mathrm{slot}}-1}$ for all k and $T_0 := \sum_{k=0}^{m-1} c_k 2^{kB_{\mathrm{slot}}}$, then the recurrence

$$r_k := T_k \bmod 2^{B_{\mathrm{slot}}} \in [0, 2^{B_{\mathrm{slot}}}), \quad c_k := \begin{cases} r_k, & r_k < 2^{B_{\mathrm{slot}}-1}, \\ r_k - 2^{B_{\mathrm{slot}}}, & r_k \geq 2^{B_{\mathrm{slot}}-1}, \end{cases} \quad T_{k+1} := \frac{T_k - c_k}{2^{B_{\mathrm{slot}}}},$$

uniquely recovers $c_0, \ldots, c_{m-1}$ in $O(mB_{\mathrm{slot}})$ bit operations.

Proof. For (i), write each $a_j \in (-2^{B_{\mathrm{slot}}-1}, 2^{B_{\mathrm{slot}}-1})$ with a borrow $\sigma_j \in \{-1, 0\}$ into the next slot: initialize $\sigma_0 := 0$, and for $j = 0, \ldots, s-1$, let $a'_j := a_j + \sigma_j$. If $a'_j \geq 0$, emit the B_{slot}-bit unsigned word $u_j := a'_j \in [0, 2^{B_{\mathrm{slot}}-1})$ and set $\sigma_{j+1} := 0$; if $a'_j < 0$, emit the B_{slot}-bit unsigned word $u_j := a'_j + 2^{B_{\mathrm{slot}}} \in$

$[2^{B_{\mathrm{slot}}-1}, 2^{B_{\mathrm{slot}}})$ and set $\sigma_{j+1} := -1$. Then $\sum_{j=0}^{s-1} a_j 2^{jB_{\mathrm{slot}}} = \sigma_s 2^{sB_{\mathrm{slot}}} + \sum_{j=0}^{s-1} u_j 2^{jB_{\mathrm{slot}}}$, where concatenating the B_{slot}-bit words u_j takes $O(sB_{\mathrm{slot}})$ bit operations. For (ii), since $T_k = c_k + 2^{B_{\mathrm{slot}}} T_{k+1}$, we have $c_k \equiv T_k \pmod{2^{B_{\mathrm{slot}}}}$, and because $[-2^{B_{\mathrm{slot}}-1}, 2^{B_{\mathrm{slot}}-1})$ contains a unique representative of each residue class modulo $2^{B_{\mathrm{slot}}}$, c_k and T_{k+1} are uniquely determined and extracted in $O(B_{\mathrm{slot}})$ bit operations per slot. □

Theorem 6.3 (Spreading by overlap-add and Kronecker substitution)**.** *At any non-leaf node ($R_\ell > 4p_\ell^2$, with $p_\ell \geq 8$) under the hypotheses of Lemma 6.1 with $p_{\mathrm{work},\ell} = p_\ell + K_\ell + 6$ and $p_{\ell+1} = p_\ell + 2K_\ell + 12$, let $x \in \widetilde{\mathbb{C}}_q^{R_\ell}$ satisfy $\|x\|_\infty \leq 2^{p_\ell}$, let $q := q_{\mathrm{child}} = p_{\ell+1} + 2$, $\tilde{W}_\ell := \min(2m_{1,\ell}+1, r_\ell) \leq r_\ell$, and*

$$B_{\mathrm{slot}} := 2q + p_\ell + \lceil \log_2(\tilde{W}_\ell + 1) \rceil + 4 = \Theta(p_{\ell+1}). \tag{31}$$

Computing the cyclic row convolutions for all $j_1 \in \{0, \ldots, t_{1,\ell} - 1\}$ with the guarded q-grid filters $\bar{H}^{(j_1)}$ via overlap-add of block length $\tilde{W}_\ell$, signed Kronecker substitution at slot width B_{slot}, exact integer overlap addition on $\widetilde{\mathbb{C}}_{2q}$, and a single final rounding per output entry to $\widetilde{\mathbb{C}}_q$ produces an array $\widehat{U} \in \widetilde{\mathbb{C}}_q^{t_{1,\ell} \times r_\ell}$ satisfying

$$\|\widehat{U} - U^\circ\|_\infty \leq (\tfrac{1}{2}\tilde{W}_\ell + 0.10) 2^{-p_{\ell+1}} \|x\|_\infty + \frac{\sqrt{2}}{8} 2^{-p_{\ell+1}} \leq 2r_\ell \, 2^{-p_{\ell+1}} \max(1, \|x\|_\infty) \tag{32}$$

in time

$$\mathcal{C}_{\mathrm{spread}}(R_\ell, p_{\ell+1}) = O\Big(\frac{R_\ell}{\tilde{W}_\ell} \, \mathsf{M}(\tilde{W}_\ell B_{\mathrm{slot}} + 1) + R_\ell p_{\ell+1} \log_2 t_{1,\ell} \Big) \tag{33}$$

bit operations, where $\mathsf{M}(N)$ denotes the bit complexity of multiplying two N-bit integers and $\tilde{W}_\ell B_{\mathrm{slot}} + 1 = \Theta(\tilde{W}_\ell p_{\ell+1})$ is the operand length of the block products in the proof.

Proof. First, by Lemma 3.3(i), de-interleaving $x \in \widetilde{\mathbb{C}}_q^{R_\ell}$ into the $t_{1,\ell}$ contiguous rows $v^{(j_1)} = (x(j_1 + t_{1,\ell} j_2))_{j_2=0}^{r_\ell - 1}$ takes $O(R_\ell p_{\ell+1} \log_2 t_{1,\ell})$ bit operations. At any non-leaf node, $R_\ell > 4p_\ell^2 \geq 256$ (since $p_\ell \geq 8$), so $K_\ell \geq 9$, $r_\ell = 2^{\lceil K_\ell/2 \rceil} \geq 32$, and $p_\ell < \frac{1}{2}\sqrt{R_\ell} \leq \frac{1}{2} r_\ell$. Under $p_{\mathrm{work},\ell} = p_\ell + K_\ell + 6$, since $K_\ell \leq 2 \log_2 r_\ell < r_\ell$ and $6 < \frac{1}{2} r_\ell$ for all $r_\ell \geq 32$, we have $p_{\mathrm{work},\ell} < 2r_\ell$, and therefore

$$\begin{aligned} \alpha_\ell = 2\sqrt{p_{\mathrm{work},\ell} \ln 2/\pi} < 2\sqrt{r_\ell} < r_\ell, \\ r - \frac{2}{\pi} \ln r \geq 32 - \frac{2}{\pi} \ln 32 > 29 > 1.62 \quad (r \geq 32), \\ \frac{2}{\pi} \ln \alpha_\ell + 2.62 < \frac{2}{\pi} \ln r_\ell + 2.62 \leq r_\ell + 1. \end{aligned}$$

Meanwhile, since $p_{\ell+1} \geq p_{\mathrm{work},\ell}$ and $\sqrt{p_{\mathrm{work},\ell} \ln 2/\pi} = \frac{1}{2}\alpha_\ell$, Lemma 6.1 gives $m_{1,\ell} \geq \max(6, \lceil \frac{1}{2}\alpha_\ell^2 \rceil)$, so $2m_{1,\ell} + 2 \geq \max(14, \alpha_\ell^2 + 2) > \frac{2}{\pi} \ln \alpha_\ell + 2.62$ for all $\alpha_\ell > 0$. Combining both branches yields $\frac{2}{\pi} \ln \alpha_\ell + 2.62 \leq \min(2m_{1,\ell} + 1, r_\ell) + 1 = \tilde{W}_\ell + 1$ across all non-leaves.

For each row j_1, partition $v^{(j_1)} \in \widetilde{\mathbb{C}}_q^{r_\ell}$ into $\lceil r_\ell / \tilde{W}_\ell \rceil \leq \frac{2r_\ell}{\tilde{W}_\ell}$ contiguous blocks of length $L_b \leq \tilde{W}_\ell$. On $\widetilde{\mathbb{C}}_q$, each scaled input coordinate $2^q \operatorname{Re}(v_{j_2}^{(j_1)}), 2^q \operatorname{Im}(v_{j_2}^{(j_1)}) \in \mathbb{Z}$ has absolute value $\leq 2^{q+p_\ell}$, and by Lemma 6.1(1),(3), the guarded filter satisfies

$$\|\bar{H}^{(j_1)}\|_1 = \|\bar{h}^{(j_1)}\|_1 \leq \|h^{(j_1),\mathrm{tr}}\|_1 + \frac{1}{2}\tilde{W}_\ell 2^{-p_{\ell+1}} \leq \|\kappa^{(j_1)}\|_1 + 1 < \frac{2}{\pi} \ln \alpha_\ell + 2.62 \leq \tilde{W}_\ell + 1,$$

so $\sum_{a=0}^{\tilde{W}_\ell - 1} |2^q \bar{H}_a^{(j_1)}| \leq (\tilde{W}_\ell + 1) 2^q$. By the ℓ_∞–ℓ_1 inequality, every integer linear-convolution coefficient between a block and $2^q \bar{H}^{(j_1)}$ has absolute value at most

$$2^{q+p_\ell} \cdot (\tilde{W}_\ell + 1) 2^q = (\tilde{W}_\ell + 1) 2^{2q+p_\ell} \leq 2^{B_{\mathrm{slot}}-4} < 2^{B_{\mathrm{slot}}-1}$$

unconditionally (whether $W_\ell \le r_\ell$ or $W_\ell > r_\ell$).

By Lemma 6.2, packing the real and imaginary parts of each block into $\mathrm{Pack}_{B_{\mathrm{slot}}}$, multiplying each by the pre-packed filter integer $\mathrm{Pack}_{B_{\mathrm{slot}}}(2^q \bar{H}^{(j_1)})$ (replayed from T_{tab} for each block, performing two integer multiplications of bit-length $\le \tilde{W}_\ell B_{\mathrm{slot}} + 1 = \Theta(\tilde{W}_\ell p_{\ell+1})$ per block), and extracting the $L_b + \tilde{W}_\ell - 1$ coefficients recovers the exact integer block convolutions at fractional precision $2q$. To accumulate the overlapping block outputs modulo r_ℓ on a multitape Turing machine, as each block b of row j_1 is unpacked into $L_b + \tilde{W}_\ell - 1 \ge \tilde{W}_\ell$ coefficients, the $(\tilde{W}_\ell - 1)$-entry carry-over tail from block $b-1$ (held on scratch tape T_1, initialized to 0 at $b = 0$) is added in lockstep to its first $\tilde{W}_\ell - 1$ coefficients. The first L_b resulting coefficients are appended sequentially to scratch tape T_2, while the remaining $\tilde{W}_\ell - 1$ coefficients overwrite T_1. After the final block of row j_1, the residual $(\tilde{W}_\ell - 1)$-entry tail on T_1 wraps cyclically modulo r_ℓ to row positions $0, \dots, \tilde{W}_\ell - 2$: rewinding T_2 to the start of row j_1 and T_1 to index 0, one pass over the first $\tilde{W}_\ell - 1$ entries adds this wrapped tail in $O(\tilde{W}_\ell B_{\mathrm{slot}}) = O(r_\ell p_{\ell+1})$ steps.

Finally, when $W_\ell \le r_\ell$, the reversed contiguous filter $\kappa^{(j_1)}_{m_{1,\ell}-a}$ $(0 \le a < W_\ell)$ offsets the cyclic convolution indices by $o = m_{1,\ell} \in \{0, \dots, r_\ell - 1\}$ (whereas when $W_\ell > r_\ell$, the modulo-r_ℓ folded filter is already indexed by $a = (-e) \bmod r_\ell$ with offset $o = 0$). During the single final rounding pass from scale $2q$ to $\widetilde{\mathbb{C}}_q$, copying the first o entries of T_2 to T_1 and streaming the remaining $r_\ell - o$ entries of T_2 followed by the o entries of T_1 rotates row j_1 into natural index order $k_2 = 0, \dots, r_\ell - 1$ in $O(r_\ell w_\ell) = O(r_\ell p_{\ell+1})$ steps while incurring a single rounding error $\le \frac{\sqrt{2}}{2} 2^{-q} = \frac{\sqrt{2}}{8} 2^{-p_{\ell+1}}$ per entry. Combining this with Lemma 6.1(3) ($\|\bar{h}^{(j_1)} - h^{(j_1),\mathrm{per}}\|_1 \le (\frac{1}{2}\tilde{W}_\ell + 0.10) 2^{-p_{\ell+1}}$) yields (32). Across all $t_{1,\ell}$ rows, there are at most $2R_\ell/\tilde{W}_\ell$ blocks, proving (33). □

Remark 6.4 (Modulo-r_ℓ folding at small non-leaves). At any non-leaf $R_\ell > 4p_\ell^2$, we have $p_\ell < \frac{1}{2}\sqrt{R_\ell} \le \frac{1}{2} r_\ell$, so $W_\ell = 2m_{1,\ell} + 1 = \Theta(p_{\ell+1}) = O(r_\ell)$. For the default radius $m_{1,\ell} = \lceil \alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi} \rceil$ under schedule (34), $W_\ell < r_\ell$ at every non-leaf reached from the root parameters (for $n > 64$ one has $p_\ell \ge 2K_\ell + 14$ at every depth, whence $W_\ell \le 1.53\, p_\ell + 3 < 2p_\ell < r_\ell$), but $W_\ell > r_\ell$ does occur inside the hypotheses of Theorem 6.3 (for the larger admissible radii, up to $m_{1,\ell} = p_{\ell+1}$, and for parameter pairs not reached from any root) and outside them (for the lower leaf cutoff of the C program, Section 9): for example, the parameters $(p_\ell, K_\ell) = (15, 10)$, which satisfy $R_\ell = 1024 > 4p_\ell^2 = 900$ but are not reached from any root, give with $m_{1,\ell} = \lceil \alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi} \rceil = 17$ the values $r_\ell = 32$ and $W_\ell = 35 > r_\ell$. Folding modulo r_ℓ (and never modulo W_ℓ) preserves the exact cyclic convolution modulo r_ℓ while keeping the effective filter length $\tilde{W}_\ell = \min(W_\ell, r_\ell) \le r_\ell$.

6.3 Forward error and the additive schedule

Lemma 6.5 (Forward error and the additive schedule)**.** *At recursion depth ℓ, let $R_\ell = 2^{K_\ell}$ $(K_\ell \ge 2)$ and let the target accuracy be $p_\ell \ge \max(8, K_\ell)$ on an input $x \in \widetilde{\mathbb{C}}^{R_\ell}_{p_\ell+2}$ with $\|x\|_\infty \le 2^{p_\ell}$. Define the additive precision schedule*

$$p_{\mathrm{work},\ell} := p_\ell + K_\ell + 6, \qquad p_{\ell+1} := p_{\mathrm{work},\ell} + K_\ell + 6 = p_\ell + 2K_\ell + 12, \qquad q_{\mathrm{child}} := p_{\ell+1} + 2, \quad (34)$$

and choose any integer window radius $m_{1,\ell} \in [\lceil \alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi} \rceil, p_{\ell+1}]$ (for instance $m_{1,\ell} := \lceil \alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi} \rceil$ or $m_{1,\ell} := p_{\ell+1}$). Assume inductively that every recursive child call at length r_ℓ and precision $p_{\ell+1}$ satisfies $\|\widehat{C}_{v_1} - F_{r_\ell}(\widehat{V}_{v_1,\cdot})\|_\infty \le 2^{-p_{\ell+1}} \max(1, \|\widehat{V}_{v_1,\cdot}\|_\infty)$. Then:

1. ***Forward error progression at a non-leaf:*** *Assume $R_\ell > 4p_\ell^2$ (non-leaf), and let $\mathcal{A} := \max(1, \|x\|_\infty)$ and $u := 2^{-p_{\ell+1}}$ (following the three-step decomposition of `FastDFT` in Algorithm 1 and Listing 1: Step 1 row spreading, Step 2 polynomial-transform 2D DFT via Steps 2a–2d, and Step 3 column deconvolution). Let $Z := F_{R_\ell} x$ be the exact 1D DFT, Z^* be the exact solution of*

the retained two-term system with ideal right-hand side $Y^\circ = F_{t_{1,\ell},r_\ell} U^\circ$, $Z^\dagger$ be the exact solution of the retained two-term system with the routed child output $\widehat{Y}$, and Z^{comp} be the output of `CyclicSolve/NyquistSolve`*. Then:*

$$\begin{aligned}\|Z^{\mathrm{comp}} - F_{R_\ell} x\|_\infty &\le \underbrace{\frac{1}{2} t_{1,\ell}\, 2^{-p_{\mathrm{work},\ell}} \mathcal{A}}_{\|Z^* - Z\|_\infty \text{ (aliases)}} + \underbrace{27 R_\ell\, u\mathcal{A}}_{\|Z^\dagger - Z^*\|_\infty \text{ (Steps 1–2)}} + \underbrace{160 R_\ell\, u\mathcal{A}}_{\|Z^{\mathrm{comp}} - Z^\dagger\|_\infty \text{ (Step 3)}} \\ &\le \frac{1}{2} t_{1,\ell}\, 2^{-p_{\mathrm{work},\ell}} \mathcal{A} + 200 R_\ell\, u\mathcal{A}.\end{aligned} \tag{35}$$

2. ***Inductive closure at non-leaves and leaves:*** *After quantizing Z^{comp} to $\widetilde{\mathbb{C}}_{p_\ell+2}$ and transposing to natural index order when $R_\ell > 4p_\ell^2$ (or after executing* `BaseRadix2FFT` *at internal precision $q_b := p_\ell + K_\ell + 4$, which* `FastDFT` *does when $R_\ell \le 4p_\ell^2$; the* `BaseRadix2FFT` *bound below holds for every length $R_\ell = 2^{K_\ell}$, $K_\ell \ge 2$, and is also used at the root length by the bottom stage of Section 7.2), the returned vector $\widehat{Z} =$* `FastDFT`*(x, R_ℓ, p_ℓ) satisfies*

$$\|\widehat{Z} - F_{R_\ell} x\|_\infty \le 2^{-p_\ell} \max(1, \|x\|_\infty). \tag{36}$$

Proof. First verify that the interval $[\lceil \alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi} \rceil,\ p_{\ell+1}]$ is non-empty and satisfies the hypotheses $p_{\mathrm{work},\ell} \ge 16 \ge 8$ and $p_{\ell+1} \ge 24 \ge \max(p_{\mathrm{work},\ell}, 20)$ of Lemma 6.1. Since $\alpha_\ell = 2\sqrt{p_{\mathrm{work},\ell} \ln 2/\pi}$ and $\frac{2\ln 2}{\pi} < 0.442 < 1$, for all $p_\ell \ge 8$ and $K_\ell \ge 2$ we have

$$\alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi} = \frac{2\ln 2}{\pi} \sqrt{p_{\mathrm{work},\ell}\, p_{\ell+1}} < 0.442\, p_{\ell+1} < p_{\ell+1}.$$

Steps 1 and 2 of Algorithm 1 (with Step 2 realized by the five steps of Proposition 4.1): $Y^\circ \to \widehat{Y}$ at a non-leaf $R_\ell > 4p_\ell^2$. By Lemma 6.1(3), $\|U^\circ\|_\infty \le \max_{j_1} \|h^{(j_1),\mathrm{per}}\|_1 \|x\|_\infty \le \sqrt{r_\ell}\mathcal{A}$, and by Theorem 6.3 (applicable since $R_\ell > 4p_\ell^2$), $\|\widehat{U} - U^\circ\|_\infty \le 2 r_\ell u \mathcal{A}$. In Step 2a (pre-twist), by (7) the 1-ulp faithful table roots $\bar{\zeta}_{k_2} \in \widetilde{\mathbb{C}}_{q_{\mathrm{child}}}$ satisfy $|\bar{\zeta}_{k_2} - \zeta^{k_2}| \le \frac{\sqrt{2}}{4} u < 0.354u$ and $|\bar{\zeta}_{k_2}| \le 1 + 0.354u$. Since $r_\ell \ge 2$ implies $\frac{0.354}{\sqrt{r_\ell}} + \frac{0.177}{r_\ell} < 0.34$ and $\frac{\log_2 t_{1,\ell}}{r_\ell} \le \frac{1}{2}$, each entry $\widehat{P}_{j_1,k_2} = \mathrm{RM}_{q_{\mathrm{child}}}(\bar{\zeta}_{k_2}, \widehat{U}_{j_1,k_2})$ differs from $P^\circ_{j_1,k_2} = \zeta^{k_2} U^\circ_{j_1,k_2}$ by at most

$$|\bar{\zeta}_{k_2}|\, |\widehat{U}_{j_1,k_2} - U^\circ_{j_1,k_2}| + |\bar{\zeta}_{k_2} - \zeta^{k_2}|\, |U^\circ_{j_1,k_2}| + \frac{\sqrt{2}}{8} u \le (1+u)(2 r_\ell u\mathcal{A}) + 0.354\sqrt{r_\ell} u\mathcal{A} + 0.177 u\mathcal{A} \le 2.6 r_\ell u \mathcal{A},$$

and $\|\widehat{P}\|_\infty \le \sqrt{r_\ell}\mathcal{A} + 2.6 r_\ell u\mathcal{A} \le 2 r_\ell \mathcal{A}$. In Step 2b (polynomial transform $G_{t_{1,\ell}}$), since $G_{t_{1,\ell}}$ is an average over $t_{1,\ell}$ monomial shifts and has coefficient-wise ℓ_∞ norm ≤ 1, Lemma 3.5 gives $\|\widehat{Q} - Q^\circ\|_\infty \le \|\widehat{P} - P^\circ\|_\infty + 0.177(\log_2 t_{1,\ell}) u \le 3 r_\ell u \mathcal{A}$ and $\|Q^\circ\|_\infty \le \sqrt{r_\ell}\mathcal{A}$. In Step 2c (post-twist by $\overline{\bar{\zeta}_{k_2}}$), the same product bound and the norm chain give

$$\begin{aligned}\|\widehat{V} - V^\circ\|_\infty &\le (1+u)(3 r_\ell u\mathcal{A}) + 0.354\sqrt{r_\ell} u\mathcal{A} + 0.177 u\mathcal{A} \le 3.6 r_\ell u\mathcal{A} < 7 r_\ell u\mathcal{A}, \\ \|V^\circ\|_\infty &= \|Q^\circ\|_\infty \le \|P^\circ\|_\infty = \|U^\circ\|_\infty \le \sqrt{r_\ell}\mathcal{A} \le r_\ell \mathcal{A}, \\ \|\widehat{V}_{v_1,\cdot}\|_\infty &\le r_\ell \mathcal{A} + 7 r_\ell u\mathcal{A} \le 2 r_\ell \mathcal{A} \le 2^{p_\ell + K_\ell + 1} < 2^{p_\ell + 2K_\ell + 12} = 2^{p_{\ell+1}}.\end{aligned}$$

Thus each child call $\widehat{C}_{v_1} =$ `FastDFT`$(\widehat{V}_{v_1,\cdot}, r_\ell, p_{\ell+1})$ in Step 2d satisfies the child input precondition $\|\widehat{V}_{v_1,\cdot}\|_\infty \le 2^{p_{\ell+1}}$, and by the induction hypothesis and $\|F_{r_\ell}\|_{\infty\to\infty} \le 1$,

$$\|\widehat{C}_{v_1} - F_{r_\ell} V^\circ_{v_1,\cdot}\|_\infty \le \|\widehat{C}_{v_1} - F_{r_\ell} \widehat{V}_{v_1,\cdot}\|_\infty + \|\widehat{V}_{v_1,\cdot} - V^\circ_{v_1,\cdot}\|_\infty \le u(2 r_\ell \mathcal{A}) + 7 r_\ell u\mathcal{A} = 9 r_\ell u \mathcal{A}.$$

Because the odd-stride permutation (14) is an index bijection, the routed array $\widehat{Y}$ satisfies $\|\widehat{Y} - Y^\circ\|_\infty \le 9r_\ell u\mathcal{A}$, and

$$\|Y^\circ\|_\infty \le \sum_{m\in\mathbb{Z}} \hat{g}(u_2/r_\ell - m)\|F_{R_\ell}x\|_\infty \le \|x\|_\infty \le \mathcal{A},$$
$$\mathcal{M}_Y = \max(1, \|\widehat{Y}\|_\infty) \le \mathcal{A} + 9r_\ell u\mathcal{A} \le 2\mathcal{A}.$$

Step 3 error ($Z \to Z^* \to Z^\dagger \to Z^{\mathrm{comp}}$). By Theorem 4.2 and Theorem 5.1(1), for every column u_2, the exact DFT Z satisfies the retained two-term equations with right-hand side $Y^\circ - \Delta^{\mathrm{alias}}$, where $\|\Delta^{\mathrm{alias}}\|_\infty \le \Psi(0)\|F_{R_\ell}x\|_\infty < 0.15 \cdot 2^{-p_{\mathrm{work},\ell}}\mathcal{A}$ (and at $u_2 = r_\ell/2$ the alternating sum A_{N} in (25) is exact on $\widetilde{\mathbb{C}}_{q_{\mathrm{child}}}$ because $q_{\mathrm{child}} = p_\ell + 2K_\ell + 14 > p_\ell + 2 + \log_2 r_\ell$). Applying the inverse norm bound $< 2.36\, t_{1,\ell} < 3t_{1,\ell}$ from Theorem 5.1(2)–(3) gives

$$\|Z^* - Z\|_\infty \le 2.36\, t_{1,\ell} \cdot 0.15 \cdot 2^{-p_{\mathrm{work},\ell}}\mathcal{A} < \frac{1}{2} t_{1,\ell}\, 2^{-p_{\mathrm{work},\ell}}\mathcal{A},$$

and perturbing the right-hand side from Y° to $\widehat{Y}$ gives

$$\|Z^\dagger - Z^*\|_\infty \le 3t_{1,\ell}\|\widehat{Y} - Y^\circ\|_\infty \le 3t_{1,\ell}(9r_\ell u\mathcal{A}) = 27R_\ell u\mathcal{A}.$$

Furthermore,

$$p_{\ell+1} - \lfloor K_\ell/2 \rfloor \ge 8 + \frac{3}{2}K_\ell + 12 \ge 23 > 20, \qquad t_{1,\ell}u \le 2^{-23} < 2^{-20},$$

so Lemma 5.2 and $\mathcal{M}_Y \le 2\mathcal{A}$ yield $\|Z^{\mathrm{comp}} - Z^\dagger\|_\infty \le 80t_{1,\ell}^2 u(2\mathcal{A}) \le 160R_\ell u\mathcal{A}$. Summing the three bounds proves (35).

Inductive closure. Under (34), $p_{\mathrm{work},\ell} - p_\ell = K_\ell + 6$ and $p_{\ell+1} - p_\ell = 2K_\ell + 12$, with $t_{1,\ell} = 2^{\lfloor K_\ell/2 \rfloor} \le 2^{K_\ell}$ and $R_\ell = 2^{K_\ell}$ ($K_\ell \ge 2$). Thus

$$\frac{1}{2} t_{1,\ell}\, 2^{-(p_{\mathrm{work},\ell} - p_\ell)} \le \frac{1}{2} 2^{K_\ell} \cdot 2^{-K_\ell - 6} = 2^{-7},$$
$$200R_\ell\, 2^{-(p_{\ell+1} - p_\ell)} = 200 \cdot 2^{-K_\ell - 12} \le 200 \cdot 2^{-14} = \frac{25}{2048} < 2^{-6}.$$

Final nearest-grid rounding $\widehat{Z} = \mathcal{Q}_{p_\ell+2}(Z^{\mathrm{comp}})$ adds at most $\frac{\sqrt{2}}{8} 2^{-p_\ell} < 0.177 \cdot 2^{-p_\ell}$, so

$$\|\widehat{Z} - F_{R_\ell}x\|_\infty \le \left(\frac{\sqrt{2}}{8} + 2^{-7} + \frac{25}{2048}\right) 2^{-p_\ell}\mathcal{A} < 0.197 \cdot 2^{-p_\ell}\mathcal{A} < \frac{1}{4} 2^{-p_\ell}\mathcal{A} < 2^{-p_\ell}\mathcal{A}.$$

For `BaseRadix2FFT` at any length $R_\ell = 2^{K_\ell}$, $K_\ell \ge 2$ (in particular at a leaf $R_\ell \le 4p_\ell^2$), `BaseRadix2FFT` bit-reverses its $R_\ell = 2^{K_\ell}$ input records and executes $K_\ell = \log_2 R_\ell$ decimation-in-time radix-2 Cooley–Tukey butterfly stages on $\widetilde{\mathbb{C}}_{q_b}$ with $q_b \ge p_\ell + K_\ell + 4$. In stage $s \in \{1, \dots, K_\ell\}$, each block of 2^s records is split into its upper and lower halves $(A_j)_{j=0}^{2^{s-1}-1}$ and $(B_j)_{j=0}^{2^{s-1}-1}$ on T_1, T_2 and paired in lockstep with the stage-s 1-ulp faithful twiddle list $(\bar{\omega}_{s,j})_{j=0}^{2^{s-1}-1}$ replayed from T_{tab} ($|\bar{\omega}_{s,j} - \omega_{s,j}| \le \sqrt{2} \cdot 2^{-q_b}$). Each pair is then updated via $W_j := \mathrm{RM}_{q_b}(\bar{\omega}_{s,j}, B_j)$ and $A_j^{\mathrm{new}}, B_j^{\mathrm{new}} := \mathcal{Q}_{q_b}(\frac{1}{2}(A_j \pm W_j))$. Inductively $\|A\|_\infty, \|B\|_\infty \le 1.01\mathcal{A}$, and each stage incurs twiddle approximation error $\le \sqrt{2} \cdot 2^{-q_b}(1.01\mathcal{A})$ plus product and halving/rounding errors $\le \sqrt{2} \cdot 2^{-q_b}\mathcal{A}$, so $e_s \le (1 + 2^{-q_b})e_{s-1} + 2.85 \cdot 2^{-q_b}\mathcal{A}$. Since $K_\ell 2^{-K_\ell} \le \frac{1}{2}$ for all $K_\ell \ge 1$, across K_ℓ stages we obtain

$$e_{K_\ell} \le 2.86K_\ell 2^{-q_b}\mathcal{A} \le 2.86K_\ell 2^{-K_\ell - 4} 2^{-p_\ell}\mathcal{A} \le 1.43 \cdot 2^{-4} 2^{-p_\ell}\mathcal{A} < 0.090 \cdot 2^{-p_\ell}\mathcal{A}$$

before final rounding to $\widetilde{\mathbb{C}}_{p_\ell+2}$ ($\le 0.177 \cdot 2^{-p_\ell}\mathcal{A}$), giving $< 0.267 \cdot 2^{-p_\ell}\mathcal{A} < 2^{-p_\ell}\mathcal{A}$. □

Remark 6.6 (Shared precision schedule). Because the additive increments $p_{\mathrm{work},\ell} - p_\ell = K_\ell + 6$ and $p_{\ell+1} - p_\ell = 2K_\ell + 12$ in (34) bound the worst-case ℓ_∞ alias amplification ($\le 2^{-7} \cdot 2^{-p_\ell}\mathcal{A}$) and the worst-case ℓ_∞ rounding and child-DFT amplification ($\le 200 \cdot 2^{-K_\ell - 12} \cdot 2^{-p_\ell}\mathcal{A} < 2^{-6} \cdot 2^{-p_\ell}\mathcal{A}$) for all $K_\ell \ge 2$ with more than a factor-of-five total margin ($< 0.197 \cdot 2^{-p_\ell}\mathcal{A}$), the exact same precision schedule (34) and root precision $p_0 = 2b + 2K_0 + 6$ (with window radius $m_{1,\ell} \in [\lceil \alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi} \rceil, p_{\ell+1}]$) are used across our written proof, our Lean 4 formalization [LeanMS26] (Section 8), and our C/GMP implementation [CGMP26] (Section 9). As noted in Section 9, the leaf threshold $R_\ell \le 4p_\ell^2$ of the written proof and of [LeanMS26] (`isBaseLeaf`, Table 3) is lowered by default to `bt` $= p_0^2$ in [CGMP26], and, as noted in Section 8, the tie-breaking rule on half-integer midpoints differs slightly across implementations.

7 The algorithm and the $O(n \log n)$ bound

7.1 Algorithm specification

Listing 1 gives the top-level specification of `Multiply` and `FastDFT`; the corresponding Turing-machine subroutines are defined in Section 3 (shared tables and breadth-first tape sweeps), Section 4 (polynomial-transform reduction and fractional-shift bridge), Section 5 (`CyclicSolve` and `NyquistSolve`), and Section 6 (overlap-add Kronecker spreading and the additive schedule (34)). Inputs of bit-length $2 \le n \le 64$ are multiplied directly by schoolbook multiplication in $O(1)$ Turing machine steps.

Listing 1: Two-call normalized `Multiply` and recursive `FastDFT` specification.

```
def Multiply(A: int, B: int, n: int) -> int:
    """
    Multiply two n-bit non-negative integers A, B in O(n log n) steps.
    Uses two-call complex forward packing and root precision p0 = 2b + 2K0 + 6.
    """
    if n <= 64:
        return A * B
    b = max(4, n.bit_length() - 1)                # Chunk size b = floor(log_2 n)
    N = (n + b - 1) // b                          # Number of chunks
    K0 = (2 * N - 1).bit_length()
    R0 = 1 << K0                                  # Smallest power of 2 >= 2N
    p0 = 2 * b + 2 * K0 + 6                       # Root precision p0 = Theta(log n)

    # 1. Normalize chunks by 2^{-b} into [0, 1) and pack A + i*B into one complex vector
    x_AB = PackNormalizedComplex(A, B, b, R0, p0)

    # 2. Build shared per-level tables once, then make two top-level calls to FastDFT
    tables = BuildSharedLevelTables(R0, p0)
    y_AB = FastDFT(x_AB, R0, p0, tables)
    y_sq_conj = [FixedPointSquare(y_AB[u], p0 + 2).conjugate() for u in range(R0)]
    z_out = FastDFT(y_sq_conj, R0, p0, tables)

    # 3. Scale imaginary part by -2^{2b - 1} * R0^2, round to nearest integers, propagate carries
    c_chunks = [RoundToNearestInt(-(1 << (2 * b - 1)) * (R0 * R0) * z.imag, p0 + 2) for z in z_out]
    return PropagateCarries(c_chunks, b)


def FastDFT(x: list, R: int, p_bits: int, tables: dict) -> list:
    """
    Recursive 1D power-of-two DFT F_R(x) returning accuracy 2^{-p_bits} * max(1, ||x||_inf).
    Uses windowed sinc-Gaussian spreading and the additive precision schedule.
    """
    if R <= 4 * p_bits * p_bits:
        return BaseRadix2FFT(x, R, p_bits, tables)

    K = R.bit_length() - 1
```

```
    t1 = 1 << (K // 2)                          # t1 = 2^{floor(K/2)}
    r  = 1 << ((K + 1) // 2)                    # r  = 2^{ceil(K/2)}, t1 | r, R = t1 * r

    # Additive precision schedule: p_child = p_bits + 2*K + 12
    p_work  = p_bits + K + 6
    p_child = p_work + K + 6
    q_child = p_child + 2

    # Save exact Nyquist spatial alternating sum A_N before modifying input layout
    A_N = ExactNyquistAlternatingSum(x, t1, r, q_child)

    # Step 1 (row spreading): sinc-Gaussian fractional shift via overlap-add and signed Kronecker packing
    x_tilde = OverlapAddSignedKroneckerRowSpread(x, t1, r, p_work, p_child, tables)

    # Step 2 (polynomial-transform 2D DFT over C[y]/(y^r + 1), Steps 2a--2d):
    # Step 2a (pre-twist)
    U_pre  = PreTwist(x_tilde, t1, r, q_child, tables)
    # Step 2b (polynomial transform)
    Q      = PolyTransform_Radix2_Mod_YrPlus1(U_pre, t1, r, q_child)
    # Step 2c (post-twist)
    Q_post = PostTwist(Q, t1, r, q_child, tables)
    # Step 2d (recursive DFTs and odd-stride permutation)
    C_child = [FastDFT(Q_post[v1], r, p_child, tables) for v1 in range(t1)]
    Y = OddStrideColumnRoute(C_child, t1, r)

    # Step 3 (column deconvolution): two-term cyclic and Nyquist solvers + output transpose
    Z_cols = SolveTwoTermBidiagonalOutput(Y, A_N, t1, r, p_work, p_child, p_bits, tables)
    return TransposeColumnMajorToRowMajor(Z_cols, t1, r)
```

7.2 Cost of the algorithm

We now restate and prove Theorem 1, establishing the explicit two-term complexity bound

$$\mathcal{C}_{\mathrm{mul}}(n) = O(n \log n) + O\big(n(\log\log n)^2\big) = O(n \log n) \tag{37}$$

via a three-stage non-circular bootstrap on a constant-tape deterministic Turing machine.

Theorem 1 (restated). *There is a deterministic multitape Turing machine and a constant $C > 0$ such that, for every $n \geq 2$ and any two n-bit integers A and B, given the binary representations of A and B on the input tape, the machine writes the $2n$-bit binary product on the output tape in at most $Cn \log_2 n$ steps.*

Its structural properties are listed after Theorem 1 in Section 1.2.

Proof. For $2 \leq n \leq 64$, schoolbook multiplication computes $A \cdot B$ in $O(1)$ Turing machine steps; assume henceforth $n > 64$ (so $b \geq 6$, $K_0 \geq 5$, and $p_0 \geq 28$).

1. Additive precision schedule and exact root integer recovery. At the root ($\ell = 0$), $b = \lfloor \log_2 n \rfloor = \Theta(\log n)$, $N = \lceil n/b \rceil$, $R_0 = 2^{K_0} = \Theta(n/\log n)$ with $2N \leq R_0 < 4N$, and $p_0 = 2b + 2K_0 + 6 = \Theta(\log n)$. At recursion depth ℓ, $K_{\ell+1} = \lceil K_\ell/2 \rceil$, so $K_\ell = \lceil K_0/2^\ell \rceil \leq K_0/2^\ell + 1$ and the recursion reaches a leaf $R_L \leq 4p_L^2$ at depth $L = O(\log K_0) = O(\log\log n)$. Summing the additive increments (34) gives

$$p_\ell = p_0 + \sum_{j=0}^{\ell-1} (2K_j + 12) \leq p_0 + 4K_0 + 14\ell = \Theta(\log n), \qquad R_0\, p_\ell = \Theta(n),$$

uniformly across all levels $\ell \in \{0, \ldots, L\}$.

At the root, the normalized chunks $\tilde{a}_j = a_j 2^{-b}, \tilde{b}_j = b_j 2^{-b} \in [0,1)$ ($0 \le j < N$, padded with zeros for $N \le j < R_0$) are represented exactly on $\widetilde{\mathbb{C}}_{p_0+2}$ since $b < p_0 + 2$. Their packed vector $\tilde{x}_{AB} = \tilde{a} + i\tilde{b}$ satisfies $\|\tilde{x}_{AB}\|_\infty < \sqrt{2}$, and because at least half of the $R_0 \ge 2N$ entries are zero,

$$\|F_{R_0}\tilde{x}_{AB}\|_\infty \le \frac{1}{R_0}\sum_{j=0}^{N-1}|\tilde{x}_{AB,j}| < \frac{N}{R_0}\sqrt{2} \le \frac{1}{\sqrt{2}}.$$

Let $\epsilon := 2^{-p_0}$. By Lemma 6.5, the first call $\hat{y}_{AB} = \texttt{FastDFT}(\tilde{x}_{AB}, R_0, p_0)$ satisfies $\|\hat{y}_{AB} - F_{R_0}\tilde{x}_{AB}\|_\infty \le \sqrt{2}\epsilon$. Pointwise squaring and conjugating on $\widetilde{\mathbb{C}}_{p_0+2}$, $\widehat{w}_u := \overline{\mathrm{RM}_{p_0+2}(\hat{y}_{AB,u}, \hat{y}_{AB,u})}$, incurs error

$$|\widehat{w}_u - \overline{(F_{R_0}\tilde{x}_{AB})_u^2}| \le 2\|F_{R_0}\tilde{x}_{AB}\|_\infty(\sqrt{2}\epsilon) + (\sqrt{2}\epsilon)^2 + \frac{\sqrt{2}}{8}\epsilon < \left(2 + 2\epsilon + \frac{\sqrt{2}}{8}\right)\epsilon < 2.19\epsilon,$$

and $\|\widehat{w}\|_\infty < (1/\sqrt{2})^2 + 2.19\epsilon < 1$, so $\max(1, \|\widehat{w}\|_\infty) = 1$. The second call $\hat{z}_{\mathrm{out}} = \texttt{FastDFT}(\widehat{w}, R_0, p_0)$ therefore adds at most $1 \cdot \epsilon$, giving

$$\left\|\hat{z}_{\mathrm{out}} - F_{R_0}\overline{((F_{R_0}\tilde{x}_{AB})^2)}\right\|_\infty < 3.19\epsilon.$$

By Lemma 3.4, the exact integer chunk convolution $c = a \circledast b$ equals $-2^{2b-1}R_0^2\,\mathrm{Im}(F_{R_0}(\overline{(F_{R_0}\tilde{x}_{AB})^2}))$. Multiplying $-\mathrm{Im}(\hat{z}_{\mathrm{out}})$ by $2^{2b-1}R_0^2 = 2^{2b+2K_0-1}$ produces each integer convolution coefficient c_j with pre-rounding error strictly less than

$$3.19 \cdot 2^{2b+2K_0-1} \cdot 2^{-p_0} = 3.19 \cdot 2^{-7} = \frac{3.19}{128} = 0.024921875 < \frac{1}{2},$$

guaranteeing exact recovery of every integer $c_j \in \mathbb{Z}$ upon rounding to the nearest integer, after which linear carry propagation in radix 2^b outputs $A \cdot B$ in $O(n)$ bit operations.

2. $(1/2)^\ell$ geometric contraction of polynomial-transform and tape-routing work ($O(n \log n)$). At depth $\ell \in \{0, \ldots, L-1\}$, there are $\frac{R_0}{R_\ell}$ subproblems of length $R_\ell = t_{1,\ell} r_\ell$, so the total number of active complex entries at depth ℓ is $\frac{R_0}{R_\ell} \cdot R_\ell = R_0$ (no grid expansion). By (5), every scalar record at depth ℓ has $O(p_{\ell+1} + H_\ell) = O(p_{\ell+1}) = O(\log n)$ bits. By Lemma 3.3 and Lemma 3.5, the $\log_2 t_{1,\ell}$-stage radix-2 polynomial transform $G_{t_{1,\ell}}$, the multitape input row de-interleaving, the odd-stride merge-sort column routing, and the final column-to-row output transpose all take $O(R_\ell p_{\ell+1}(1 + \log_2 t_{1,\ell}))$ bit operations per node. Since $\log_2 t_{1,\ell} = \lfloor K_\ell/2 \rfloor \le K_0/2^{\ell+1} + 1$ and $R_0 p_{\ell+1} = \Theta(n)$ uniformly across all ℓ, summing over all $\frac{R_0}{R_\ell}$ nodes at depth ℓ yields the geometric work envelope

$$\mathcal{C}_{\mathrm{syn+route}}(\ell) = \frac{R_0}{R_\ell} \cdot O(R_\ell p_{\ell+1}(1 + \log_2 t_{1,\ell})) = O\left(n\left(\frac{K_0}{2^{\ell+1}} + 1\right)\right).$$

Using the telescoping sum

$$\sum_{\ell=0}^{L-1} \log_2 t_{1,\ell} = K_0 - K_L \le K_0 = \log_2 R_0 = O(\log n),$$

summing over $\ell = 0, \ldots, L-1$ gives

$$\mathcal{C}_{\mathrm{syn+route}} = \sum_{\ell=0}^{L-1} \mathcal{C}_{\mathrm{syn+route}}(\ell) = O(nK_0 + nL) = O(n \log n). \tag{38}$$

3. $O(n(\log\log n)^2)$ **secondary work and constant-tape discipline via three-stage bootstrap.** By (8) and Appendix A.2, shared per-level table generation for the outer call takes $o(n)$ Turing machine steps and sequential level-table replay (including per-stage leaf twiddle list replay) takes $O(n\log\log n)$ steps. Every non-table integer multiplication inside the root pointwise squaring, Steps 1, 2a, 2c, 3, and the leaf `BaseRadix2FFT` at depth ℓ has operand bit-length at most

$$N_\ell = O(\tilde{W}_\ell p_{\ell+1}) = O(p_L^2) = O((\log n)^2) \ll n.$$

To establish (37) without circularity in the asymptotic constant and with a fixed constant number of tapes, we define a three-stage hierarchy of multiplier procedures ($\texttt{Multiply}_2 \to \texttt{Multiply}_1 \to$ `BasicMul`), allocating a separate constant bank of work tapes to each stage ($3 \times O(1) = O(1)$ tapes in total). Whenever an outer sweep invokes an inner N-bit multiplier, it copies the two N-bit operands onto the callee stage's input tape, runs the callee on its dedicated tape bank, copies the $2N$-bit product back, and erases and rewinds the callee's tapes in $O(\text{space}) \le O(\mathcal{C}_{\text{callee}}(N))$ steps:

- *Bottom stage (`BasicMul`):* Multiplies two N-bit integers using the two-call root reduction directly to `BaseRadix2FFT` of length $R_0 = \Theta(N/\log(2N))$ at precision $q_b = \Theta(\log(2N))$ (its accuracy is the `BaseRadix2FFT` clause of Lemma 6.5(2), valid for every length 2^K with $K \ge 2$; for $N \le 64$, `BasicMul` multiplies by schoolbook as in Listing 1). All butterfly twiddle products, root pointwise squarings, and root table evaluations are performed by schoolbook multiplication and division. By Lemma 3.2, evaluating the $R_0 - 1 < R_0$ stage twiddles at $D = \Theta(\log(2N))$ in $O(D^3)$ steps each costs $O\big(R_0(\log(2N))^3\big) = O\big(N(\log(2N))^2\big)$ steps. By Appendix A.2, executing $\log_2 R_0$ breadth-first `BaseRadix2FFT` stages of $R_0/2$ schoolbook $O(q_b)$-bit butterfly products and $O(R_0 q_b)$-bit scans costs $O\big(R_0(\log R_0)q_b^2\big) = O\big(N(\log(2N))^2\big)$ steps, so its total step count satisfies

$$M_0(N) = O\big(N(\log(2N))^2\big).$$

- *Middle stage ($\texttt{Multiply}_1$):* Runs the recursive algorithm of Listing 1 on its dedicated tape bank, invoking `BasicMul` (M_0) for all inner integer products at the root, in Steps 1–3, and at the leaves. By (8), building $\texttt{Multiply}_1$'s internal shared tables for an N-bit input takes $O\big(\sqrt{N}(\log(2N))^{7/2}\log\log(4N)\big) = o(N)$ steps, while root chunk packing and final scaling, rounding, and carry propagation take $O(N)$ steps. Moreover, the $R_0(N) = \Theta(N/\log(2N))$ root pointwise squarings and conjugations via M_0 take $R_0(N)\cdot O(M_0(p_0(N))) = O\big(N(\log\log(4N))^2\big)$ steps. At each level $\ell < L(N)$ of an N-bit call to $\texttt{Multiply}_1$, the inner products have bit-length $O((\log(2N))^2)$, so $M_0(O(\tilde{W}_\ell p_{\ell+1})) = O(\tilde{W}_\ell p_{\ell+1}(\log\log(4N))^2)$. By Theorem 6.3 and (38), summing over $L(N) = O(\log\log(4N))$ levels plus the root and leaf levels gives

$$\mathcal{C}_1(N) = O(N\log(2N)) + O\big(N(\log\log(4N))^3\big) = O\big(N\log(2N)\big).$$

- *Top stage ($\texttt{Multiply} := \texttt{Multiply}_2$):* Runs the recursive algorithm on the n-bit input (A, B), invoking the already-proved $\texttt{Multiply}_1$ ($\mathcal{C}_1(N) = O(N\log(2N))$, which includes $\texttt{Multiply}_1$'s internal table-building and tape-reset work) for all inner integer products:

  (a) *Root packing, pointwise squaring, and carry propagation:* `PackNormalizedComplex` and `PropagateCarries` take $O(R_0 p_0 + n) = O(n)$ steps, and the $R_0 = \Theta(n/\log n)$ root pointwise squarings and conjugations on $\widetilde{\mathbb{C}}_{p_0+2}$ via $\texttt{Multiply}_1$ cost $R_0 \cdot O(\mathcal{C}_1(p_0)) = O(R_0 p_0 \log p_0) = O(n\log\log n)$ steps.

  (b) *Per-level multiplications ($\ell = 0, \ldots, L-1$):* At level ℓ, the $O(R_\ell p_{\ell+1}\log_2 t_{1,\ell})$ row de-interleaving scan is already counted in $\mathcal{C}_{\text{syn+route}}$ in Part 2. By Theorem 6.3 with $\mathsf{M} = \mathcal{C}_1$,

the packing, Kronecker multiplication, and overlap-add work in Step 1 across all $\frac{R_0}{R_\ell}$ nodes costs

$$\frac{R_0}{R_\ell} \cdot O\Big(\frac{R_\ell}{\tilde{W}_\ell} \cdot \tilde{W}_\ell p_{\ell+1} \log(\tilde{W}_\ell p_{\ell+1})\Big) = O(R_0 p_{\ell+1} \log p_{\ell+1}) = O(n \log\log n)$$

steps. Likewise, the $2R_\ell$ pre/post-twists and the $\le 3R_\ell$ real-by-complex solver products per node each multiply $O(p_{\ell+1})$-bit integers via $\texttt{Multiply}_1$ in $O(p_{\ell+1} \log p_{\ell+1})$ steps, costing $O(R_0 p_{\ell+1} \log p_{\ell+1}) = O(n \log\log n)$ per level. Summing over all $L = O(\log\log n)$ levels gives $\mathcal{C}_{\text{ovh}} = O\big(n(\log\log n)^2\big)$.

(c) *Leaf base-case FFTs ($\ell = L$, $R_L \le 4p_L^2$):* At depth L, we have

$$1 + K_L = 1 + \log_2 R_L = O(\log p_L) = O(\log\log n), \qquad q_b = \Theta(\log n).$$

Executing `BaseRadix2FFT` across all $\frac{R_0}{R_L}$ leaves costs

$$\mathcal{C}_{\text{base}} = \frac{R_0}{R_L} \cdot O\big(R_L(1 + K_L)\, \mathcal{C}_1(q_b)\big) = O\big(R_0(\log\log n)\, p_L \log p_L\big) = O\big(n(\log\log n)^2\big).$$

Combining $\mathcal{C}_{\text{syn+route}} + \mathcal{C}_{\text{ovh}} + \mathcal{C}_{\text{base}} + \mathcal{C}_{\text{tables}}$ proves (37) and Theorem 1. □

8 Formal verification in Lean 4

Scope and structure. We formalize Theorem 2 in the library `IntMul` (`IntMul.lean` and 26 modules under `IntMul/`), comprising 27 files and 82,377 lines of Lean 4 [dMU21] on Mathlib [Mat20]. The headline theorem `IntMul.tmIntMul_evalsToInTime_n_log_n` (`IntMul/TMEndToEnd.lean`) is proved for one concrete, parameter-free multi-stack machine `tmIntMul` in Mathlib's `Turing.FinTM2` model. We quote the bit-encoding definitions from `IntMul/Defs.lean` and the machine and complexity declarations from `IntMul/TMEndToEnd.lean` verbatim (definitions with their bodies, theorems as statements only; docstrings, proofs, and the body of `eightStacksEquivFourBiInfiniteTapes` are omitted):

```
def encodeNatBits (w : ℕ) (x : ℕ) : List Bool :=
  List.ofFn (fun i : Fin w => x.testBit i.val)

def encodeInputBits (n A B : ℕ) : List Bool :=
  encodeNatBits n A ++ encodeNatBits n B

def tmIntMul : Turing.FinTM2 :=
  (bmRunRaw bmIntMulCore).toFinTM2

theorem isUniformFinTM2_tmIntMul :
    IsUniformFinTM2 tmIntMul (bmRunRaw bmIntMulCore).numLabels 32 2

theorem tmIntMul_evalsToInTime_n_log_n :
    ∃ C > 0, ∀ n : ℕ, 2 ≤ n → ∀ A B : ℕ, A < 2 ^ n → B < 2 ^ n →
      ∃ t ≤ C * n * Nat.log 2 n,
        Turing.EvalsToInTime tmIntMul.step
          (Turing.initList tmIntMul (encodeInputBits n A B))
          (some (Turing.haltList tmIntMul (encodeNatBits (2 * n) (A * B)))) t
```

Here `encodeNatBits w x` (equivalently `(List.range w).map (x.testBit ·)`) is the fixed-width w-bit little-endian binary list of x, `encodeInputBits n A B` concatenates the two n-bit inputs on stack 0 with all other stacks initially empty (`Turing.initList`), and `Turing.haltList tmIntMul s` is the halting configuration (label `none`, register reset to `initialState`, `s` on the output stack, and all other stacks empty). The witness for C in the proof is the closed arithmetic expression `C_tm_total` (`C_tm_d2 + 500`, where `C_tm_d2` multiplies the tier-1 constant `C_tm_d1` $\approx 3.1 \cdot 10^{22}$ by $1.5 \cdot 10^{13}$ and by twice the counted-operation constant `C_nested_total` $\approx 5.4 \cdot 10^{56}$); it evaluates to about $5.0 \cdot 10^{92}$. No attempt was made to optimize it. Mathlib's `Turing.EvalsToInTime f a b m` is a `Type`-valued structure (a run from `a` reaching `b` together with its step count, at most `m`); we quote verbatim the `CoeDep` instance and uniformity predicate from `IntMul/TMPrimitives.lean` together with the elementary-operation counter and predicate from `IntMul.lean`:

```
instance instCoeDepEvalsToInTimeProp {σ : Type u} (f : σ → Option σ) (a : σ) (b : Option σ)
    (m : ℕ) : CoeDep Type (Turing.EvalsToInTime f a b m) Prop where
  coe := Nonempty (Turing.EvalsToInTime f a b m)

def IsUniformFinTM2 (tm : Turing.FinTM2) (maxLabels maxStates maxAlpha : ℕ) : Prop :=
  Fintype.card tm.Λ ≤ maxLabels ∧
  Fintype.card tm.σ ≤ maxStates ∧
  Fintype.card (tm.Γ tm.k₀) ≤ maxAlpha ∧
  Fintype.card tm.K ≤ 8 ∧
  ∃ inst : ∀ k : tm.K, Fintype (tm.Γ k),
    ∀ k : tm.K, @Fintype.card (tm.Γ k) (inst k) ≤ maxAlpha

def stmtMaxElemOps {K : Type*} {Γ : K → Type*} {Λ σ : Type*} :
    Turing.TM2.Stmt Γ Λ σ → ℕ
  | .push _ _ q => 1 + stmtMaxElemOps q
  | .peek _ _ q => 1 + stmtMaxElemOps q
  | .pop _ _ q => 1 + stmtMaxElemOps q
  | .load _ q => 1 + stmtMaxElemOps q
  | .branch _ q₁ q₂ => 1 + max (stmtMaxElemOps q₁) (stmtMaxElemOps q₂)
  | .goto _ => 1
  | .halt => 1

def FinTM2BoundedElemOps (M : Turing.FinTM2) (C_step : ℕ) : Prop :=
  ∀ l : M.Λ, stmtMaxElemOps (M.m l) ≤ C_step
```

Machine model and elementary operations. The uniformity theorem `isUniformFinTM2_tmIntMul` thus certifies at most eight stacks, stack alphabets of size at most 2, at most 32 register states, and at most `(bmRunRaw bmIntMulCore).numLabels` control labels (a fixed natural number determined by the construction; the library does not state its value, but the check file `checks/EvalProven.lean` distributed with the snapshot proves `(bmRunRaw bmIntMulCore).numLabels = 1041232178` from the library's definitions, so the finite control has about 10^9 labels), all independent of n, A, B. By construction `tmIntMul` has exactly eight `Bool` stacks (its stack index type `TapeIdx` has eight elements by `rfl`) and the register type `DelimReg` with `Fintype.card DelimReg = 32` (theorem `card_delimReg`). In Mathlib's `Turing.TM2` semantics, one `TM2.step` executes the statement tree of one control label; using `stmtMaxElemOps` and `FinTM2BoundedElemOps` quoted above, we lift the step bound to total elementary stack operations alongside a data-level pairing of the eight stacks into four list pairs (`IntMul.lean`):

```
theorem tmIntMul_elementary_stack_ops_n_log_n :
    ∃ C_step C_time C_ops : ℕ, 0 < C_step ∧ 0 < C_time ∧ 0 < C_ops ∧
```

```
      FinTM2BoundedElemOps tmIntMul C_step ∧
      ∀ n : ℕ, 2 ≤ n → ∀ A B : ℕ, A < 2 ^ n → B < 2 ^ n →
        ∃ t ≤ C_time * n * Nat.log 2 n,
          C_step * t ≤ C_ops * n * Nat.log 2 n ∧
          Turing.EvalsToInTime tmIntMul.step
            (Turing.initList tmIntMul (encodeInputBits n A B))
            (some (Turing.haltList tmIntMul (encodeNatBits (2 * n) (A * B)))) t

def eightStacksEquivFourBiInfiniteTapes :
    (Fin 8 → List Bool) ≃ (Fin 4 → List Bool × List Bool)
```

The same check file proves `finTM2MaxStmtElemOps tmIntMul = 16`, so the general lifting lemma's witness $C_{\text{step}} =$ `finTM2MaxStmtElemOps tmIntMul`$+1$ equals 17 (`finTM2BoundedElemOps_self`; the $+1$ makes the witness positive for every machine, and 16 would also satisfy `FinTM2BoundedElemOps`). Both values are obtained by folding the label counts and operation maxima over the machine's syntax tree, because a direct `#eval` would enumerate the 10^9 labels; the check file is a small metaprogram, run separately after `lake build` (`lake env lean checks/EvalProven.lean`), whose proof-generating elaborator uses `unsafe`, `implemented_by`, and `opaque`, while the two theorems it adds are checked by the kernel and its own `#print axioms` commands report only the three standard axioms for them. `eightStacksEquivFourBiInfiniteTapes` only pairs the eight stacks into four (left, right) lists; it is not a simulation theorem. A k-stack machine over finite alphabets and a multitape Turing machine simulate each other with constant-factor slowdown (one tape per stack), and the formal statement is about the stack machine.

Value-level cost model and analytic library. In addition to `tmIntMul`, the library proves value-level correctness and bottom-up counted $O(n \log n)$ bit-operation bounds for the executable functional multiplier (`computableNatMulTracked_spec` and `computableNatMulBitsTracked_spec` in `IntMul/EndToEnd.lean`), packaged with `isUniformFinTM2_tmIntMul` and `tmIntMul_evalsToInTime_n_log_n` in `intMul_complete_endToEnd_summary`, and with the uniform `FinTM2` primitive and composition suites in `intMul_endToEnd_unconditional_summary`. The analytic and fixed-point bounds on which the formal proof rests (Poisson identity, passband, stopband and conditioning bounds, spreading, multi-level accuracy, root recovery) are proved inside the same library; Table 3 lists the correspondence we established and the places where the formal constants are looser than the printed ones.

Axioms and compile-time checks. The 27 library files (`checks/EvalProven.lean` is not among them and is not part of the `lake build` target) contain no `sorry`, `admit`, `axiom`, `native_decide`, `implemented_by`, `extern`, `opaque`, `unsafe`, or `partial` declarations. The umbrella file `IntMul.lean` ends with compile-time `#verify_standard_axioms` commands (a command defined in `IntMul.lean` itself on top of `Lean.collectAxioms`) that cause compilation to fail if the transitive axiom closure of `tmIntMul`, `isUniformFinTM2_tmIntMul`, `tmIntMul_evalsToInTime_n_log_n`, `tmIntMul_elementary_stack_ops_n_log_n`, `intMul_complete_endToEnd_summary`, or `intMul_endToEnd_unconditional_summary` contains anything beyond `propext`, `Classical.choice`, and `Quot.sound`, followed by `#print axioms` commands for the same six declarations.

Departures from the written text. The formal construction departs from the written text in the following details (none of which affects the $O(n \log n)$ form of Theorem 2), in addition to the variant elementary-function evaluations recorded in Table 3 (π via $\arctan \frac{1}{2} + \arctan \frac{1}{3}$; exp, $e^{i\theta}$, sinc, and $\int_0^z e^{-t^2}\,dt$ via direct rational Taylor polynomials without range reduction).

Table 3: Correspondence between paper statements and Lean 4 declarations in namespace `IntMul`.

Paper statement	Lean 4 declarations and constant notes
Definition 3.1, schedule (34)	`cSpec`, `pWorkEll`, `pChildEll`, `qChildEll`, `isBaseLeaf`, `roundShiftInt` (`Defs.lean`; identical schedule and leaf predicate $2^{K_\ell} \le 4p_\ell^2$ (`isBaseLeaf`), floor-shifted tie rule $\lfloor 2^q v + \frac{1}{2} \rfloor$)
Lemma 3.2 (elementary series)	`abs_approxPiRat_sub_pi_le`, `abs_approxLogTwoRat_sub_log_two_le`, `abs_approxSqrtRat_sub_sqrt_le`, `abs_expTaylorRat_sub_exp_le`, `norm_expITaylorRat_sub_exp_le`, `abs_sincTaylorRat_sub_sinc_le`, `abs_erfIntegralTaylorRat_sub_integral_le` (`ComputableSchedule.lean`; rational series and integer-square-root bounds, with $\frac{\pi}{4} = \arctan\frac{1}{2} + \arctan\frac{1}{3}$)
Lemma 3.3 (transpose & odd-stride)	`transposeMatrix_spec`, `oddColGather_oddColScatter`, `oddColScatter_oddColGather` (`TMPrimitives.lean`, value-level); `bmTransposeMatrix_computesInTime` (`TMSpreadFFT.lean`), `bmOddColScatterCols_computesInTime` (`TMKronecker.lean`, stack-machine time bounds)
Lemma 3.4, Lemma 3.5	`twoCallDFTConv_re_eq_linearConvCoeff` (`FastDFT.lean`; `twoCallDFTConv` evaluates $R\sum_u \frac{(F_R z)_u^2}{2i} e^{-2\pi i u k/R} = R^2\,\overline{F_R\big(\overline{(F_R z)^2/(2i)}\big)}_k$ and reads the real part, which equals $-\frac{R^2}{2}\operatorname{Im} F_R\big(\overline{(F_R z)^2}\big)_k$, the identity (10), and states it for linear chunk convolutions, which coincide with the cyclic ones because $R_0 \ge 2N$); `radix2SyntheticFFTExact_eq_syntheticFFT`, `syntheticFFT_rounding_error_le` (`SyntheticFFT.lean`; bounds stage rounding by $\le k_{1,\ell}\frac{\sqrt{2}}{4}2^{-p_{\ell+1}} \le 0.18K_\ell 2^{-p_{\ell+1}}$ vs. $0.177k_{1,\ell}2^{-p_{\ell+1}}$)
Proposition 4.1, Theorem 4.2	`twisted_fourier_duality_reduction` (`SyntheticFFT.lean`); `gHat_eq_sub`, `untruncatedFilterResponse_eq_untruncatedPoissonModeSum`, `poisson_fractional_shift_identity` (`PoissonKernel.lean`)
Theorem 5.1 (passband, two-term, Nyquist)	`tsum_gHat_sub_int_eq_one`, `gHat_passband_max_ge` (≥ 0.49999999), `gHat_stopband_alias_lt` ($< 0.15 \cdot 2^{-p_{\mathrm{work}}}$ vs. $< 0.120 \cdot 2^{-p_{\mathrm{work}}}$), `gHat_ratio_pow_t1_le_015` ($a^{t_1} \le 0.15$ vs. < 0.14; conditioning $\le 2.36\,t_1$ in `gHat_deconvolution_conditioning_le_236_t1`); `nyquist_alternating_sum_eq`, `nyquistColumnSolve_spec`, `vecInfNorm_nyquistColumnSolve_le` ($t_1/c_0 \le 2.36\,t_1$ vs. $(t_1-1)/(2c) < 1.0000001\,t_1$)
Lemma 6.1, Lemma 6.2, Theorem 6.3	`filter_l1_norm_le` ($\le \frac{2}{\pi}\ln\alpha + 1.62$ vs. $< \frac{2}{\pi}\ln\alpha + 1.444$), `spatial_truncation_tail_le` ($< 0.4 \cdot 2^{-p_{\ell+1}}$ vs. $< 0.077 \cdot 2^{-p_{\ell+1}}$); `overlapAdd_eq_cyclicConv`, `kronecker_spread_exact_of_bounds`, `kronecker_spread_complexity_bound` (`KroneckerSpread.lean`)
Lemma 5.2, Lemma 6.5	`level_error_budget_le`, `base_leaf_error_budget_le` (formal statements prove the total $\le 2^{-p_\ell}$; the paper's margins are 0.197 and 0.267), `inductive_accuracy_closure` (`Deconvolution.lean`); `vecInfNorm_fastDFTRecursive_sub_normDFT1D_le`, `vecInfNorm_fastDFTRecursive_sub_normDFT1D_le_two_zpow` (`FastDFT.lean`)
Theorems 1 and 2	`reconstructProductViaFastDFT_eq_mul` (root margin $\le \frac{1}{8} < \frac{1}{2}$ vs. $< 3.19 \cdot 2^{-7}$); `computableNatMulTracked_spec`, `computableNatMulBitsTracked_spec` (`EndToEnd.lean`); `tmIntMul_evalsToInTime_n_log_n`, `tmIntMul_elementary_stack_ops_n_log_n`, `intMul_complete_endToEnd_summary`

(1) *Rounding.* Rounding in Lean uses the floor-shifted tie rule $\lfloor 2^q v + \frac{1}{2} \rfloor$ (`roundShiftInt`), whereas the C program uses MPFR's nearest rounding (`MPFR_RNDN`) when quantizing precomputed tables and $\lfloor m/2^s + \frac{1}{2} \rfloor$ in fixed-point shifts, while Definition 3.1 allows any fixed nearest tie rule.

(2) *Bootstrap.* `tmIntMul` uses a two-tier self-recursive bootstrap with binary shift-add schoolbook multiplication at the bottom, with secondary work $O(n(\log\log n)^4)$, whereas Section 7 uses a three-stage bootstrap with secondary work $O(n(\log\log n)^2)$. The $(\log\log n)^4$ bound is an

intermediate bound of the formal proof: `nestedSubMulBitOps_le_log_cube` bounds the middle-tier multiplier by $O(q(\log q)^3)$ per q-bit product and `log_rootK0_add_one_fourth_le` absorbs the resulting factor into $O(n \log n)$, both in `IntMul/EndToEnd.lean`.

(3) *Input encoding.* Inputs use the fixed-width encoding `encodeInputBits n A B`, the list `encodeNatBits n A ++ encodeNatBits n B` (bit i of each operand at position i, least significant first) on the input stack; the output is `encodeNatBits (2 * n) (A * B)`.

(4) *Small inputs.* `tmIntMul` handles every $n \geq 2$ uniformly without the $n \leq 64$ shortcut of the pseudocode.

(5) *Window radius.* The formal development uses the admissible window radius `computeM1Ell K p` $= \lfloor \sqrt{p_{\mathrm{work},\ell}\, p_{\ell+1}} \rfloor + 1$ (integer square root; theorems `m1Ell_le_computeM1Ell` and `computeM1Ell_le_pChildEll` place it in the interval of Lemma 6.5) instead of the default radius $\lceil \alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi} \rceil$ of Remark 6.4 and of the C program (the Turing machine of Appendix A.2 computes it as $\lceil \tilde{x} + 2^{-s-20} \rceil \in \{\lceil \cdot \rceil, \lceil \cdot \rceil + 1\}$; every statement holds for every admissible radius).

Separately, the companion paper's development [CLW26] contains its own kernel-checked multiplier for this algorithm.

Toolchain and distributed snapshot. The development was carried out under Lean 4.27.0 and Mathlib revision `a3a10db0`. To build under Lean 4.33.1 / Mathlib `0df444a3` the port adjusted imports, instance resolution and proof scripts; in 18 files it sets the compatibility option `backward.isDefEq.respectTransparency false` (restoring the pre-4.3x unifier behaviour for implicit arguments) and raises `maxHeartbeats`/`maxRecDepth` limits. These options affect elaboration only; every proof term is re-checked by the kernel, and no theorem statement was changed (we verified that the declaration headers recorded in `SIGNATURES.txt` are byte-identical to those of the original development, which is not part of the snapshot). The distributed snapshot `82b659a08376` [LeanMS26] compiles with `lake build` (exit 0) and `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound` for the headline theorems.

9 Implementation and measurements

We measure the C/GMP implementation `intmul_fastdft` (`algo=1`, about 1,700 lines of C [CGMP26]) of the algorithm of this paper, with the windowed sinc-Gaussian kernel and the additive schedule, and with inner integer products delegated to GMP (it does not implement the companion paper's d-dimensional algorithm [CLW26]). Compared with the Turing machine specification in Sections 3–7, `intmul_fastdft.c` uses the identical additive schedule (34) ($p_{\mathrm{work},\ell} = p_\ell + K_\ell + 6$, $p_{\ell+1} = p_\ell + 2K_\ell + 12$, $p_0 = 2b + 2K_0 + 6$), evaluates precomputed tables with MPFR nearest rounding, packs signed Kronecker slots into non-negative GMP `mpz_mul` operands via a precomputed bias offset, performs direct fixed-point division in the two-term column solver, and lowers the default leaf cutoff to `bt` $= p_0^2$ evaluated at the root p_0 (rather than $4p_\ell^2$). At $n = 10^9$ bits ($b = 29$, $K_0 = 27$, $R_0 = 2^{27}$, $p_0 = 118$), `bt` $= 13{,}924 < R_1 = 2^{14} = 16{,}384$, so the 10^9-bit run exercises two recursion levels ($2^{27} \to 2^{14} \to 2^7 = 128$). At that level $R_1 = 2^{14} < 4p_1^2 = 135{,}424$, so these nodes lie outside the non-leaf hypothesis of Theorem 6.3 and Lemma 6.5(1) (and exercise the folding case $W_1 = 191 > r_1 = 128$ of Remark 6.4); the bit-exact agreement observed at 10^9 bits is therefore an empirical observation, not a consequence of the proofs. The runs at 10^7, 10^8, and $5 \cdot 10^8$ bits treat level 1 as a leaf and stay within the parameter regime of the proofs (the implementation differences listed above remain).

All measurements were run on an AMD EPYC 7B13 (Zen 3) virtual machine under KVM (64 vCPUs, 117 GiB RAM, Linux x86_64), compiled with `gcc` 15.2.0 (`-O2 -march=native -fopenmp -DNDEBUG`) against GMP 6.3.0 and MPFR 4.2.2 [CGMP26]. The measured binary was compiled from a copy of `intmul_fastdft.c` that differs from the shipped file only in a header comment and two preprocessor-guard blocks (an unused include branch under a build-internal macro, removed from the shipped file, and a guard added so that the shipped file also compiles without OpenMP; under the stated flags neither changes the compiled code), with both SHA-256 digests recorded in `bench/MACHINE.txt`; the host was lightly shared (load average 2–5 on 64 vCPUs) during the single-threaded runs, and 'one core' means one vCPU of the KVM guest. Each run (one random operand pair per size) invoked `intmul_fastdft mul <n> 0 7 <algo>` (`seed` 7, random n-bit operands with top bit set) and compared the result bit for bit against single-threaded GMP `mpz_mul` on the same operands in the same process. The program's built-in forced-deep self-test at 4,096 and 65,536 bits reported `SELFTEST: ALL PASSED` (every product `EXACT`, both kernels), and every benchmark run from 10^7 to 10^9 bits printed `EXACT MATCH` (bit-exact on every tested input).

Table 4: Single-threaded measurements (`OMP_NUM_THREADS=1`) of the C/GMP implementation `intmul_fastdft` (`algo=1`, windowed sinc-Gaussian kernel and additive schedule; inner products by GMP) against single-threaded GMP `mpz_mul` on the same operands (`seed` 7, AMD EPYC 7B13). Every run printed `EXACT MATCH`. Slowdown ratios are those printed by the program from the unrounded timings; recomputing them from the rounded timings shown here can differ by one in the last digit.

n (bits)	Chain (R_ℓ, p_ℓ)	Core (s)	End-to-end (s)	Tables (s)	`mpz_mul` (s)	Slowdown	Peak RSS
10^7	$2^{20}\,(92) \to 1024\,(144)$	9.63	9.92	1.82	0.0478	207×	0.56 GiB
10^8	$2^{23}\,(104) \to 4096\,(162)$	90.00	92.51	5.46	0.6059	153×	4.4 GiB
$5 \cdot 10^8$	$2^{26}\,(114) \to 8192\,(178)$	786.57	805.37	19.44	3.4417	234×	33.8 GiB
10^9	$2^{27}\,(118) \to 2^{14}\,(184) \to 128\,(224)$	2574.67	2612.65	25.95	8.0969	323×	67.3 GiB

Table 4 gives the single-threaded timings (`OMP_NUM_THREADS=1`). Here "core" is the time in the two top-level `FastDFT` calls, "end-to-end" includes chunk unpacking, both transforms, pointwise squaring, and carry assembly, and "tables" is the one-time filter and twiddle setup for that size (including table setup, the single-threaded slowdowns relative to `mpz_mul` are 245×, 162×, 240×, and 326× as reported in `bench/BENCHMARK.md`; recomputed from the rounded timings of Table 4 the first is 246×). Single-threaded on one Zen 3 core, the C/GMP program multiplies two 10^8-bit integers in 92.5 s, about 153 times slower than GMP's `mpz_mul` (0.606 s) on the same operands; it agrees bit for bit with GMP at every size tested, up to 10^9 bits. The 10^9-bit run needs about 72 GB of RAM (peak RSS 67.3 GiB). In one 64-thread run at 10^8 bits (`OMP_NUM_THREADS=64`, same binary and operands), `intmul_fastdft` (`algo=1`) took 4.90 s in `FastDFT` core and 6.31 s end-to-end (4.4 GiB peak RSS), which is 10.4× slower than single-threaded GMP `mpz_mul` (0.6056 s).

In single runs on the same machine and operands, the plain-Gaussian variant with the multiplicative schedule (`algo=0`) took 1.87× and 1.77× as long and used 3.6× and 2.4× the memory at 10^7 and 10^8 bits. GMP's `mpz_mul` uses a Schönhage–Strassen FFT at these sizes; the constant factors of spreading, twisting, and deconvolution dominate at every tested size, and no crossover is expected at any physically storable size. Accordingly, no practical speedup is claimed.

10 Discussion

Within the two-dimensional algorithm of Theorem 1, every scalar and polynomial multiplication across the computation (root pointwise squaring, overlap-add Kronecker spreading, pre- and post-twists, two-term and Nyquist column deconvolution, and leaf radix-2 FFT butterflies) executes in $O(n(\log\log n)^2)$ steps on a multitape Turing machine, while shared per-level table generation takes $o(n)$ steps. The only $\Theta(n\log n)$ term in the upper bound is $\mathcal{C}_{\mathrm{syn+route}}$ in (38): the multiplication-free polynomial transform $G_{t_{1,\ell}}$ over $\mathbb{C}[y]/(y^{r_\ell}+1)$ together with the per-level matrix transposes and odd-stride tape routing. Known $\Omega(n\log n)$ lower bounds for integer multiplication are conditional, resting on the network-coding conjecture (for Boolean circuits) [AFKL19] or on the conjectured $\Omega(n^2\log n)$ multitape Turing machine complexity of $n\times n$ binary matrix transposition [HvdH25]. Whether the $\Theta(n\log n)$ polynomial-transform and tape-routing term in a fixed-dimension algorithm can be reduced unconditionally remains open, although [OAI26] gives, and [CLW26] claims, $o(n\log n)$ bounds by letting the dimension grow with n, and no lower bound is claimed in this paper.

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A Elementary functions, shared tables, and tape schedule

A.1 Proof of Lemma 3.2

Proof of Lemma 3.2. We establish the bit-width, accuracy, and time bounds for primitives (i)–(v) in turn, and then verify the composite table-evaluation contract (7).

(i) Series for π and $\ln 2$. This block bounds the truncation tails, loop-termination rule, integer divisors, and $\widetilde{\mathbb{R}}_S$-summation error for the approximations $\tilde{\pi}$, $\tilde{\ell}_2$, and $\tilde{h}$. Recall $S = s + 15D + 64$ and $N_{\text{terms}} = 2S + 36D$ with $D = \max(s, K_0) \geq 8$. The denominator base power satisfies

$$5^{2N_{\text{terms}}+1} > 3^{2N_{\text{terms}}} = 9^{N_{\text{terms}}} > 8^{2S+36D} = 2^{6S+108D} \gg 20 \cdot 2^{S+16}.$$

Consequently, the combined alternating series tails for π from both $\arctan(1/5)$ and $\arctan(1/239)$ and the geometric series tail for $\ln 2$ satisfy, respectively,

$$\left|\pi - 4\sum_{k=0}^{N_{\text{terms}}-1}(-1)^k \frac{4\cdot 5^{-2k-1} - 239^{-2k-1}}{2k+1}\right| \leq 20(2N_{\text{terms}}+1)^{-1}5^{-(2N_{\text{terms}}+1)} < 2^{-S-16},$$

$$2\sum_{k=N_{\text{terms}}}^{\infty} \frac{1}{(2k+1)3^{2k+1}} \leq \frac{3}{4(2N_{\text{terms}}+1)}3^{-2N_{\text{terms}}} < 2^{-S-16}.$$

In each series loop for $b \in \{3, 5, 239\}$, the machine stops incrementing b^{2k+1} as soon as $b^{2k+1} > 32 \cdot 2^S$. Every subsequent term with $k < N_{\text{terms}}$ has exact magnitude below $\frac{16}{32\cdot 2^S} = \frac{1}{2}2^{-S}$ and therefore rounds via $\mathcal{Q}_S$ to 0 on $\widetilde{\mathbb{R}}_S$ identically. For all $D \geq 8$, every integer divisor formed before termination satisfies

$$(2k+1)b^{2k+1} \leq (2N_{\text{terms}}+1)\cdot 239^2 \cdot 32 \cdot 2^S < 2^{S+23+\lceil \log_2(169D)\rceil} \ll 2^{2S+4D+16}.$$

Summing the non-zero rounded terms on $\widetilde{\mathbb{R}}_S$ incurs rounding error at most $4N_{\text{terms}}2^{-S} < 2^{-s-D-32}$, defining the approximations and error bounds

$$\tilde{\pi},\, \tilde{\ell}_2 \in \widetilde{\mathbb{R}}_S, \qquad \tilde{h} := \mathcal{Q}_S(\tilde{\pi}/2),$$
$$|\tilde{\pi} - \pi|,\, |\tilde{\ell}_2 - \ln 2| \leq (4N_{\text{terms}}+1)2^{-S}, \qquad |\tilde{h} - \pi/2| \leq (2N_{\text{terms}}+1)2^{-S}.$$

(ii) Square root. This block defines the scaled integer radicand $N_{\tilde{x}}$ and its integer square root m and bounds the fixed-point and input-perturbation errors on $[2^{-D}, 2^{2D}]$. Since both x and $\tilde{x}$ lie in $[2^{-D}, 2^{2D}]$, define

$$N_{\tilde{x}} := 2^{2S}\tilde{x} \in \mathbb{Z}_{\geq 1}, \qquad m := \lfloor \sqrt{N_{\tilde{x}}} \rfloor.$$

The integer $N_{\tilde{x}}$ has bit-length at most $2S + 2D + 1 = O(D)$. Computing m via digit-by-digit binary integer square root takes $O(D^2)$ steps and satisfies $|2^{-S}m - \sqrt{\tilde{x}}| < 2^{-S} < 2^{-s-D-32}$. On $[2^{-D}, 2^{2D}]$, the input perturbation satisfies

$$|\sqrt{\tilde{x}} - \sqrt{x}| \leq \frac{1}{2\sqrt{2^{-D}}}\delta_{\text{in}} = 2^{D/2-1}\delta_{\text{in}}.$$

(iii) Exponential. This block verifies the underflow cutoff, defines the quotient-aligned range-reduction variables $m, \tilde{r}, r_m$ and Taylor terms $\tilde{T}_k$, and bounds exp on $[-\pi(D+2)^2, 0]$. Whenever the exact $\widetilde{\mathbb{R}}_S$ cutoff $\tilde{x} < -(S+16)\tilde{\ell}_2$ holds, the bound $|\tilde{\ell}_2 - \ln 2| \le (4N_{\text{terms}}+1)2^{-S}$ implies $\tilde{x} < -(S+16)\ln 2 + 2^{-100}$ and $e^{\tilde{x}} < 2^{-S-15} < 2^{-s-D-32}$, so returning 0 is valid.

Otherwise $-(S+16)\tilde{\ell}_2 \le \tilde{x} \le 0$, and integer division of $2^S(-\tilde{x})$ by $2^S\tilde{\ell}_2$ defines

$$\begin{aligned} m &:= \lfloor (-\tilde{x})/\tilde{\ell}_2 \rfloor \in \{0, \dots, S+16\}, \\ \tilde{r} &:= -\tilde{x} - m\tilde{\ell}_2 \in \widetilde{\mathbb{R}}_S \cap [0, \tilde{\ell}_2), \qquad r_m := -\tilde{x} - m\ln 2. \end{aligned}$$

Because r_m is defined with that same integer m, we have $e^{\tilde{x}} = 2^{-m}e^{-r_m}$ identically, and comparing $\tilde{r}$ to r_m incurs no quotient-boundary jump. Their difference satisfies

$$|\tilde{r} - r_m| = m|\tilde{\ell}_2 - \ln 2| \le (S+16)(4N_{\text{terms}}+1)2^{-S} < 2^{-100},$$

with $r_m \in [-2^{-100}, \ln 2 + 2^{-100}]$ and $|(e^{-t})'| \le e^{2^{-100}} < 1 + 2^{-99}$ between $\tilde{r}$ and r_m.

The N_{terms} Taylor terms for $\exp(-\tilde{r})$ are generated on $\widetilde{\mathbb{R}}_S$ by the recurrence

$$\tilde{T}_0 := 1, \qquad \tilde{T}_k := \mathrm{RD}_S\big(\mathrm{RM}_S(\tilde{T}_{k-1}, -\tilde{r}),\, k\big) \quad (1 \le k < N_{\text{terms}}).$$

Since $\tilde{r}/k < 0.70$, each term satisfies $|\tilde{T}_k - T_k| < 3.34 \cdot 2^{-S}$, and every intermediate integer stays within $2S + 4D + 16$ bits. Moreover, the truncation tail satisfies

$$\frac{|\tilde{r}|^{N_{\text{terms}}}}{N_{\text{terms}}!} \le \left(\frac{e}{N_{\text{terms}}}\right)^{N_{\text{terms}}} < 2^{-4S} < 2^{-S-16}.$$

After summing the N_{terms} terms on $\widetilde{\mathbb{R}}_S$, the machine applies a rounded m-bit right shift $\mathcal{Q}_S(2^{-m}\cdot)$ onto $\widetilde{\mathbb{R}}_S$ for 2^{-m}, adding at most $\frac{1}{2}2^{-S}$. For all $D \ge 8$, the total error relative to $e^{\tilde{x}}$ is at most

$$((S+18)(4N_{\text{terms}}+1)+1)2^{-S} < 2^{14}D^2 2^{-S} < 2^{-s-D-32}.$$

(iv) Sine and cosine. This block defines the $\pi/2$ range-reduction quantities $m, \tilde{r}, r_m$ and recurrence terms $\tilde{T}_k$ on $|x| \le \pi(D+2)$, followed by the direct sinc series on $[-1/2, 1/2]$. Using $\tilde{h} = \mathcal{Q}_S(\tilde{\pi}/2)$ with $|\tilde{h} - \pi/2| \le (2N_{\text{terms}}+1)2^{-S}$, define

$$m := \lfloor \tilde{x}/\tilde{h} + 1/2 \rfloor \in \mathbb{Z}, \qquad \tilde{r} := \tilde{x} - m\tilde{h} \in \widetilde{\mathbb{R}}_S, \qquad r_m := \tilde{x} - m(\pi/2).$$

Here $|m| \le 2(D+2)+1 < S+16$, and the reduced remainder satisfies

$$\begin{aligned} |\tilde{r}| &\le \frac{\tilde{h}}{2} < 0.79 < 1, \\ |\tilde{r} - r_m| &= |m|\,|\tilde{h} - \pi/2| \le (S+16)(4N_{\text{terms}}+1)2^{-S} < 2^{-100}. \end{aligned}$$

Using that same integer m in r_m avoids any quotient-boundary jump and gives $\sin\tilde{x} = \sin(m\frac{\pi}{2} + r_m)$ and $\cos\tilde{x} = \cos(m\frac{\pi}{2} + r_m)$ identically via $m \bmod 4$, while $|(\sin t)'|, |(\cos t)'| \le 1$.

For $\cos\tilde{r}$ and $\sin\tilde{r}$, the N_{terms} Taylor terms are generated on $\widetilde{\mathbb{R}}_S$ by

$$\begin{aligned} &\tilde{T}_0 := 1 \quad (\text{for } \cos\tilde{r},\ \text{divisor } 2k(2k-1)), \qquad \tilde{T}_0 := \tilde{r} \quad (\text{for } \sin\tilde{r},\ \text{divisor } 2k(2k+1)), \\ &\tilde{T}_k := \mathrm{RD}_S\Big(\mathrm{RM}_S\big(\tilde{T}_{k-1}, -\mathrm{RM}_S(\tilde{r}, \tilde{r})\big),\, 2k(2k \mp 1)\Big) \quad (1 \le k < N_{\text{terms}}). \end{aligned}$$

Because $\tilde{r}^2/2 < 0.32$, each term satisfies $|\tilde{T}_k - T_k| < 1.48 \cdot 2^{-S}$, and all intermediate integers stay within $2S + 4D + 16$ bits. For all $D \geq 8$, the alternating tail bound of Lemma 3.2(iv) and the total error $|\tilde{y} - f(\tilde{x})|$ for $f \in \{\sin, \cos\}$ satisfy

$$\frac{|\tilde{r}|^{2N_{\text{terms}}}}{(2N_{\text{terms}})!} \leq \frac{|\tilde{r}|^{N_{\text{terms}}}}{N_{\text{terms}}!} < 2^{-S-16},$$
$$|\tilde{y} - f(\tilde{x})| \leq ((S+18)(4N_{\text{terms}}+1)+1)2^{-S} < 2^{14}D^2 2^{-S} < 2^{-s-D-32}.$$

Claim: Direct sinc **series on** $[-1/2, 1/2]$**.** For $|t| \leq 1/2$, define the scaled argument $\tilde{u}$ and Taylor terms $\tilde{T}_k$ on $\widetilde{\mathbb{R}}_S$ by

$$\tilde{u} := \mathrm{RM}_S(\tilde{\pi}, \tilde{t}), \qquad \tilde{T}_0 := 1,$$
$$\tilde{T}_k := \mathrm{RD}_S\Big(\mathrm{RM}_S(\tilde{T}_{k-1}, -\mathrm{RM}_S(\tilde{u}, \tilde{u})),\, 2k(2k+1)\Big) \quad (1 \leq k < N_{\text{terms}}).$$

Then $|\tilde{u} - \pi\tilde{t}| \leq (2N_{\text{terms}}+1)2^{-S}$ and $|\tilde{u}| < 1.58$. Every step ratio satisfies $\tilde{u}^2/(2k(2k+1)) < 0.42 < 1/2$, and every term satisfies $|\tilde{T}_k| \leq 1$. The alternating tail bound of Lemma 3.2(iv) and the derivative on $|u| \leq \pi/2$ satisfy

$$\frac{(\pi/2)^{2N_{\text{terms}}}}{(2N_{\text{terms}}+1)!} \leq \left(\frac{\pi e}{4N_{\text{terms}}}\right)^{2N_{\text{terms}}} < 2^{-4S} < 2^{-S-16},$$
$$\left|\frac{d}{du}\left(\frac{\sin u}{u}\right)\right| \leq \frac{4}{\pi^2} < 0.41 \leq 1 \quad (|u| \leq \pi/2),$$

with $|\operatorname{sinc}'(t)| \leq \frac{4}{\pi} < 1.28$ on $[-1/2, 1/2]$. Summing $\tilde{T}_0, \ldots, \tilde{T}_{N_{\text{terms}}-1}$ on $\widetilde{\mathbb{R}}_S$ therefore gives

$$|\tilde{y} - \operatorname{sinc}(\tilde{t})| \leq (6N_{\text{terms}}+2)2^{-S} + 2^{-S-16} < 2^{-s-D-32}.$$

(v) Error function erf**.** This block analyzes the early-exit cutoff, the power-series recurrence $\tilde{T}_n$ with auxiliary quantities $\Sigma(\tilde{z}), w_{\text{sq}}, V, T_n, \delta_k$, and the Gaussian prefactor multiplication on $[0, 3\sqrt{D}]$.

Claim: Early-exit branch of erf**.** Comparing $\mathrm{RM}_S(\tilde{z}, \tilde{z})$ against $(s+D+24)\tilde{\ell}_2$ on $\widetilde{\mathbb{R}}_S$ tests the cutoff within $O(D^2 2^{-S})$. Whenever $\mathrm{RM}_S(\tilde{z}, \tilde{z}) > (s+D+24)\tilde{\ell}_2$, we have $\tilde{z}^2 > (s+D+24)\ln 2 - 2^{-100}$, and $s + D + 24 \geq 40$ gives $\tilde{z} > 5.26$. Hence returning 1 incurs error

$$\operatorname{erfc}(\tilde{z}) \leq \frac{e^{-\tilde{z}^2}}{\sqrt{\pi}\tilde{z}} < 2^{-s-D-26} < 2^{-s-D-23}.$$

Claim: Series branch and recurrence rounding for erf**.** On the series branch, $\tilde{z} \in \widetilde{\mathbb{R}}_S \cap [0, 3\sqrt{D}]$ satisfies $\tilde{z}^2 \leq (s+D+24)\ln 2 + 2^{-100}$, so $\tilde{z}^2 \leq 9D$ and $e^{\tilde{z}^2} < 1.01 \cdot 2^{s+D+24} \leq 1.01 \cdot 2^{2D+24}$. Define the positive series

$$\Sigma(\tilde{z}) := \tilde{z} \sum_{n=0}^{\infty} \frac{(2\tilde{z}^2)^n}{(2n+1)!!} = \frac{\sqrt{\pi}}{2} e^{\tilde{z}^2} \operatorname{erf}(\tilde{z}).$$

Since $e^9 < 8104 < 2^{13}$, $\Sigma(\tilde{z})$ satisfies $\Sigma(\tilde{z}) < 2^{s+D+24} \leq 2^{2D+24}$ and $\Sigma(\tilde{z}) \leq 3\sqrt{D}\, e^{9D} < 2^{14D}$. Using $(2n+1)!! \geq 2^n n! \geq 2^n (n/e)^n$, for every index n in the tail we have

$$n \geq N_{\text{terms}} = 2S + 36D > 66D, \qquad \frac{\tilde{z}^2 e}{n} < \frac{9eD}{66D} < 0.371 < 2^{-1.43},$$

so the series tail for $n \geq N_{\text{terms}}$ is bounded by $6\sqrt{D} \cdot 2^{-1.43 N_{\text{terms}}} < 2^{-S-16}$.

To bound the fixed-point rounding error on $\widetilde{\mathbb{R}}_S$, define

$$w_{\mathrm{sq}} := \mathrm{RM}_S(2\tilde{z}, \tilde{z}), \qquad V := \max(2\tilde{z}^2, w_{\mathrm{sq}}) \le 2\tilde{z}^2 + 1,$$

so $|w_{\mathrm{sq}} - 2\tilde{z}^2| \le \frac{1}{2}2^{-S}$, and compute the recurrence

$$\tilde{T}_0 := \tilde{z}, \qquad \tilde{T}_n := \mathrm{RD}_S(\mathrm{RM}_S(\tilde{T}_{n-1}, w_{\mathrm{sq}}),\, 2n+1) \quad (1 \le n < N_{\mathrm{terms}}).$$

Define the exact n-th term T_n and the step residual δ_k by

$$\begin{aligned} T_n &:= \frac{(2\tilde{z}^2)^n \tilde{z}}{(2n+1)!!}, \qquad |\delta_0| = |\tilde{T}_0 - T_0| = 0, \\ \delta_k &:= (\tilde{T}_k - T_k) - \frac{w_{\mathrm{sq}}}{2k+1}(\tilde{T}_{k-1} - T_{k-1}) \\ &= \left(\tilde{T}_k - \frac{w_{\mathrm{sq}}}{2k+1}\tilde{T}_{k-1}\right) + \frac{w_{\mathrm{sq}} - 2\tilde{z}^2}{2k+1} T_{k-1} \quad (k \ge 1). \end{aligned}$$

Each residual satisfies

$$|\delta_k| \le \frac{T_{k-1}}{2k+1}|w_{\mathrm{sq}} - 2\tilde{z}^2| + \left(\frac{1}{2(2k+1)} + \frac{1}{2}\right)2^{-S} < \frac{T_{k-1}}{2(2k+1)}2^{-S} + 2^{-S}.$$

Unrolling the linear recurrence gives $\tilde{T}_n - T_n = \sum_{k=1}^{n} \delta_k \prod_{j=k+1}^{n} \frac{w_{\mathrm{sq}}}{2j+1}$. Using $(2k-1)!!(2k+1)\prod_{j=k+1}^{n}(2j+1) = (2n+1)!! \ge 3\cdot 2^{n-1}(n-1)!$ and $\prod_{j=k+1}^{n}(2j+1) \ge 2^{n-k}(n-k)!$, the propagation products satisfy

$$\begin{aligned} \frac{T_{k-1}}{2k+1}\prod_{j=k+1}^{n}\frac{w_{\mathrm{sq}}}{2j+1} &\le \frac{\tilde{z}V^{n-1}}{(2n+1)!!} \le \sqrt{D}\, e^{V/2}, \\ \prod_{j=k+1}^{n}\frac{w_{\mathrm{sq}}}{2j+1} &\le \frac{(V/2)^{n-k}}{(n-k)!} \le e^{V/2} \le e^{9D+1/2} = \sqrt{e}\,(e^9)^D < 2\cdot(2^{13})^D = 2^{13D+1}. \end{aligned}$$

Thus each step $k \le n$ contributes at most $(\sqrt{D}+1)e^{V/2}2^{-S} \le 8D\cdot 2^{13D+1-S}$ to $|\tilde{T}_n - T_n|$. For all $D = \max(s, K_0) \ge 8$, we have $S = s + 15D + 64 \le 24D$, $D^3 2^{-D} \le 8^3 2^{-8} = 2$, and

$$N_{\mathrm{terms}} = 2S + 36D \le 84D < 2^7 D, \qquad \frac{N_{\mathrm{terms}}(N_{\mathrm{terms}}+1)}{2} < 2^{12}D^2.$$

Summing over all pairs (k, n) with $1 \le k \le n$ and $n < N_{\mathrm{terms}}$ and adding the truncation tail $< 2^{-S-16}$ yields

$$\begin{aligned} |\widetilde{\Sigma}(\tilde{z}) - \Sigma(\tilde{z})| &\le \frac{N_{\mathrm{terms}}(N_{\mathrm{terms}}+1)}{2}\cdot 8D\cdot 2^{13D+1-S} + 2^{-S-16} \\ &< 2^{16}D^3 2^{-2D}\cdot 2^{-s-64} + 2^{-S-16} \le 2^{-s-D-47} + 2^{-S-16} < 2^{-s-D-24}. \end{aligned}$$

Claim: Prefactor multiplication and final erf **error.** For $\tilde{z} \in [0, 3\sqrt{D}]$, forming $-\mathrm{RM}_S(\tilde{z}, \tilde{z})$ perturbs $-\tilde{z}^2$ by at most $\frac{1}{2}2^{-S}$, so by (i)–(iii) the prefactor $\frac{2}{\sqrt{\pi}}e^{-\tilde{z}^2} \in (0, \frac{2}{\sqrt{\pi}}] \subset (0, 1.13]$ is computed on $\widetilde{\mathbb{R}}_S$ with error $< 2^{15}D^2 2^{-S}$. Because $|\widetilde{\Sigma}(\tilde{z})| < 2^{2D+25}$, the scaled integer product between this prefactor and $\widetilde{\Sigma}(\tilde{z})$ has bit-length at most $2S + 2D + 27 < 2S + 4D + 16$. For all $D \ge 8$, multiplying by $\widetilde{\Sigma}(\tilde{z})$ amplifies the prefactor error by at most

$$2^{2D+25}\cdot 2^{15}D^2 2^{-S} = D^2 2^{-12D-24}\cdot 2^{-s-D} < 2^{-s-D-110}.$$

Rounding the product onto $\widetilde{\mathbb{R}}_S$ adds at most $\frac{1}{2}2^{-S}$, which gives

$$|\tilde{y} - \mathrm{erf}(\tilde{z})| \le 1.13\cdot 2^{-s-D-24} + 2^{-s-D-110} + \frac{1}{2}2^{-S} < 2^{-s-D-23} < 2^{-s-30}$$

together with the Lipschitz bound $|\mathrm{erf}(\tilde{z}) - \mathrm{erf}(z)| \le \frac{2}{\sqrt{\pi}}|\tilde{z} - z|$.

Composite table expressions and faithful rounding. This block verifies domain membership, derivative amplification, repeated-squaring stability for $\theta(u_2)$, the 1-ulp rounding bound (7), and the $O(D^3)$ step count across all table expressions. The real constants and composite table expressions evaluated in our algorithm are as follows.

- First are the width parameter $\alpha_\ell = 2\sqrt{p_{\mathrm{work},\ell} \ln 2/\pi}$, the window-radius expression $\alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi}$ (Appendix A.2(a)), and the filter samples $g(t)$ at exact dyadic rational arguments $t = d + j_1/t_{1,\ell} \in \widetilde{\mathbb{R}}_{K_0} \subseteq \widetilde{\mathbb{R}}_S$ with $|d| \le m_{1,\ell} \le D$, using the direct $\mathrm{sinc}(t)$ series on $|t| \le 1/2$ and dividing by πt on $|t| > 1/2$ where $|\pi t| > \pi/2 > 1.57$.
- Second are the twist roots $e^{\pi i k_2/r_\ell}$ and leaf butterfly roots $e^{2\pi i j \cdot 2^{K_L - s}/R_L}$.
- Third are the passband coefficients $c_0(u_2), c_1(u_2) \in [0,1]$, the Nyquist reciprocal $c(r_\ell/2)^{-1}$, and for $u_2 \ne r_\ell/2$ the non-Nyquist column solver scalars $M(u_2)^{-1}$, $\rho(u_2)$, and $\theta(u_2) = (1 - \rho(u_2)^{t_{1,\ell}})^{-1}$, where $M(u_2) > 0.499$ and $1 - \rho(u_2)^{t_{1,\ell}} > 0.86$ for all $u_2 \ne r_\ell/2$.

Each expression composes $O(1)$ evaluations of (i)–(v) on $\widetilde{\mathbb{R}}_S$ with all divisions bounded away from zero by at least 0.42, plus at most $\log_2 t_{1,\ell} \le D/2$ squarings on $\widetilde{\mathbb{R}}_S$.

Claim: Call-site domain membership. At every call site in the algorithm, both the exact argument and the computed $\widetilde{\mathbb{R}}_S$-input lie inside the stated domain of Lemma 3.2. Every $\sqrt{x}$ evaluation, namely $\sqrt{\pi}$, $\alpha_\ell = 2\sqrt{p_{\mathrm{work},\ell} \ln 2/\pi}$, and $\sqrt{p_{\ell+1} \ln 2/\pi}$, satisfies

$$8 \le p_{\mathrm{work},\ell} \le p_{\ell+1} < s \le D, \qquad x, \tilde{x} \in [1, 2^{2D}].$$

On $[1, 2^{2D}]$, $\sqrt{\cdot}$ has local Lipschitz constant at most $\frac{1}{2}$ rather than $2^{D/2-1}$, and its inputs from (i) have $\widetilde{\mathbb{R}}_S$-error $O(D2^{-S})$. Likewise, every negated-square exp evaluation has $x, \tilde{x} \in [-\pi(D+2)^2, 0]$, every $\sin, \cos$ evaluation has $|x|, |\tilde{x}| \le \pi(D+2)$, and every erf evaluation has $z, \tilde{z} \in [0, 3\sqrt{D}]$ by using $\mathrm{erf}(-z) = -\mathrm{erf}(z)$ on $y - 1/2 \le 0$ for $0 \le u_2 \le r_\ell/2$.

Claim: Non-θ composite error bound. From (i) and (ii) we have $|\tilde{\pi} - \pi|, |\tilde{\ell}_2 - \ln 2|, |\tilde{\alpha}_\ell - \alpha_\ell| \le O(D^2 2^{-S}) < 2^{-s-D-100}$. Across all non-$\theta$ composite expressions, the partial derivatives along the compositions satisfy the following five bounds.

- For $(\alpha_\ell, \pi) \mapsto -\pi t^2/\alpha_\ell^2 \le 0$, the bound is $\frac{2\pi(D+1)^2}{\alpha_\ell^3} + \frac{(D+1)^2}{\alpha_\ell^2} < \frac{1}{2}(D+1)^2$, since $|t| < D+1$ and $\alpha_\ell > 2.65$.
- For the product $\alpha_\ell \cdot \sqrt{p_{\ell+1} \ln 2/\pi}$ of two factors in $[1, 2\sqrt{D}]$, the bound is $2\sqrt{D} < \frac{1}{2}(D+1)^2$.
- For $\alpha_\ell \mapsto \mathrm{erf}(\sqrt{\pi}\alpha_\ell(y \pm \frac{1}{2}))$, the bound is $2|y \pm \frac{1}{2}| \le 3$.
- For each reciprocal of a quantity ≥ 0.42, the bound is $1/0.42^2 < 5.7$, and for $(c_{\min}, M) \mapsto -c_{\min}/M$ with $0 \le c_{\min} \le M$ and $M(u_2) > 0.499$, it is $2/M(u_2) < 4.1 < 5.7$.
- For exp on $(-\infty, 0]$, sin, cos, and each factor of a product on $[-1,1]^2$, the bound is at most 1, while exact $t \in \widetilde{\mathbb{R}}_S$ gives $|\widetilde{\mathrm{sinc}}(t) - \mathrm{sinc}(t)| < 2^{-s-D-23}$ directly.

With primitive errors $< 2^{-s-D-23}$ from (i)–(v) and $\widetilde{\mathbb{R}}_S$ arithmetic rounding $O(D^2 2^{-S}) \ll 2^{-s-D-100}$, every non-$\theta$ table expression has error at most

$$5.7\big(\tfrac{1}{2}(D+1)^2 + 4\big) \cdot 2^{-s-D-23} < 2^{D+3} \cdot 2^{-s-D-23} = 2^{-s-20} \quad (D \ge 8).$$

At $D = 8$, this inequality holds because

$$5.7 \times 44.5 = 253.65 < 2048 = 2^{11}.$$

In particular, the passband coefficients and recurrence ratio satisfy

$$|\tilde{c}_0(u_2) - c_0(u_2)|,\ |\tilde{c}_1(u_2) - c_1(u_2)| < 2 \cdot 2^{-s-D-23} = 2^{-s-D-22},$$
$$|\tilde{\rho}(u_2) - \rho(u_2)| < 12 \cdot 2^{-s-D-23} < 2^{-s-D-19}.$$

Claim: Repeated squaring for $\theta(u_2)$ and 1-ulp rounding. For $\theta(u_2) = (1-\rho(u_2)^{t_{1,\ell}})^{-1}$ at $u_2 \neq r_\ell/2$, we have $M(u_2) > 0.499$, $t_{1,\ell} \le 2^{D/2}$, and

$$|c_0(u_2) - c_1(u_2)| > \frac{0.424}{t_{1,\ell}} \ge 0.424 \cdot 2^{-D/2} \gg 2^{-s-D-20}.$$

Together with $|\tilde{c}_0(u_2) - c_0(u_2)|, |\tilde{c}_1(u_2) - c_1(u_2)| < 2^{-s-D-22}$, this ensures $|\tilde{\rho}(u_2) - \rho(u_2)| < 2^{-s-D-19}$ and $|\tilde{\rho}(u_2)| < 1$. Forming $\widetilde{(\rho^{t_{1,\ell}})}$ by $\log_2 t_{1,\ell} \le D/2$ repeated squarings on $\widetilde{\mathbb{R}}_S$ therefore satisfies

$$\begin{aligned}\left|\widetilde{(\rho^{t_{1,\ell}})} - \rho(u_2)^{t_{1,\ell}}\right| &\le t_{1,\ell} 2^{-S} + t_{1,\ell} \max(|\tilde{\rho}(u_2)|, |\rho(u_2)|)^{t_{1,\ell}-1} |\tilde{\rho}(u_2) - \rho(u_2)| \\ &\le 2^{D/2-S} + 2^{D/2} \cdot 2^{-s-D-19} < 2^{-s-22},\end{aligned}$$

and dividing $(1 - \widetilde{(\rho^{t_{1,\ell}})})^{-1}$ yields $|\tilde{\theta}(u_2) - \theta(u_2)| < 2^{-s-20}$ unconditionally for all $s \ge 8$ and $D = \max(s, K_0)$.

At $s = q + 10$, every pre-rounded table value $\tilde{v} \in \widetilde{\mathbb{R}}_S$ satisfies $|\tilde{v} - v| < 2^{-s-20} < 2^{-q-10}$, so rounding $\bar{v} := \mathcal{Q}_q(\tilde{v})$ gives

$$|\bar{v} - v| \le |\mathcal{Q}_q(\tilde{v}) - \tilde{v}| + |\tilde{v} - v| \le \left(\frac{1}{2} + 2^{-10}\right) 2^{-q} < 2^{-q},$$

proving (7). The reused error-budget thresholds across (i)–(v) and the composite expressions are summarized below:

Error component	Bound	Where used
Series truncation tail (N_{terms} terms)	$< 2^{-S-16}$	π, $\ln 2$, exp, sin / cos, sinc, erf
Perturbation induced by $\tilde{\ell}_2, \tilde{h}$ (range reduction $\lvert\tilde{r} - r_m\rvert$, cutoff slack)	$< 2^{-100}$	exp, sin / cos, erf cutoff
Primitive $\widetilde{\mathbb{R}}_S$-evaluation error $\lvert\tilde{y} - f(\tilde{x})\rvert$	$< 2^{-s-D-32}$ (and $< 2^{-s-D-23}$)	(i)–(iv) (and (v) erf)
Pre-rounded composite table error $\lvert\tilde{v} - v\rvert$	$< 2^{-s-20} < 2^{-q-10}$ ($s = q + 10$)	all level and leaf tables
Final $\mathcal{Q}_q$ table entry error $\lvert\bar{v} - v\rvert$	$< 2^{-q}$ (real), $< \sqrt{2} \cdot 2^{-q}$ (complex)	1-ulp contract (7)

Finally, each primitive evaluates $N_{\text{terms}} = \Theta(D)$ terms using $O(1)$ schoolbook multiplications and divisions of $O(S) = O(D)$-bit integers per term, taking $O(D^3)$ bit operations. □

A.2 Register width inventory and breadth-first tape schedule

Scratch and accumulator register widths. Complementing the uniform record width $w_\ell = 2w_{\text{coord},\ell}$ with $w_{\text{coord},\ell} = q_{\text{child}} + H_{\ell+1} + 3 = \Theta(\log n)$ in (6), all intermediate scratch and accumulator registers across the algorithm have explicitly bounded widths:

- *Unrounded products and shifted dividends in* $\mathrm{RM}_q, \mathrm{RD}_q$*:* Multiplying two $w_{\text{coord},\ell}$-bit signed integers in RM_q before rounding from fractional precision $2q$ to q, and shifting a stored numerator $N_z = 2^q z$ by 2^q in RD_q, use scratch product and dividend registers of respective bit widths

$$2w_{\text{coord},\ell} + 2 = \Theta(\log n), \qquad w_{\text{coord},\ell} + q + 2 = \Theta(\log n).$$

- *Nyquist spatial alternating sum* $A_{\mathrm{N}}^{(\ell,i)}$ *and* `NyquistSolve` *scratch sum* $\tilde{S}$*:* Accumulating the unscaled alternating sum $\sum_{j_2=0}^{r_\ell - 1} (-1)^{j_2} x^{(\ell,i)}(t_{1,\ell}/2 + t_{1,\ell} j_2)$ uses a scratch accumulator before the exact $k_{2,\ell}$-bit shift r_ℓ^{-1} places $A_{\mathrm{N}}^{(\ell,i)}$ into a standard w_ℓ-bit record on T_{stack}. Similarly, accumulating $\tilde{S} = \sum_{i=0}^{t_{1,\ell}-1} (-1)^i \tilde{P}_i$ inside `NyquistSolve` before dividing by $t_{1,\ell} = 2^{k_{1,\ell}}$ and rounding to $\widetilde{\mathbb{C}}_{q_{\text{child}}}$ uses a second scratch accumulator, with respective bit widths

$$w_{\text{coord},\ell} + k_{2,\ell} + 1 = \Theta(\log n), \qquad w_{\text{coord},\ell} + k_{1,\ell} + 1 = \Theta(\log n).$$

- *Step 1 Kronecker integers and overlap-add accumulators:* Each packed block or filter integer in Theorem 6.3 has at most $\tilde{W}_\ell B_{\mathrm{slot}} + 1$ bits. Their integer product and each unpacked overlap-add slot accumulator at fractional precision $2q_{\mathrm{child}}$ have respective bit widths at most

$$2\tilde{W}_\ell B_{\mathrm{slot}} + 2 = \Theta(\tilde{W}_\ell p_{\ell+1}), \qquad B_{\mathrm{slot}} + 1 = \Theta(p_{\ell+1}).$$

- *Leaf `BaseRadix2FFT` records and root carry registers:* At depth L with $q_b := p_L + K_L + 4$, leaf records on $\widetilde{\mathbb{C}}_{q_b}$ and signed integer registers for root chunk extraction and radix-2^b carry propagation have respective bit widths

$$w_{\mathrm{leaf}} := 2(q_b + H_L + 3) = \Theta(\log n), \qquad 2b + K_0 + 3 = \Theta(\log n).$$

Contents of the shared per-level table tape T_{tab}. Before starting the recursive transforms, the outer algorithm writes once per level $\ell \in \{0, \dots, L\}$ a sequential level-table segment onto T_{tab} containing four components.

(a) For $\ell < L$, the machine first computes the Lemma 3.2 approximation $\tilde{x} \in \widetilde{\mathbb{R}}_S$ of $\alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi}$ at $s = q_{\mathrm{child}} + 10$ and sets the integer window radius

$$m_{1,\ell} := \lceil \tilde{x} + 2^{-s-20} \rceil, \qquad \tilde{W}_\ell := \min(2m_{1,\ell} + 1, r_\ell).$$

Since $|\tilde{x} - \alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi}| < 2^{-s-20}$, this value equals $\lceil \alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi} \rceil$ or that integer plus one. It lies in the admissible interval of Lemma 6.5 because

$$\alpha_\ell \sqrt{p_{\ell+1} \ln 2/\pi} \le \frac{2 \ln 2}{\pi} p_{\ell+1} < 0.45\, p_{\ell+1}.$$

The machine then writes the $t_{1,\ell}$ guarded modulo-r_ℓ folded filter tables $\bar{H}^{(j_1)} \in \widetilde{\mathbb{R}}_{q_{\mathrm{child}}}^{\tilde{W}_\ell}$ of length $\tilde{W}_\ell$ and their Kronecker-packed integers $\mathrm{Pack}_{B_{\mathrm{slot}}}(2^{q_{\mathrm{child}}} \bar{H}^{(j_1)})$ of $O(\tilde{W}_\ell p_{\ell+1})$ bits each.

(b) For $\ell < L$, it writes the r_ℓ 1-ulp faithful twist roots $\bar{\zeta}_{k_2} \in \widetilde{\mathbb{C}}_{q_{\mathrm{child}}}$ approximating $e^{\pi i k_2 / r_\ell}$ in natural order $k_2 = 0, \dots, r_\ell - 1$. Each root is evaluated via Lemma 3.2 at $s = q_{\mathrm{child}} + 10$ and rounded via $\mathcal{Q}_{q_{\mathrm{child}}}$.

(c) For $\ell < L$, it writes the r_ℓ 1-ulp faithful column solver tuples $(\mu(u_2), \hat{\rho}(u_2), \hat{\theta}(u_2)) \in \widetilde{\mathbb{R}}_{q_{\mathrm{child}}}^3$ in natural order $u_2 = 0, \dots, r_\ell - 1$. For $0 \le u_2 < r_\ell/2$, we have $c_0(u_2) > c_1(u_2)$, $M(u_2) = c_0(u_2) > 0.49999999$, $\rho(u_2) = -c_1(u_2)/c_0(u_2) \in (-1, 0)$, and $1 - \rho(u_2)^{t_{1,\ell}} > 0.86$. On this half, the pre-rounded approximations $\tilde{c}_0(u_2), \tilde{c}_1(u_2) \in \widetilde{\mathbb{R}}_S$ are evaluated via Lemma 3.2 at $s = q_{\mathrm{child}} + 10$ to form $(\mu(u_2), \hat{\rho}(u_2), \hat{\theta}(u_2))$ via $\mathcal{Q}_{q_{\mathrm{child}}}$. These three rounded scalars approximate $(M(u_2)^{-1}, \rho(u_2), \theta(u_2))$ within $\frac{1}{4}u$ each, and only those three rounded scalars are written to T_{tab}. For $r_\ell/2 < u_2 < r_\ell$, the tuple is set by reflection symmetry to

$$(\mu(u_2),\, \hat{\rho}(u_2),\, \hat{\theta}(u_2)) := (\mu(r_\ell - u_2),\, \hat{\rho}(r_\ell - u_2),\, \hat{\theta}(r_\ell - u_2)).$$

At the Nyquist column $u_2 = r_\ell/2$, `NyquistSolve` reads only the Nyquist reciprocal $\mu_{\mathrm{N}} := \mu(r_\ell/2) \approx c^{-1}$, evaluated via Lemma 3.2 at $s = q_{\mathrm{child}} + 10$ and rounded via $\mathcal{Q}_{q_{\mathrm{child}}}$, with placeholder values $(\hat{\rho}(r_\ell/2), \hat{\theta}(r_\ell/2)) := (-1, 0)$.

(d) At the leaf level $\ell = L$ with $R_L = 2^{K_L} \le 4p_L^2$, for each butterfly stage $s \in \{1, \dots, K_L\}$ of `BaseRadix2FFT`, it writes the contiguous stage-s 1-ulp faithful twiddle list $(\bar{\omega}_{s,j})_{j=0}^{2^{s-1}-1} \in \widetilde{\mathbb{C}}_{q_b}^{2^{s-1}}$

approximating $e^{2\pi i j \cdot 2^{K_L - s}/R_L}$ at precision $q_b := p_L + K_L + 4$. Each twiddle entry is evaluated via Lemma 3.2 at target precision $q_b + 10$ followed by $\mathcal{Q}_{q_b}$, streaming

$$\sum_{s=1}^{K_L} 2^{s-1} = R_L - 1 < R_L$$

total twiddle records sequentially without strided rescans.

Breadth-first downward, leaf, and upward tape sweeps. To execute the recursion tree in $O(1)$ tapes without recursion-stack tape growth or head shuttling between parent and child regions, the machine uses, besides the input and output tapes, two alternating data tapes $T^{(0)}_{\text{data}}, T^{(1)}_{\text{data}}$, four scratch tapes $T_1, \ldots, T_4$, the table tape T_{tab}, and a level-segmented stack tape T_{stack}. It also allocates $O(1)$ dedicated work tapes for the nested inner multiplier (Section 7.2):

- *Downward sweep for $\ell = 0, \ldots, L-1$:* At the start of level ℓ, $T^{(\ell \bmod 2)}_{\text{data}}$ holds the R_0/R_ℓ node input vectors $x^{(\ell,i)} \in \widetilde{\mathbb{C}}^{R_\ell}_{p_\ell+2}$ for $0 \le i < R_0/R_\ell$ contiguously in natural node order as uniform fixed-width records of width w_ℓ, occupying $R_0 w_\ell = \Theta(n)$ bits. Positioning T_{tab} at the level-ℓ table segment and $T^{(\ell+1 \bmod 2)}_{\text{data}}$ at its start, the machine processes each node $i = 0, \ldots, R_0/R_\ell - 1$ sequentially in three steps.

  First, it scans $x^{(\ell,i)}$ on $T^{(\ell \bmod 2)}_{\text{data}}$ using a $\log_2 t_{1,\ell}$-bit counter to accumulate the exact Nyquist spatial alternating sum $A^{(\ell,i)}_{\text{N}} \in \widetilde{\mathbb{C}}_{q_{\text{child}}}$ from the stride-$t_{1,\ell}$ offset-$t_{1,\ell}/2$ entries (25). Appending $A^{(\ell,i)}_{\text{N}}$ to the level-ℓ segment of T_{stack} costs $O(R_\ell w_\ell)$ steps per node and writes $O((R_0/R_\ell)w_\ell) = O(n)$ stack bits per level.

  Second, it de-interleaves $x^{(\ell,i)}$ into $t_{1,\ell}$ rows of length r_ℓ (Lemma 3.3(i)) and runs Step 1 row spreading (Theorem 6.3). Row j_1's pre-packed filter integer of $O(\tilde{W}_\ell p_{\ell+1})$ bits is replayed from T_{tab} once for each of the $\lceil r_\ell/\tilde{W}_\ell \rceil$ blocks in row j_1, taking $O(r_\ell p_{\ell+1})$ table-replay steps per row and $O(R_\ell p_{\ell+1})$ per node.

  Third, it runs Step 2a pre-twist by replaying the r_ℓ twist roots from T_{tab} once per row in $O(R_\ell p_{\ell+1})$ steps per node. It then runs the Step 2b polynomial transform $G_{t_{1,\ell}}$ (Lemma 3.5), whose output polynomials $\widehat{Q}_{v_1}$ arrive in natural order $v_1 = 0, \ldots, t_{1,\ell} - 1$, followed by Step 2c post-twist. The resulting $t_{1,\ell}$ child input vectors $\widehat{V}_{v_1,\cdot}$, padded to width $w_{\ell+1}$, are appended sequentially to $T^{(\ell+1 \bmod 2)}_{\text{data}}$. At the end of level ℓ, rewinding $T^{(\ell+1 \bmod 2)}_{\text{data}}$ takes $O(R_0 w_{\ell+1}) = O(n)$ steps, and replaying the level-ℓ table across all R_0/R_ℓ nodes takes $O(R_0 p_{\ell+1}) = O(n)$ steps.

- *Leaf evaluation at $\ell = L$:* The machine reads the R_0/R_L contiguous leaf vectors of length $R_L \le 4p_L^2$ sequentially from $T^{(L \bmod 2)}_{\text{data}}$, writes the evaluated leaf outputs sequentially back to $T^{(L \bmod 2)}_{\text{data}}$, and rewinds at the end in $O(n)$ steps. On each leaf vector, `BaseRadix2FFT` first bit-reverses the R_L records in $K_L = \log_2 R_L$ sequential even/odd split passes in $O(R_L(1+K_L)q_b)$ steps. In each stage $s \in \{1, \ldots, K_L\}$, it splits each block of 2^s records into its upper and lower halves of length 2^{s-1} on T_1, T_2 while replaying the 2^{s-1}-entry stage-s twiddle list from T_{tab} once per block. Across all K_L stages and all R_0/R_L leaves, leaf table replay and tape movement take

$$\frac{R_0}{R_L} \cdot O(R_L(1+K_L)q_b) = O(n \log\log n)$$

  Turing machine steps.

- *Upward sweep for $\ell = L-1$ down to 0:* At the start of level ℓ, $T_{\text{data}}^{(\ell+1 \bmod 2)}$ holds the $(R_0/R_\ell)t_{1,\ell}$ child output vectors grouped into R_0/R_ℓ contiguous node groups of $t_{1,\ell}$ rows of length r_ℓ in natural node order $i = 0, \ldots, R_0/R_\ell - 1$. The machine rewinds T_{stack} across the level-ℓ segment in $O(n)$ steps to its first entry $i = 0$ and rewinds $T_{\text{data}}^{(\ell \bmod 2)}$ to its start. Processing the node groups sequentially in natural forward order $i = 0, \ldots, R_0/R_\ell - 1$, it first reads node i's $t_{1,\ell} \times r_\ell$ child outputs from $T_{\text{data}}^{(\ell+1 \bmod 2)}$ and routes them into column-major order (u_2, u_1) via Lemma 3.3(ii). Next, it reads $A_{\text{N}}^{(\ell,i)}$ sequentially from the level-ℓ segment of T_{stack} and replays the r_ℓ solver tuples from T_{tab} sequentially as $u_2 = 0, \ldots, r_\ell - 1$ in $O(r_\ell p_{\ell+1})$ table steps per node to run `CyclicSolve`/`NyquistSolve` and quantize to $\widetilde{\mathbb{C}}_{p_\ell+2}$. Finally, it transposes $(\widehat{Z}_{u_1,u_2})$ from column-major order (u_2, u_1) back to natural frequency order $u_2 + r_\ell u_1$ via Lemma 3.3(i) and appends the solved parent vector, re-padded to width w_ℓ, sequentially to $T_{\text{data}}^{(\ell \bmod 2)}$. At the end of level ℓ, it rewinds $T_{\text{data}}^{(\ell \bmod 2)}$ and erases and rewinds the level-ℓ segment of T_{stack} in $O(n)$ steps.

Summing over all $L = O(\log \log n)$ levels, total table replay and level-sweep coordination take $O(n \log \log n) = o(n \log n)$ Turing machine steps.

B Multitape routing proofs

Proof of Lemma 3.3. We first record the amortized cost of binary record counters and then prove parts (i) and (ii) in turn.

Counter amortization. Incrementing or decrementing a binary counter on a dedicated counter tape N times in succession flips bit b at most $\lceil N/2^b \rceil$ times. Across all N operations, this costs $O(N + \log N)$ total head steps, or amortized $O(1)$ steps per increment.

(i) Matrix transposition. A $t_1 \times r$ array of fixed-width w-bit records occupies

$$t_1 r w = \Theta(t_1 r q)$$

tape cells. Here row-major order (j_1, j_2) consists of $t_1 = 2^{k_1}$ contiguous rows of length r, and column-major order (j_2, j_1) consists of r contiguous columns of length t_1 (cf. [HvdH21, Section 2.1]). To transpose from row-major order (j_1, j_2) to column-major order (j_2, j_1), execute $k_1 = \log_2 t_1$ sequential passes $s = 1, \ldots, k_1$. At the start of pass s, the tape holds $t_1/2^{s-1}$ blocks, each consisting of r contiguous sub-columns of 2^{s-1} records. In a single scan over the tape, copy even-indexed blocks to auxiliary tape T_1 and odd-indexed blocks to auxiliary tape T_2. A $(k_2 + s - 1)$-bit record counter tracks the block boundaries during this scan. Next, rewind T_1 and T_2, and scan T_1 and T_2 in lockstep to interleave each pair of 2^{s-1}-record sub-columns into a 2^s-record sub-column, producing $t_1/2^s$ blocks of r sub-columns of length 2^s. After k_1 passes, the single block holds r contiguous columns of length t_1.

Reversing the k_1 passes transposes column-major order (j_2, j_1), indexed by $j = j_1 + t_1 j_2$, back into row-major order (j_1, j_2). Specifically, each reversed pass splits every length-2^s sub-column into its first and second halves of length 2^{s-1} on T_1, T_2 and concatenates them block-wise. Each pass scans $O(t_1 r w)$ cells, totaling

$$O\big(t_1 r q (1 + \log_2 t_1)\big)$$

Turing machine steps.

(ii) Odd-stride column routing. First transpose $(\widehat{C}_{v_1,u_2})$ from row-major order (v_1, u_2) to column-major order (u_2, v_1) by (i) in $O(t_1 rq(1+\log_2 t_1))$ steps. For each column $u_2 \in \{0, \dots, r-1\}$, its t_1 records now occupy a contiguous block in natural order $v_1 = 0, \dots, t_1 - 1$. In one sequential scan over all columns $u_2 = 0, \dots, r-1$, maintain the column stride and destination tag

$$s(u_2) := (2u_2 - 1) \bmod t_1, \qquad u_1 = (2u_2 - 1)v_1 \bmod t_1.$$

A k_1-bit accumulator is initialized to 0 at the start of each column and adds $s(u_2)$ modulo t_1 as v_1 increments from 0 to $t_1 - 1$. Prepending the k_1-bit destination tag u_1 to each record takes $O(k_1) \le O(q)$ steps per record.

Since each tagged record has fixed width $w + k_1 = \Theta(q)$, execute $k_1 = \log_2 t_1$ passes of standard bottom-up merge sort within all r columns simultaneously. In pass $s \in \{0, \dots, k_1 - 1\}$, bit s of a k_1-bit record counter detects alternating sorted runs of length 2^s within each column and distributes them to T_1 and T_2. Merging each pair of length-2^s runs on T_1 and T_2 by comparing their k_1-bit tags produces a sorted run of length 2^{s+1}. Each of the k_1 passes scans $O(t_1 rq)$ tape cells, taking

$$O(t_1 rq(1 + \log_2 t_1))$$

steps in total. □

C Deferred error and filter calculations

C.1 Proof of Lemma 5.2

Proof of Lemma 5.2. By (7), on $\widetilde{\mathbb{C}}_{q_{\text{child}}}$ each real table entry has error $\le \frac{1}{4}u = 0.25u$, while each complex rounded product $\mathrm{RM}_{q_{\text{child}}}$ has error $\le \frac{\sqrt{2}}{8}u < 0.177u$. For $u_2 \ne r/2$, by reflection symmetry between $u_2 < r/2$ and $u_2 > r/2$, it suffices to consider $u_2 < r/2$. Then $M(u_2) > 0.49999999$ implies $M(u_2)^{-1} < 2.0000001$ and $|\mu| \le 2.0001$. Setting $B_i^\dagger := M(u_2)^{-1}\widehat{Y}_i$, we have $|B_i^\dagger| \le 2.0001\mathcal{M}_Y$ and

$$e_i^{(B)} := |\tilde{B}_i - B_i^\dagger| \le |\mu - M(u_2)^{-1}|\,|\widehat{Y}_i| + \frac{\sqrt{2}}{8}u \le \left(\frac{1}{4} + \frac{\sqrt{2}}{8}\right) u\mathcal{M}_Y < 0.427u\mathcal{M}_Y.$$

In the prefix sweep, $P_0^\dagger = B_0^\dagger$ and $P_i^\dagger = B_i^\dagger + \rho P_{i-1}^\dagger$ with $|\rho| = a < 1$, so $|P_i^\dagger| \le 2.0001(i+1)\mathcal{M}_Y$. Since $|\hat{\rho}| \le 1 + \frac{1}{4}u$, the prefix error $e_i^{(P)} := |\tilde{P}_i - P_i^\dagger|$ satisfies $e_0^{(P)} = e_0^{(B)} \le 0.427u\mathcal{M}_Y$ and, for $1 \le i \le t_1 - 1$,

$$e_i^{(P)} \le |\hat{\rho}|e_{i-1}^{(P)} + |\hat{\rho} - \rho|\,|P_{i-1}^\dagger| + e_i^{(B)} + \frac{\sqrt{2}}{8}u \le \left(1 + \frac{1}{4}u\right) e_{i-1}^{(P)} + (0.604 + 0.501i)u\mathcal{M}_Y.$$

Unrolling this linear recurrence from $i = 0$ to $t_1 - 1$, using $(1 + \frac{1}{4}u)^{t_1} \le e^{t_1 u/4} < 1.0001$ since $t_1 u \le 2^{-20}$ and the triangular sum $\sum_{i=1}^{t_1-1} i = \frac{t_1(t_1-1)}{2} < \frac{1}{2}t_1^2$, gives for $t_1 \ge 2$:

$$e_{t_1-1}^{(P)} \le 1.0001\left(0.604t_1 + 0.501\frac{t_1(t_1-1)}{2}\right) u\mathcal{M}_Y < (0.605t_1 + 0.251t_1^2)u\mathcal{M}_Y < 0.56t_1^2 u\mathcal{M}_Y.$$

Next, in the wrap-around closure $Z_{t_1-1}^{\text{comp}} = \mathrm{RM}_{q_{\text{child}}}(\hat{\theta}, \tilde{P}_{t_1-1})$ versus $Z_{t_1-1}^\dagger = \theta P_{t_1-1}^\dagger$, where $\theta := (1 - \rho^{t_1})^{-1} < 1.163$, $|\hat{\theta}| < 1.164$, and $|P_{t_1-1}^\dagger| \le 2.0001t_1\mathcal{M}_Y$, we have

$$\begin{aligned} e_{t_1-1}^{(Z)} := |Z_{t_1-1}^{\text{comp}} - Z_{t_1-1}^\dagger| &\le |\hat{\theta}|e_{t_1-1}^{(P)} + |\hat{\theta} - \theta|\,|P_{t_1-1}^\dagger| + \frac{\sqrt{2}}{8}u \\ &< (0.652t_1^2 + 0.501t_1 + 0.177)u\mathcal{M}_Y < 0.95t_1^2 u\mathcal{M}_Y. \end{aligned}$$

Finally, in the second recurrence pass $Z_i^{\mathrm{comp}} = \tilde{B}_i + \mathrm{RM}_{q_{\mathrm{child}}}(\hat{\rho}, Z^{\mathrm{comp}}_{(i-1) \bmod t_1})$ versus $Z_i^\dagger = B_i^\dagger + \rho Z^\dagger_{(i-1) \bmod t_1}$ for $i = 0, \dots, t_1 - 2$, Theorem 5.1(2) bounds the exact solution uniformly by $\|Z^\dagger\|_\infty \le \|B^{-1}\|_{\infty\to\infty}\|\widehat{Y}\|_\infty < 2.36 t_1 \mathcal{M}_Y$. Hence at every step of the second pass, $|\hat{\rho} - \rho|\,|Z^\dagger_{(i-1) \bmod t_1}| \le \frac{1}{4}u(2.36 t_1 \mathcal{M}_Y) = 0.590 t_1 u \mathcal{M}_Y$, so each of the $t_1 - 1$ steps adds local error at most $(0.427 + 0.177 + 0.590t_1)u\mathcal{M}_Y = (0.604 + 0.590t_1)u\mathcal{M}_Y$. Across at most $t_1 - 1$ steps with amplification $(1 + \frac{1}{4}u)^{t_1 - 1} < 1.0001$, we obtain

$$\|Z^{\mathrm{comp}} - Z^\dagger\|_\infty \le 1.0001\Big(e^{(Z)}_{t_1-1} + (t_1 - 1)(0.604 + 0.590t_1)u\mathcal{M}_Y\Big) < 1.85\, t_1^2\, u\, \mathcal{M}_Y \le 80\, t_1^2\, u\, \mathcal{M}_Y.$$

For $u_2 = r/2$, $|\mu_{\mathrm{N}}| \le 2.0001$, $\|\tilde{D}_i - c^{-1}\widehat{Y}_i\|_\infty \le 0.427 u\mathcal{M}_Y$, and the alternating prefix sum $\tilde{P}_i = \tilde{D}_i - \tilde{P}_{i-1}$ is exact on $\widetilde{\mathbb{C}}_{q_{\mathrm{child}}}$, so $\|\tilde{P}_i - P_i^\dagger\|_\infty \le 0.427 i u \mathcal{M}_Y < 0.427 t_1 u \mathcal{M}_Y$. The alternating sum $\tilde{S}$ is exact on $\widetilde{\mathbb{C}}_{q_{\mathrm{child}}}$, and the single rounding of $(A_{\mathrm{N}} - \tilde{S})/t_1$ adds $\le 0.177u$, yielding

$$\|Z^{\mathrm{comp}} - Z^\dagger\|_\infty \le (0.854 t_1 + 0.177)u\mathcal{M}_Y \le 5 t_1 u \mathcal{M}_Y \le 80 t_1^2 u \mathcal{M}_Y.$$

Since $5t_1 \le 2.5\, t_1^2$ for $t_1 \ge 2$, both cases are below $2.5\, t_1^2\, u\, \mathcal{M}_Y$. We state the generous constant 80, which is the value used in Lemma 6.5. □

C.2 Proof of Lemma 6.1

Proof of Lemma 6.1. **Part 1.** For $j_1 = 0$, $\kappa_0^{(0)} = 1$ and $\kappa_d^{(0)} = 0$ for all $d \ne 0$, so $\|\kappa^{(0)}\|_1 = 1$. For $1 \le j_1 < t_{1,\ell}$, let $\delta := j_1/t_{1,\ell} \in (0,1)$. For $d \in \{0, -1\}$, $|\kappa_0^{(j_1)}| + |\kappa_{-1}^{(j_1)}| \le \mathrm{sinc}(\delta) + \mathrm{sinc}(1-\delta) = \frac{\sin(\pi\delta)}{\pi\delta(1-\delta)}$. On $[0,1]$, let $h(\delta) := \sin(\pi\delta) - 4\delta(1-\delta)$. By symmetry $h(\delta) = h(1-\delta)$, it suffices to consider $\delta \in [0, 1/2]$, where $h(0) = h(1/2) = 0$, $h'(0) = \pi - 4 < 0$, $h'(1/2) = 0$, and $h''(\delta) = -\pi^2 \sin(\pi\delta) + 8$ decreases from $8 > 0$ at $\delta = 0$ to $8 - \pi^2 < 0$ at $\delta = 1/2$, vanishing at a unique $\delta_* \in (0, 1/2)$. On $[\delta_*, 1/2]$, h is concave with $h(1/2) = h'(1/2) = 0$, so $h(\delta) \le h(1/2) + h'(1/2)(\delta - 1/2) = 0$, and in particular $h(\delta_*) \le 0$. On $[0, \delta_*]$, h is convex, so $h(\delta) \le \max(h(0), h(\delta_*)) \le 0$. Hence $\sin(\pi\delta) \le 4\delta(1-\delta)$ on $[0,1]$ and the two central terms sum to at most $4/\pi$. For $d \ge 1$ and $d \le -2$, pairing $d = k \ge 1$ with $d = -1 - k$ gives $|\kappa_k^{(j_1)}| + |\kappa_{-1-k}^{(j_1)}| \le \frac{2}{\pi k} e^{-\pi k^2/\alpha_\ell^2}$. Since $k \mapsto \frac{1}{k} e^{-\pi k^2/\alpha_\ell^2}$ is decreasing on $[1, \infty)$,

$$\sum_{k=1}^{\infty} \frac{2}{\pi k} e^{-\pi k^2/\alpha_\ell^2} \le \frac{2}{\pi} e^{-\pi/\alpha_\ell^2} + \frac{2}{\pi}\int_1^\infty \frac{e^{-\pi t^2/\alpha_\ell^2}}{t}\, dt = \frac{2}{\pi} e^{-\pi/\alpha_\ell^2} + \frac{1}{\pi}\int_{\pi/\alpha_\ell^2}^\infty \frac{e^{-u}}{u}\, du.$$

For the integral over $[\pi/\alpha_\ell^2, 1]$ and $x = \pi/\alpha_\ell^2 \in (0, 0.445)$, we have

$$\int_{\pi/\alpha_\ell^2}^1 \frac{e^{-u}}{u}\, du \le \int_{\pi/\alpha_\ell^2}^1 \frac{1 - u + u^2/2}{u}\, du = \ln(\alpha_\ell^2/\pi) - (1 - \pi/\alpha_\ell^2) + \frac{1}{4}(1 - \pi^2/\alpha_\ell^4),$$

$$2e^{-x} + x - 1 + \frac{1}{4}(1 - x^2) \le 2(1 - x + x^2/2) + x - 1 + \frac{1}{4}(1 - x^2) = \frac{5}{4} - x\left(1 - \frac{3}{4}x\right) < \frac{5}{4} = 1.25.$$

Combining these bounds gives $\sum_{k=1}^\infty \frac{2}{\pi k} e^{-\pi k^2/\alpha_\ell^2} \le \frac{2}{\pi}\ln\alpha_\ell + \frac{1.25 - \ln\pi + E_1(1)}{\pi}$. Since $(1+v)(1 - v/2 + v^2/3) = 1 + \frac{v}{6}(3 - v + 2v^2) \ge 1$ for all $v \ge 0$, we have $\frac{1}{1+v} \le 1 - \frac{v}{2} + \frac{v^2}{3}$ and hence

$$E_1(1) = \int_1^\infty \frac{e^{-u}}{u}\, du = e^{-1}\int_0^\infty \frac{e^{-v}}{1+v}\, dv \le e^{-1}\int_0^\infty e^{-v}\left(1 - \frac{v}{2} + \frac{v^2}{3}\right) dv = \frac{7}{6e} < 0.42920.$$

Using $\ln \pi > 1.14472$, the total constant is at most

$$\frac{4 + 1.25 - 1.14472 + 0.42920}{\pi} = \frac{4.53448}{\pi} < 1.444 < 1.62.$$

Part 2. For $d \geq m_{1,\ell}+1$, since $\delta \in [0,1)$, we have $d+\delta \geq d \geq m_{1,\ell}+1$, so $\sum_{d=m_{1,\ell}+1}^{\infty} |g(d+\delta)| \leq \sum_{d=m_{1,\ell}+1}^{\infty} \frac{e^{-\pi d^2/\alpha_\ell^2}}{\pi d} \leq \int_{m_{1,\ell}}^{\infty} \frac{e^{-\pi t^2/\alpha_\ell^2}}{\pi t}\, dt$. For $d \leq -m_{1,\ell} - 1$, let $k := -d \geq m_{1,\ell} + 1$, so that

$$|d + \delta| = k - \delta > k - 1 \geq m_{1,\ell}.$$

Separating the single negative boundary term $k = m_{1,\ell} + 1$, which satisfies $|g(-m_{1,\ell} - 1 + \delta)| < \frac{e^{-\pi m_{1,\ell}^2/\alpha_\ell^2}}{\pi m_{1,\ell}} \leq \frac{2^{-p_{\ell+1}}}{\pi m_{1,\ell}}$, and bounding $\sum_{k=m_{1,\ell}+2}^{\infty} \frac{e^{-\pi(k-1)^2/\alpha_\ell^2}}{\pi(k-1)} \leq \int_{m_{1,\ell}}^{\infty} \frac{e^{-\pi t^2/\alpha_\ell^2}}{\pi t}\, dt$ gives

$$\tau \leq \frac{2^{-p_{\ell+1}}}{\pi m_{1,\ell}} + 2\int_{m_{1,\ell}}^{\infty} \frac{e^{-\pi t^2/\alpha_\ell^2}}{\pi t}\, dt \leq \left(\frac{1}{\pi m_{1,\ell}} + \frac{1}{\pi p_{\ell+1} \ln 2}\right) 2^{-p_{\ell+1}},$$

proving (28). Since $p_{\mathrm{work},\ell} \geq 8$ and $p_{\ell+1} \geq 20$ imply $m_{1,\ell} \geq \lceil \frac{2\ln 2}{\pi}\sqrt{160} \rceil = 6$, we have

$$\frac{1}{\pi m_{1,\ell}} + \frac{1}{\pi p_{\ell+1} \ln 2} \leq \frac{1}{6\pi} + \frac{1}{20\pi \ln 2} < 0.077 < 0.10.$$

At all non-leaves, $m_{1,\ell} \geq 7$, giving $\frac{1}{\pi m_{1,\ell}} + \frac{1}{\pi p_{\ell+1} \ln 2} \leq \frac{1}{7\pi} + \frac{1}{20\pi \ln 2} < 0.069$.

Part 3. By Poisson summation, substituting $s = t + j_1/t_{1,\ell}$ in $\int_{-\infty}^{\infty} g(t + j_1/t_{1,\ell}) e^{2\pi i t(u_2/r_\ell - m)}\, dt$, the discrete Fourier coefficients of $h^{(j_1),\mathrm{per}} \in \mathbb{C}^{r_\ell}$ are

$$\widehat{h^{(j_1),\mathrm{per}}}_{u_2} = \frac{1}{r_\ell} \sum_{m \in \mathbb{Z}} \hat{g}\left(\frac{u_2}{r_\ell} - m\right) e^{-2\pi i (j_1/t_{1,\ell})(u_2/r_\ell - m)}.$$

In contrast, U° in Theorem 4.2 evaluates $h^{(j_1),\mathrm{per}}$ at reversed index $j_2 - k_2$, flipping the phase sign. Since $\hat{g} \geq 0$ and $\sum_{m\in\mathbb{Z}} \hat{g}(u_2/r_\ell - m) = 1$, we have $|\widehat{h^{(j_1),\mathrm{per}}}_{u_2}| \leq \frac{1}{r_\ell}$ for all u_2. By Parseval's identity on $\mathbb{Z}_{r_\ell}$,

$$\|h^{(j_1),\mathrm{per}}\|_2^2 = r_\ell \sum_{u_2=0}^{r_\ell - 1} |\widehat{h^{(j_1),\mathrm{per}}}_{u_2}|^2 \leq r_\ell \cdot r_\ell \cdot \frac{1}{r_\ell^2} = 1,$$

and Cauchy–Schwarz gives $\|h^{(j_1),\mathrm{per}}\|_1 \leq \sqrt{r_\ell}\|h^{(j_1),\mathrm{per}}\|_2 \leq \sqrt{r_\ell}$. By the triangle inequality, $\|h^{(j_1),\mathrm{tr}} - h^{(j_1),\mathrm{per}}\|_1 \leq \sum_{|e|>m_{1,\ell}} |\kappa_e^{(j_1)}| = \tau$. Finally, at guard precision v_{tap}, summing at most $\lceil W_\ell/r_\ell \rceil \leq W_\ell$ terms per residue class incurs guard error $\leq W_\ell 2^{-v_{\mathrm{tap}}} \leq 2^{-p_{\ell+1}-10}$ before the final rounding to $\widetilde{\mathbb{R}}_{q_{\mathrm{child}}}$, which adds at most $\frac{1}{2}2^{-q_{\mathrm{child}}} = \frac{1}{8}2^{-p_{\ell+1}}$. Thus for each $a \in \{0, \ldots, \tilde{W}_\ell - 1\}$, $|\bar{H}_a^{(j_1)} - h^{(j_1),\mathrm{tr}}_{(o-a) \bmod r_\ell}| \leq (\frac{1}{8} + 2^{-10})2^{-p_{\ell+1}} < \frac{1}{2}2^{-p_{\ell+1}}$. Since $h_d^{(j_1),\mathrm{tr}} = \bar{h}_d^{(j_1)} = 0$ outside the $\tilde{W}_\ell$ indices $d = (o-a) \bmod r_\ell$, $\|\bar{h}^{(j_1)} - h^{(j_1),\mathrm{tr}}\|_1 = \sum_{a=0}^{\tilde{W}_\ell - 1} |\bar{H}_a^{(j_1)} - h^{(j_1),\mathrm{tr}}_{(o-a) \bmod r_\ell}| \leq \tilde{W}_\ell 2^{-p_{\ell+1}-1}$. □